Measuring Geopolitical Alignment and Economic Growth

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We introduce an event-based measure of bilateral geopolitical alignment from 373,020 events across 193 countries over 1960–2024, compiled using large language models and validated with external benchmarks and a human audit. Aggregating alignment with major nations summarizes each country's external geopolitical environment. In local projections using within-country variation, a sustained one-standard-deviation improvement in this environment is followed by about 10 percent higher GDP per capita after 25 years. This relationship is similar across major-nation partners and historical periods, and is accompanied by greater stability, investment, trade openness, and productivity. Accounting exercises map observed realignments into substantial GDP differences. (*JEL*: F50, F51, O11, O40, C80, P16)

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1. Introduction

Research on the wealth of nations has focused on domestic determinants. Early empirical contributions by Barro (1991) and Mankiw, Romer, and Weil (1992) identified physical and human capital accumulation as key drivers of cross-country income differences. Subsequent research has examined technology, institutions, geography, culture, and financial systems as fundamental determinants, with Acemoglu et al. (2019) providing causal evidence that democratization increases GDP per capita by approximately 20 percent in the long run. Yet large cross-country differences in growth and income remain unexplained.

Countries develop not only through these domestic factors, but also within an external environment. This paper studies one dimension of this environment: geopolitical alignment, the state of a country's diplomatic and strategic relationships with the major nations. Cooperative relations with major nations can open access to their markets, capital, and technology, while conflictual relations can bring sanctions, isolation, and security threats. The recent fragmentation of international relations makes understanding the economic consequences of these relationships especially pressing. We construct and validate a measure of geopolitical alignment and show that improvements in a country's external geopolitical environment are followed by large and persistent GDP gains: a sustained one-standard-deviation improvement is followed by about 10 percent higher GDP per capita after 25 years.

To characterize this external geopolitical environment, we introduce a new event-based measure of bilateral geopolitical alignment that captures both the timing and intensity of changes in relations between countries. We first fix a taxonomy of consequential geopolitical events, 25 event types in six issue areas: economic relations, diplomatic and political relations, security and defense, legal and territorial issues, multilateral governance, and other bilateral interactions. A large language model with web search then compiles, for each country pair and year, the events of these types that affect or indicate the state of their relationship.

The model codes each event following established conventions: it classifies the event using the Conflict and Mediation Event Observations (CAMEO) framework and assigns a Goldstein score, from -10 (maximum conflict) to $+10$ (maximum cooperation), that measures the conflictual or cooperative intensity of the event.[1] The resulting database contains 373,020 events in the bilateral relationships between each of the 193 United Nations member states and the 24 major nations from 1960 to 2024.

We validate the database against independent evidence. We first benchmark it against 4,670 documented bilateral developments in eight classes, ranging from war onsets and sanctions to new diplomatic ties and alliances. Our database records an event in the expected conflict or cooperation direction for 92 percent of them within each class's matching window, comparable to the rates of machine-coded databases such as ICEWS (Boschee et al. 2015), GDELT (Leetaru and Schrodt 2013), and Phoenix (Althaus et al. 2020), even though they record many times more events. The key

[1] CAMEO is a framework for coding international political events into cooperation–conflict categories. See Schrodt and Yilmaz (2012) for CAMEO, Goldstein (1992) for the Goldstein scale, and Appendix C.3 for our coding and scoring rules.

difference is that, by focusing on consequential events rather than averaging over large volumes of routine reports, our annual scores better capture the dynamics of international relations: relative to each pair's usual level, they fall more in years of documented conflict and rise more in years of documented cooperation. Finally, in a human audit of 200 dyad-years, trained coders first reconstruct the events independently with the model's codebook but without seeing its output. Coders' individual annual scores correlate with the model's at 0.80, compared with 0.76 between the two coders. The audit also finds only three fabricated records among the model's 856 events.

Building on this event database, we construct a *bilateral geopolitical alignment score* for each country pair and year. The score is a weighted average of current and past event scores, with the weight on past events depreciating at 30 percent per year, so that it tracks how each relationship evolves over time. These scores capture well-known episodes in international relations, such as US–Soviet détente in the 1970s, the India–Pakistan wars of 1965 and 1971, and the deterioration of Greece–Turkey relations after the 1974 Cyprus invasion. At the country level, the *geopolitical alignment index* averages each country's bilateral alignment scores with the 24 major nations, weighted by their shares of world GDP, and so summarizes the external geopolitical environment each country faces. Our measure spans six decades, from Cold War bipolarity through post-1990 globalization to the recent era of geopolitical fragmentation.

In our main application, improvements in the index are followed by GDP gains that emerge within a few years, peak after about a decade, and remain positive two decades later. The estimates come from local projections that compare each country with itself over time, absorb shocks common to its region in each year, and control for past GDP and alignment. A sustained one-standard-deviation improvement, about the size of the deterioration in Russia's relations through the Crimean crisis (2005–2015), is followed by about 10 percent (10 log points) higher GDP per capita after 25 years. Because the index summarizes a country's relations with all major nations, this estimate reflects the combined changes in access and risk that accompany more cooperative relations.

Alternative measures of alignment do not reproduce this relationship. Measures based on UN voting similarity, or built the same way from machine-coded event databases, show little or no detectable relationship with subsequent growth. Adding sanctions or formal alliances leaves our estimates nearly unchanged, while the sanctions and alliance coefficients shrink and are no longer statistically distinguishable from zero.

The relationship is similar across partners and historical periods: GDP gains follow improved relations with the United States, with other Western nations, and with other major nations, and appear both during and after the Cold War. It also holds across issue areas: GDP responses are positive for improvements in diplomatic and security relations, not only in economic relations. It holds as well when the index uses only formal acts that contemporaneous sources record regardless of later outcomes, such as agreements, sanctions, and uses of force, a result that weighs against look-ahead bias in which events the model records and in how it scores them. The estimates are robust to alternative fixed effects and to controlling for the past paths of trade openness, domestic unrest, war exposure, institutions, and a broader set of economic fundamentals, factors that could

drive both international relations and growth. Our long-run estimates are also robust to dynamic panel estimation and to an unsmoothed index (Section 5).

Shifts in relations with the major nations often coincide with domestic reforms and regime changes that may affect growth on their own. We address this concern in two ways. First, we use only the changes in relations around leadership transitions that follow deaths in office or close elections in the major nations. These estimates are positive and similar to or larger than the baseline magnitude, though less precise. Second, as a conservative check, we exclude variation from all years around major domestic reforms and regime changes. The estimated gain remains three-fifths of the baseline or more over most of the first two decades.

Improvements in the index are accompanied by changes in growth fundamentals, each with its own timing: domestic unrest falls and trade openness rises immediately, investment and productivity rise within a few years, and the capital stock builds over the following decade. This sequence is consistent with an early reduction in risk followed by gains from better access to markets and technology. We then revisit the democracy–growth relationship in light of geopolitical alignment. Democracy and geopolitical alignment are related but distinct: democratization is followed by closer relations with Western nations, and controlling for alignment lowers its estimated growth effect, most at short and medium horizons, while its long-run benefits persist.

Applying the estimated responses to observed changes in alignment, we find that each decade's realignment from the 1970s through the 1990s implied a positive median long-run GDP gain, peaking at about 4.6 log points for the 1980s, while fragmentation in the 2010s turned the median slightly negative, at −1.2 log points. Across countries, realignment implies illustrative growth differences of about −31 to +31 log points during and after the Cold War, and alignment in 2024 implies income differences of about −29 to +12 log points relative to the median.

Related Literature. We contribute to the measurement of geopolitical alignment, the study of growth determinants, and geoeconomics.

Measuring Bilateral Geopolitical Alignment. Our primary contribution is a measure of bilateral geopolitical alignment built from consequential geopolitical events. Existing measures differ in how they conceptualize alignment (Voeten 2026). The most widely used, UN General Assembly (UNGA) voting similarity (Signorino and Ritter 1999; Bailey, Strezhnev, and Voeten 2017), treats alignment as agreement and captures countries' positions in multilateral forums rather than the state of their bilateral relations. Measures that treat alignment as a relationship, including strategic rivalries (Thompson 2001), sanctions (Felbermayr et al. 2021), military alliances (Leeds et al. 2002), and bilateral treaties (Broner et al. 2025), provide discrete classifications that record changes in a relationship's status but not its evolving intensity. Our measure follows the relational view and tracks the intensity of cooperation and conflict in each relationship continuously, at annual frequency, across six decades.

Our measure is closest to event data. Machine-coded event databases such as ICEWS, GDELT, and Phoenix also track bilateral interactions continuously, but they extract events automatically from

large volumes of news reports. We instead compile only consequential events and validate them against independent evidence. Our construction draws on the text-as-data framework (Gentzkow, Kelly, and Taddy 2019) and its applications to economic and political risks (Baker, Bloom, and Davis 2016; Hassan et al. 2019; Caldara and Iacoviello 2022; Hassan et al. 2024), as well as the growing use of large language models for data construction in economics (Clayton et al. 2025; Dell 2025; Asirvatham, Mokski, and Shleifer 2026). As Ludwig, Mullainathan, and Rambachan (2026) and Dell and Rambachan (2026) emphasize, such measurement requires a validation sample, and our external benchmarks and human audit serve that role.

Economic Growth Determinants. We contribute to the literature on cross-country growth determinants (Levine and Renelt 1992; Barro 1997; Durlauf, Johnson, and Temple 2005). Existing research has identified fundamental drivers internal to countries—institutions (Acemoglu, Johnson, and Robinson 2001; Dell 2010), human capital (Barro 1991; Mankiw, Romer, and Weil 1992), culture (Nunn 2008; Tabellini 2010), and geography (Gallup, Sachs, and Mellinger 1999; Dell, Jones, and Olken 2012)—as well as external economic integration through trade (Frankel and Romer 1999; Alcalá and Ciccone 2004) and cross-border technology diffusion (Parente and Prescott 1994; Coe and Helpman 1995). Yet the role of countries' external political environment remains underexplored. We measure this environment and, building on the empirical framework of Acemoglu et al. (2019), use local projections (Jordà 2005) to estimate its dynamic relationship with growth.

Geoeconomics. Our research bridges classical economic statecraft (Hirschman 1945; Baldwin 1985) with contemporary geoeconomic analysis (Clayton, Maggiori, and Schreger 2024, 2026; Fernández-Villaverde, Mineyama, and Song 2024; Kleinman, Liu, and Redding 2024; Gopinath et al. 2025; Liu and Yang 2025). This literature has studied economic coercion and the fragmentation of trade and investment, and related work studies individual instruments of cooperation and conflict separately, including aid (Alesina and Dollar 2000; Nunn and Qian 2014), sanctions (Morgan, Syropoulos, and Yotov 2023), and wars (Martin, Mayer, and Thoenig 2008; Federle et al. 2026). Our measure places these instruments on a single cooperation–conflict scale and lets us relate a country's overall geopolitical environment to its aggregate growth. A companion paper (Fan, Wo, and Xiang 2025) extends the database to all country pairs to study how geopolitical relations shape trade globalization.

Road Map. Section 2 introduces and validates our event-based measure of geopolitical alignment. Section 3 estimates the dynamic relationship between the index and GDP per capita, assesses its robustness, and discusses its interpretation. Section 4 examines accompanying changes in growth fundamentals, the relationship with democracy, and the implied GDP magnitudes. Section 5 reports additional robustness checks and comparisons with alternative measures. Section 6 concludes.

2. Measuring Geopolitical Alignment and the External Environment

We use a large language model with web search to compile and score 373,020 geopolitical events from 1960 to 2024 involving at least one of 24 major nations.[2] These events cover all 193 UN member states through their relations with the 24 major nations, denoted $\mathcal{N}$.[3] Using these events, we construct bilateral geopolitical alignment scores that vary by country pair and year, then aggregate them into a country-level geopolitical alignment index. The resulting scores track well-known historical episodes, agree with externally documented events and a blind human audit, and co-move with bilateral political and economic activity such as trade, aid, and sanctions (Section 2.3).

2.1. A Global Database of Geopolitical Events, 1960–2024

Bilateral geopolitical events—diplomatic agreements, economic sanctions, military actions, and territorial disputes—are prominent features of the historical record, extensively documented across news archives, diplomatic records, government publications, and scholarly databases. Unlike granular economic transactions or social interactions that may leave limited historical traces, these events are salient and widely corroborated across independent sources, making them well-suited for systematic extraction by large language models (LLMs) trained on such corpora. We use Gemini 2.5 Pro, an LLM with web search, to collect, verify, and analyze bilateral geopolitical events. Figure 1 illustrates our LLM-based analysis procedure. Appendices A.2 and C provide the complete analytical framework and prompt specifications.



FIGURE 1. LLM Geopolitical Event Analysis Procedure

Notes: For each country pair and year, the model verifies the political entities (for example, the Soviet Union for Russia before 1992), compiles the year's bilateral events, assigns each event its CAMEO codes, and scores it on the Goldstein scale from −10 to +10. Appendix C reports the prompt.

The prompt instructs the LLM to perform four tasks in sequence. It first verifies the historical political entities for each country pair and year, accounting for state succession (e.g., Soviet Union to Russian Federation). It then searches its knowledge base and the internet for events of the 25 types across six categories: economic relations, diplomatic and political relations, security and defense, legal and territorial issues, multilateral governance, and other bilateral interactions (Table A.1 in Appendix A.2). It records the events that affect or indicate the state of the relationship, verified against authoritative sources. The prompt states in advance what clears this threshold, including regular high-level meetings, formal agreements, and diplomatic communications (Appendix C.2).

[2] In the companion paper, Fan, Wo, and Xiang (2025), we extend the database to all bilateral pairs among the 193 UN member states over 1950–2024, yielding 833,485 events with complete CAMEO and Goldstein coding.

[3] The 24 major nations are Argentina, Australia, Belgium, Brazil, Canada, Switzerland, People's Republic of China, Germany, Denmark, Spain, France, United Kingdom, Indonesia, India, Italy, Japan, Republic of Korea, Mexico, Netherlands, Poland, Russian Federation (Soviet Union), Saudi Arabia, Turkey, and United States. Appendix A.1 discusses these choices. Collectively, these 24 nations account for 82–90 percent of global GDP over the study period.

The model next classifies each event using the CAMEO framework and finally assigns a Goldstein score from −10 (maximum conflict) to +10 (maximum cooperation) based on event intensity.[4]

The design thus separates two stages: the taxonomy and threshold fixed in the prompt define *which* events enter the database, while the CAMEO and Goldstein frameworks guide *how* the recorded events are classified and scored. This separation matters for the validation below: what distinguishes the resulting database from machine-coded event databases is the first stage, which restricts the database to consequential events.

TABLE 1. US–Russia Bilateral Geopolitical Events in 2022: LLM Analysis Results

Event Name	**Event Description**	**CAMEO Class.**	**Category**	**Goldstein Score**
US Military Assistance to Ukraine	US committed billions in security assistance through Presidential Drawdown Authority and Ukraine Security Assistance Initiative	Material Conflict (17-170)	Security & Defense	−8.0
Sweeping Sanctions on Russia	Following Russia's invasion of Ukraine, US coordinated with G7 and EU to impose extensive sanctions	Material Conflict (16-163)	Economic	−7.5
US Leads Russia's Suspension from UN Human Rights Council	Following reports of civilian deaths in Bucha, US led successful campaign to suspend Russia from UN Human Rights Council by 93–24 vote	Material Conflict (16-166)	Multilateral	−6.5
Biden Labels Putin a "War Criminal"	President Biden publicly called President Putin a "war criminal" for actions during the Ukraine invasion, a label the Kremlin called "unacceptable and unforgivable"	Verbal Conflict (11-112)	Diplomatic & Political	−5.5
US Leads Diplomatic Condemnation at UN	US led effort resulting in UN General Assembly Resolution ES-11/1, which passed 141–5 deploring Russia's aggression and demanding immediate withdrawal from Ukraine	Verbal Conflict (11-113)	Multilateral	−4.5
Griner–Bout Prisoner Exchange	US and Russia conducted prisoner swap in Abu Dhabi, exchanging Russian arms dealer Viktor Bout for American basketball player Brittney Griner after months of negotiations	Material Coop. (08-084)	Legal & Territorial Issues	+7.0

Notes: The table lists all six events the model records for the US–Russia pair in 2022, ordered from most conflictual to most cooperative. Event names and descriptions are abridged from the model output. The CAMEO column gives the quad class with the root and event codes (Appendix C.3), and the category column gives the event category (Table A.1). Goldstein scores range from −10 (most conflictual) to +10 (most cooperative).

Table 1 shows the coded US–Russia bilateral events for 2022, the year Russia launched its full-scale war on Ukraine. Five of the six events represent conflict, with Goldstein scores ranging from −4.5 (UN General Assembly condemnation) to −8.0 (US military assistance to Ukraine). Events that involve third countries are scored by their implications for the focal pair: US military assistance

[4]Appendix C.3 describes the coding conventions and anchored cases that the LLM uses with the CAMEO and Goldstein frameworks to calibrate each score to the event's context and severity.

to Ukraine, for example, enters as a conflictual US action toward Russia. The sole cooperative event, the Griner–Bout prisoner exchange, scores +7.0. Table A.2 in Appendix A.2 provides a contrasting example from the 1972 détente period, when all nine US–Soviet events are cooperative with scores ranging from +5.0 to +9.0.

TABLE 2. Geopolitical Events Summary by Decade, 1960–2024

	1960s	1970s	1980s	1990s	2000s	2010s	2020s	Total
Cooperation Events	22,011	26,872	29,657	40,537	59,230	82,878	48,545	309,730
Conflict Events	7,278	7,353	9,390	7,989	9,649	13,297	8,334	63,290
Mean Goldstein Score	2.86	3.28	2.92	3.98	4.09	3.88	3.73	3.67

Notes: The table covers the 373,020 events of 1960–2024 in country pairs involving at least one of the 24 major nations. Cooperation events include verbal and material cooperation (CAMEO classes 1–2). Conflict events include verbal and material conflict (CAMEO classes 3–4). Goldstein scores range from −10 (most conflictual) to +10 (most cooperative). The 2020s column covers 2020–2024.

Coverage and Descriptive Statistics. Table 2 summarizes the 373,020 events of 1960–2024 by decade. Cooperation accounts for 75–79 percent of events in the 1960s–1980s and 84–86 percent in each subsequent decade. Mean Goldstein scores rise from 2.86–3.28 in the first three decades to 3.98–4.09 in the 1990s and 2000s, then decline to 3.88 in the 2010s, as lower-scored verbal cooperation gains share, and to 3.73 over 2020–2024, as the conflict share also rises. Appendix A.2.2 provides detailed statistics on event classification and temporal patterns.[5]

2.2. Bilateral Geopolitical Alignment Scores

Dynamic Bilateral Geopolitical Alignment. Let $\left\{s^n_{ij,t}\right\}_{n=1}^{\tilde{N}_{ij,t}}$ denote the Goldstein scores for the set of events between countries i and j in year t, where $\tilde{N}_{ij,t}$ is the total number of events. The annual event score, normalized to the $[-1, +1]$ interval, is:

$$\tilde{S}_{ij,t} = \frac{1}{\tilde{N}_{ij,t}} \sum_{n=1}^{\tilde{N}_{ij,t}} s^n_{ij,t}/10$$

To capture both the immediate impact of political events and the institutional memory that characterizes international relationships, we then construct a dynamic geopolitical alignment score for each country pair as a weighted moving average:

$$S_{ij,t} = \left(1 - \phi_{ij,t}\right) \cdot S_{ij,t-1} + \phi_{ij,t} \cdot \tilde{S}_{ij,t} \tag{1}$$
$$\phi_{ij,t} = \tilde{N}_{ij,t}/N_{ij,t}, \quad N_{ij,t} = (1-\delta)\, N_{ij,t-1} + \tilde{N}_{ij,t}$$

[5]The event record becomes denser over time: the share of major-nation dyad-years with at least one recorded event rises from 32 percent in the 1950s to 80 percent in the 2020s. Appendix A.2.2 documents this coverage gradient. Section 2.3 benchmarks detection against externally documented events.

where $\phi_{ij,t}$ represents the updating weight, $N_{ij,t}$ is the effective cumulative number of events, and δ is the depreciation rate for past observations. We set $\delta = 0.3$, which gives the weight on past events a half-life of about two years and allows the measure to capture the persistence consistent with a typical four-year electoral cycle.[6] The dynamic geopolitical alignment score is normalized to range from −1 (maximum conflict) to +1 (maximum cooperation). Throughout the paper, we refer to $S_{ij,t}$ as the *bilateral geopolitical alignment score* and to the GDP-weighted aggregate p_{ct} (defined below) as the *geopolitical alignment index*. Appendix A.3 examines the dynamic properties of both measures.

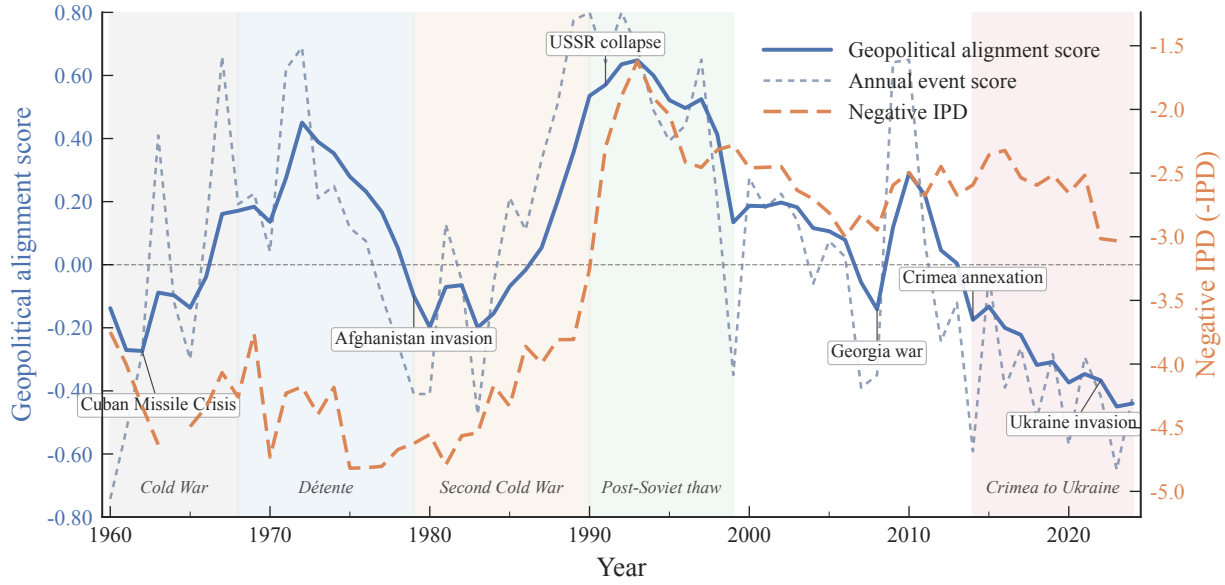


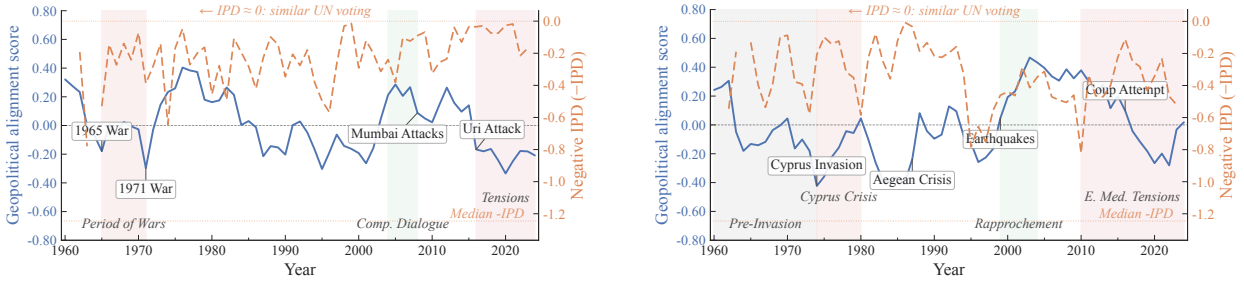
FIGURE 2. Geopolitical Alignment Scores Between the United States and Russia (Soviet Union)

Notes: Time-series comparison of bilateral geopolitical alignment measures between the United States and Russia, 1960–2024. The blue solid line shows our dynamic geopolitical alignment score (left axis), the light blue dashed line shows the annual (unsmoothed) event score, and the orange dashed line displays the negative Ideal Point Distance (−IPD) from UN voting data (right axis). Shaded regions indicate five geopolitical eras: Cold War (1960–1968), Détente (1968–1979), Second Cold War (1979–1990), Post-Soviet thaw (1990–1999), and Crimea to Ukraine (2014–2024). Key geopolitical events are annotated.

Figure 2 shows the dynamic score for a familiar bilateral relationship. The measure tracks major historical episodes: the Cuban Missile Crisis marking peak Cold War tensions, the détente period (1968–1979), deterioration following the Soviet invasion of Afghanistan, improvement beginning with Gorbachev's reforms in the mid-1980s, the post-Soviet peak in 1993, and subsequent decline culminating in the Crimean crisis. The 2014–2024 period records the lowest values in the sample. The Ideal Point Distance (IPD) measure from UN voting data (orange dashed line) is less responsive to these bilateral shifts: it changes little during détente and after the Crimean crisis despite the

[6]When no events are observed ($\tilde{N}_{ij,t} = 0$), we let both the effective event count and the score decay toward zero: $N_{ij,t} = (1-\delta)N_{ij,t-1}$ and $S_{ij,t} = (1-\delta)S_{ij,t-1}$. For the initial year of each bilateral pair, $S_{ij,t_0} = \tilde{S}_{ij,t_0}$ and $N_{ij,t_0} = \tilde{N}_{ij,t_0}$. Section 5.1 considers $\delta = 1$ (an unsmoothed index). Holding the score unchanged in years without events instead gives a similar GDP path and a 25-year sustained-path coefficient of 10.0 log points (Appendix Figure B.1c). Because the local projections include lags of geopolitical alignment, they use current alignment variation after conditioning on its lagged history.

sustained deterioration in bilateral relations. Figure 3 provides two further historical illustrations involving non-superpower relationships.



(a) India and Pakistan

(b) Greece and Turkey

FIGURE 3. Geopolitical Alignment Scores: India–Pakistan and Greece–Turkey

Notes: Time-series comparison of bilateral geopolitical alignment measures for India–Pakistan (panel (a)) and Greece–Turkey (panel (b)), 1960–2024. Blue solid lines show our dynamic geopolitical alignment score (left axis). Orange dashed lines display the negative Ideal Point Distance (−IPD) from UN voting data (right axis). The dotted horizontal line marks the pooled sample median of negative IPD. Shaded regions highlight key periods of conflict and cooperation. Boxed labels mark major bilateral events.

Panel (a) of Figure 3 traces India–Pakistan relations through six decades of recurrent conflict. Our measure tracks the wars of 1965 and 1971, the Mumbai terrorist attacks (2008), and the Uri attack and subsequent surgical strikes (2016), with scores falling from +0.32 in 1960 to −0.30 during the 1971 war. The measure captures repeated cycles of conflict and cautious engagement: scores recover to +0.40 in the late 1970s, rise again during the Composite Dialogue period (2004–2008) to approximately +0.30, and collapse after the Mumbai attacks. The IPD measure, by contrast, shows these persistent rivals as increasingly aligned because both countries vote similarly on postcolonial issues at the UN.

Panel (b) shows a similar pattern in Greece–Turkey relations: our measure registers the sharp deterioration following the 1974 Cyprus invasion, the 1987 Aegean crisis, the brief rapprochement after the 1999 earthquakes, and the sustained decline during Eastern Mediterranean tensions (2010–2024). The IPD measure shows these NATO allies as consistently close throughout, reflecting their similar voting patterns within the Western bloc despite bilateral disputes over Cyprus, the Aegean, and maritime boundaries. In both cases, negative IPD remains above the pooled sample median (dotted line): similar UN voting positions can coexist with substantial bilateral conflict.[7] Appendix A.4 provides additional historical illustrations for eight bilateral relationships, both those involving major nations (Figure A.4) and regional rivalries (Figure A.5). Geopolitical maps appear in Appendix A.5.

We also use the cross-sectional matrix of bilateral geopolitical alignment scores, $[S_{ij,t}]$, to characterize the topology of the international system. Figure 4 compares this topology in 1980 and

[7] Fan, Wo, and Xiang (2025) show formally that UNGA voting similarity correlates with the low-dimensional bloc structure of our bilateral alignment scores—the distance over the first three principal components—but misses the higher-dimensional bilateral variation associated with economic outcomes.

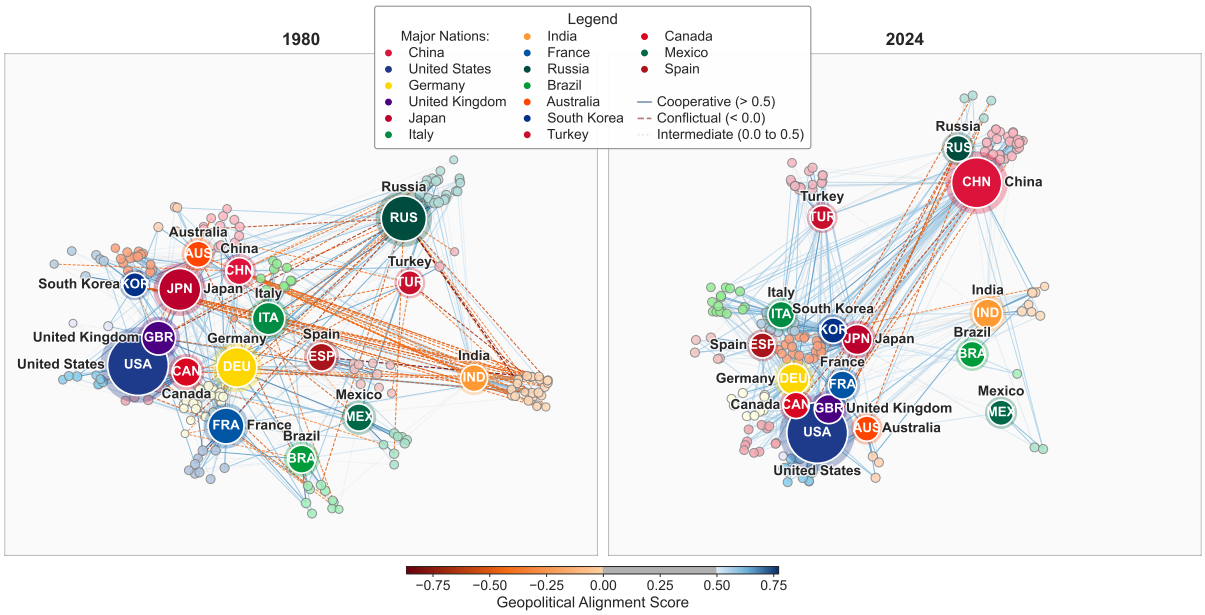


FIGURE 4. Geopolitical Alignment Topology, 1980 and 2024

Notes: Positions of 16 selected major nations come from multidimensional scaling of correlation distances between their bilateral-alignment profiles with all 193 countries. Closer positions indicate more similar global relationship profiles. Major-nation circle sizes scale with GDP shares. Smaller circles denote other countries, colored by the major nation with which they have the highest alignment score and positioned using their major-nation alignment profiles. Links show each smaller country's three largest major-nation relationships in absolute value: blue solid lines indicate scores above 0.5, red dashed lines scores below 0, and gray dotted lines scores from 0 to 0.5.

2024. In 1980, the United States and the Soviet Union anchor distinct clusters. China lies between them but closer to the Western cluster, consistent with the Sino-American rapprochement following Nixon's opening and shared opposition to Soviet expansion.

By 2024, this bipolar structure has given way to a more fragmented system. The United States, United Kingdom, Germany, France, Japan, Australia, and South Korea form a tightly integrated cluster, while China and Russia lie apart from it. India occupies an intermediate position, and "connector" states such as Turkey, Brazil, and Mexico bridge the clusters.

2.3. Validation

Beyond the case studies in Section 2.2, we assess the measure in five ways. We benchmark the event record against independently documented bilateral developments, compare it with machine-coded event databases, test stability across models and pipelines, conduct an independent human audit, and examine associations with bilateral economic outcomes. Section 5.2 separately compares alternative measures within the growth application.

Validation against Documented Events. A first concern is that uneven historical source coverage may cause the database to miss developments in earlier periods or involving poorer countries (Fariss 2014). We benchmark the database, including the 1950s records that initialize the scores, against

4,670 bilateral developments over 1950–2023 from eight externally compiled event classes.[8] Within the class-specific matching windows, our database records some event for 97.6 percent of these developments, an event in the expected conflict or cooperation direction for 92.1 percent, and an event of the expected type for 82.6 percent. Matching on event type does not vary systematically with country income and rises only mildly over time, with the increase concentrated in two of the eight event classes (Appendix Figure A.8). Matching also does not vary with the country's growth over the following decade, suggesting that the model does not record documented developments selectively by subsequent growth (Appendix Table A.6).

We then examine whether the recorded intensity of bilateral relations moves with these documented developments. We regress the unsmoothed annual event score jointly on the eight documented-event indicators, absorbing dyad and country-by-year fixed effects. All eight conditional annual-score differences have the expected sign and seven are statistically distinguishable from zero, with the largest negative difference for war onsets and the largest positive difference for alliance formations. Appendix A.6 reports the detailed benchmark analysis.

Comparison with Machine-Coded Event Databases. We next compare our database with three machine-coded event databases: ICEWS (Boschee et al. 2015), GDELT (Leetaru and Schrodt 2013), and the Cline Center Historical Phoenix database (Althaus et al. 2020). These databases automatically extract events from large volumes of news reports and code them using the CAMEO framework. The comparison separates two questions: whether each event database records independently documented developments and how its annual score differs during documented conflict and cooperation years. Appendix A.7 reports the detailed comparison.

TABLE 3. Detection and Conditional Annual-Score Differences across Event Databases

	Our database	ICEWS	GDELT	Phoenix
Panel A. Documented events, 1995–2018 (N = 2,038)				
Mean events in documented-event year	3.0	371.7	4,083.0	61.3
Any event recorded	0.982	0.907	0.980	0.844
Event in expected direction	0.917	0.869	0.972	0.794
Panel B. Common observed dyad-years (N = 34,412)				
Documented conflict	−1.467 (0.179)	−0.475 (0.078)	−0.120 (0.054)	−0.295 (0.084)
Documented cooperation	+0.475 (0.112)	+0.194 (0.060)	+0.033 (0.051)	−0.133 (0.096)

Notes: Panel A uses the 2,038 documented developments over 1995–2018 that all four databases cover after actor harmonization; detection uses the matching windows in Appendix Table A.4, and median events per documented-event year are 3, 34, 288.5, and 6, respectively. Panel B regresses each database's unsmoothed annual mean Goldstein score, in raw points, on indicators for any documented conflict and any documented cooperation, with dyad and country-by-year fixed effects, on identical dyad-years; standard errors, clustered by dyad, are in parentheses. Appendix A.7 reports the specification and complete results.

[8]The four conflict classes are interstate war onsets, militarized interstate disputes involving force short of war, sanction impositions, and diplomatic severances. The four cooperative classes are sanction removals, diplomatic establishments, regional trade-agreement entries, and alliance formations. An event of the expected type has the expected event category or CAMEO root (a broad event-family match in Appendix A.6).

Table 3 shows that, in the common 1995–2018 sample, our database records some event around externally documented developments as often as the other event databases, while containing only a fraction of their event volume. The key distinction lies in the sensitivity of the annual score to documented events: on identical dyad-years and under a common fixed-effects specification, our database exhibits the largest negative conditional difference during documented conflict and the largest positive difference during documented cooperation. These results suggest that compiling a few consequential geopolitical events per pair and year, instead of averaging over many routine reports, produces a more differentiated annual conflict–cooperation signal. Section 5.2 shows that these databases capture little growth-relevant variation under the same construction.

Reliability and Replicability. We also test whether annual event scores are stable across models and repeated runs. Cross-model comparisons of Gemini 2.5 Pro, GPT 5.4, Claude Opus 4.6, and Claude Sonnet 4.6 on 300 randomly sampled dyad-years yield a pooled correlation of 0.88. Repeated runs of each model give a pooled within-model correlation of 0.93.

A separate exercise changes both the LLM and the prompt structure. An alternative LLM pipeline constructed primarily with Gemini 2.5 Flash (preview) and an earlier prompt produces a country-year index that correlates at 0.87 with the baseline. It also yields very similar GDP coefficient paths (Appendix A.8).

Human Audit. Cross-model agreement does not establish accuracy because models may share errors in interpreting and scoring events. We therefore conduct a staged human audit of a stratified sample of 200 eventful dyad-years, with two coder assessments per case. Six trained research assistants from Peking University first reconstruct and score bilateral events without seeing model output or using generative AI. Like the model, they work on one dyad-year at a time and see only the country pair, the year, and the codebook the model follows. After locking these records, they review and score the events described by the model while its classifications and scores remain concealed. Neither the model nor the coders are given economic data, and both search only for events with geopolitical consequences for the pair. Because both work from the current historical record, we test for look-ahead bias using formal acts only (Section 3.3.2).

Blind human and model annual scores correlate at 0.80, compared with 0.76 between human coders. For events that coders identified independently and that match a model event, Goldstein scores correlate at 0.92. After review, correlations are 0.92 for recoded model events and 0.89 for annual scores. The audit identifies three fabricated records among the 856 model-generated events, less than 0.5 percent. The audit thus supports the annual conflict–cooperation signal and points to the set of recorded events as an important source of the remaining differences. Adding simulated error of the size implied by the audit to the annual event scores lowers the peak GDP response estimated in Section 3 by 14 percent and the 25-year sustained response by 6 percent on average. Appendix A.9 reports the detailed audit design and results.

Bilateral Outcomes. Finally, the bilateral alignment score co-varies with bilateral economic outcomes. In standard gravity specifications, more cooperative relations are associated with greater bilateral trade, more official development assistance, and a lower probability of sanctions. These patterns persist in specifications with dyad fixed effects, which exploit within-pair changes over time, and, with smaller coefficients, after rebuilding the score without events directly linked to each outcome. Appendix A.10 reports the full results, including migration.

2.4. The External Geopolitical Environment

In international relations, "alignment" is a relational and directional concept: countries align *with* particular partners and *against* others. Our dyadic score respects this usage: it measures the cooperative-versus-conflictual state of one bilateral relationship. The country-level aggregate we now construct is a different object. It does not record which side a country has chosen. It summarizes the overall state of a country's relations with the major nations, what we call the *external geopolitical environment* facing that country. The political form of an improvement in this environment differs across international systems: under Cold War bipolarity it often reflected side-switching or normalization with the dominant economic bloc, while in the post-Cold War system it more often reflects broad-based cooperative engagement with multiple poles. The index does not identify a universal foreign-policy strategy. It places these distinct political changes on a common scale according to the resulting state of relations with major nations.

We aggregate bilateral geopolitical alignment scores $S_{cj,t}$ into a country-level index using GDP weights:

$$p_{ct} = \sum_{j \in \mathcal{N} \setminus \{c\}} S_{cj,t} \times \text{GDP share}_{jt} \tag{2}$$

Here, GDP share_{jt} denotes country j's share of world nominal GDP in current US dollars. We interpret nominal GDP as a proxy for geopolitical influence. The resulting *geopolitical alignment index* p_{ct} is our primary measure of a country's external geopolitical environment. Higher values of p_{ct} indicate more cooperative relations with major nations and hence a more favorable external environment. Lower values indicate more conflictual relations.

Why should this environment matter for growth? More cooperative relations with major nations may expand a country's *access* to export markets and imported inputs, capital and technology, aid and official finance, and the security guarantees and diplomatic legitimacy that underpin long-horizon investment. More conflictual relations may increase *risk* and exclusion through sanctions and embargoes, technology and financial restrictions, military threats, diplomatic isolation, and the uncertainty that accompanies all of these. The index summarizes bilateral conditions on both margins, not a single policy intervention. As a check on the conflict side of the index, Appendix A.12 shows that the GDP-weighted conflict component of the underlying events co-moves with the Geopolitical Risk Index of Caldara and Iacoviello (2022), particularly during major military confrontations.

The Global Distribution, 1960–2024. The cross-country distribution of the index evolves through three periods: Cold War bipolarity (1960–1991), post-Cold War convergence (1992–2009), and contemporary fragmentation (2010–2024). These periods provide descriptive context for the within-country changes analyzed in Section 3.

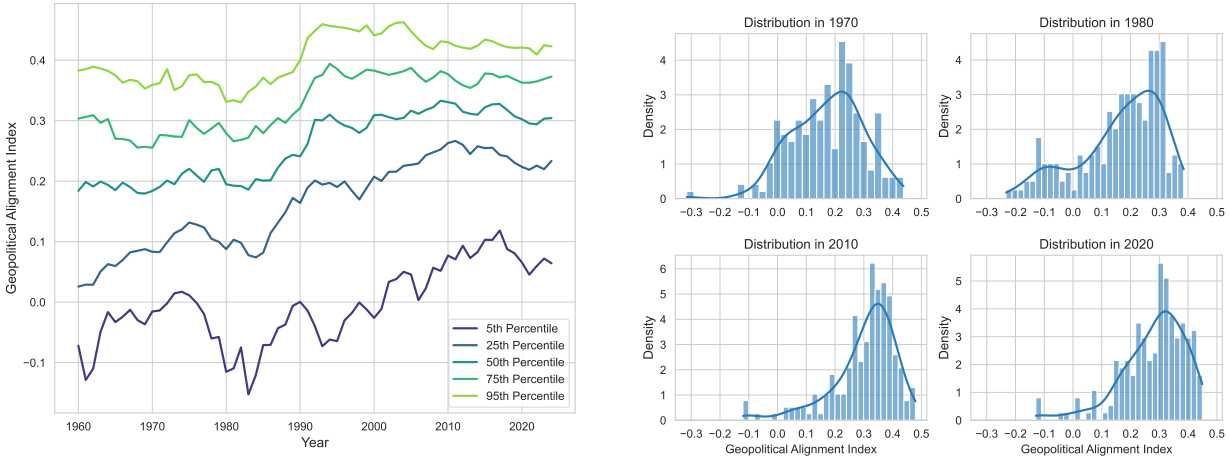


FIGURE 5. Evolution of the Distribution of the Geopolitical Alignment Index

Notes: The left panel shows the 5th, 25th, 50th, 75th, and 95th percentiles of the country-level geopolitical alignment index from 1960 to 2024. The right panels show histograms and kernel-density estimates for 1970, 1980, 2010, and 2020, highlighting the shift from Cold War bipolarity through post-Cold War convergence to contemporary fragmentation.

Figure 5 shows these shifts. During the Cold War era (1960–1991), the 5th percentile fluctuated between −0.15 and 0.02, reflecting the volatile position of countries caught in superpower competition, while upper percentiles remained relatively stable. The 1970 and 1980 distributions are left-skewed—with a primary peak around 0.2 and considerable mass extending into negative territory—capturing the division into opposing geopolitical camps.

During 1992–2009, the distribution moved toward broad convergence. By 2010, the 5th percentile had improved from −0.04 to 0.08, the median had risen from 0.30 to 0.33, and the distribution was concentrated tightly around 0.33 with minimal mass below 0.0. Over 2010–2024, the distribution widened modestly: the 5th percentile declined from 0.08 to 0.06, variance increased slightly, and the 2020 distribution shows a flatter peak with a left tail extending below 0.0.[9] The measure is constructed without reference to these periods.[10]

To benchmark magnitudes, Table 4 reports changes between selected historical endpoints in index units and standard deviations. One pooled-sample standard deviation is 0.141. An interquartile move is 0.186 units, or 1.32 standard deviations. Russia's 2005–2015 deterioration and Indonesia's 1996–2004 improvement are each about one standard deviation, while the changes in South Africa, Iran, and Iraq span several standard deviations.

[9]The 5th–95th percentile spread narrowed from roughly 0.51 (ranging from −0.13 to 0.39) during peak Cold War to 0.35 (0.08 to 0.43) by 2010, before widening again to 0.36 (0.06 to 0.42) by 2024.

[10]Summary statistics by decade, including means and standard deviations, appear in Table A.10 (Appendix A.11).

TABLE 4. Historical Realignments on the Index Scale

Benchmark or episode	Δ index (SD)	Benchmark or episode	Δ index (SD)
Interquartile shift (25th→75th)	0.186 (1.32)	Iraq: Gulf War and sanctions (1989–94)	−0.58 (−4.1)
South Africa: apartheid ends (1986–96)	+0.69 (+4.9)	Vietnam: normalization with the US (1990–2000)	+0.19 (+1.4)
Iran: Islamic Revolution (1977–83)	−0.48 (−3.4)	Russia: deterioration through Crimea (2005–15)	−0.14 (−1.0)
Poland: post-communist transition (1988–98)	+0.27 (+1.9)	Indonesia: post-Suharto reorientation (1996–2004)	+0.13 (+1.0)
China: reform and opening (1976–86)	+0.15 (+1.1)	Egypt: Camp David-era realignment (1973–81)	+0.05 (+0.3)

Notes: Δ is the end-year minus start-year value for each selected window. Parentheses report the change in standard deviations, using the full-sample pooled $\sigma = 0.141$. The interquartile benchmark is the shift from the 25th percentile (0.154) to the 75th percentile (0.340). Episode labels provide historical context, and examples are chosen for recognizability rather than as a ranking.

3. Geopolitical Alignment and Economic Growth: Dynamic Estimates

Improvements in a country's external geopolitical environment, measured by p_{ct}, are followed by a hump-shaped rise in GDP per capita that peaks about a decade later. This section presents these local-projection estimates and their robustness, and probes the causal interpretation of the dynamic relationship.

3.1. Economic Data and Sample

We use a panel of 193 countries from 1960 to 2024, of which 184 have GDP data from the World Development Indicators. The main estimates use a long-coverage sample of 148 countries with at least 40 years of GDP data. We report the full available unbalanced panel alongside the baseline. The primary outcome variable is log GDP per capita in constant US dollars from the World Development Indicators, which provides broad country coverage. We build upon the dataset constructed by Acemoglu et al. (2019), expanding coverage and incorporating additional variables from the Penn World Table (Feenstra, Inklaar, and Timmer 2015) and Acemoglu et al. (2025).[11]

We also use growth determinants from several categories: augmented Solow fundamentals (physical capital, investment rates, population growth, education), institutional measures (democracy indices, governance quality), trade openness, and additional correlates identified in the growth literature. These variables allow us to examine whether changes in the external geopolitical environment are accompanied by changes in growth fundamentals.

Distinctiveness from Economic Fundamentals. Table B.2 in Appendix B.2 reports within-year cross-sectional correlations between the geopolitical alignment index and standard economic fundamentals. These correlations are small and unstable across decades, typically below 0.2 in absolute value. In contrast, human capital, capital stock, and investment exhibit strong and stable correlations with GDP per capita.

[11]Table B.1 in Appendix B.1 provides detailed information on country coverage and data availability for each variable.

3.2. Empirical Specification and Main Estimates

We estimate how GDP per capita evolves at subsequent horizons following changes in the geopolitical alignment index. For each horizon h, we estimate the following local projection (LP) (Jordà 2005):

$$y_{c,t+h} = \alpha_h^{\mathrm{LP}} p_{ct} + \gamma_h' \mathbf{x}_{ct} + \varepsilon_{c,t+h}, \quad h = 0, 1, \ldots, H \tag{3}$$

where $y_{c,t+h}$ is log GDP per capita ($\times$ 100), p_{ct} is the geopolitical alignment index, and $\mathbf{x}_{ct}$ is a vector of controls. The coefficient α_h^{LP} is the horizon-specific population linear-projection coefficient on current alignment. Throughout, we report α_h^{LP} and all other coefficients on the index per one standard deviation of the index (0.141), so that each summarizes the conditional change in future GDP per capita associated with a one-standard-deviation increase in the index, holding fixed the included controls.[12] The control vector is

$$\mathbf{x}_{ct} = \left\{ \{y_{c,t-\ell}, p_{c,t-\ell}\}_{\ell=1}^{J}, \eta_c, \eta_{r(c)t}, \boldsymbol{m}_{ct} \right\},$$

where $y_{c,t-\ell}$ and $p_{c,t-\ell}$ are lagged values of GDP and geopolitical alignment, respectively, η_c denotes country fixed effects, $\eta_{r(c)t}$ represents region-year fixed effects, and $\boldsymbol{m}_{ct}$ includes time-varying country-specific controls employed in robustness checks. In our baseline specification, we include four lags ($J = 4$) of both GDP and geopolitical alignment, following Acemoglu et al. (2019), to capture growth dynamics and support inference in the presence of serial correlation (Montiel Olea and Plagborg-Møller 2021).

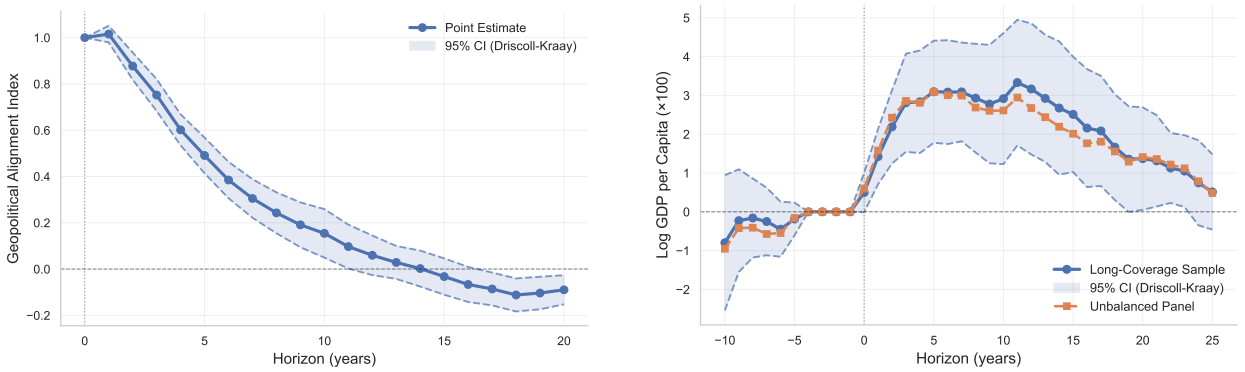
Country fixed effects η_c account for time-invariant heterogeneity across countries, so the coefficient is estimated from within-country variation over time (Durlauf, Johnson, and Temple 2005). Region-year fixed effects $\eta_{r(c)t}$ control for regional and global shocks, including shared temporal dynamics such as regional conflicts, financial crises, and globalization waves.[13] Section 3.5 discusses the assumptions under which these conditional dynamic responses admit a causal interpretation. Figure 6 reports horizon-specific coefficients from local projections of future geopolitical alignment and GDP per capita on current alignment.

Dynamics of Geopolitical Alignment and Economic Growth. To characterize the conditional dynamics of the index, we estimate the same local-projection specification with future geopolitical alignment, $p_{c,t+h}$, as the dependent variable. Panel (a) of Figure 6 shows that alignment is persistent but mean-reverting: the coefficient declines from one at $h = 0$ to 0.60 at year 4 and 0.15 at year 10. It reaches zero by year 14 and turns slightly negative thereafter.[14] Accordingly, the estimated GDP dynamics reflect the response both to the initial improvement in alignment and to the index's subsequent changes.

[12] Coefficients are in log points (GDP per capita $\times$ 100) per standard deviation of the index, the pooled standard deviation across all 12,545 country-years. Estimation uses the index in its natural units, and the reported coefficients and standard errors are multiplied by 0.141.

[13] Our estimates are robust to using only year fixed effects, as shown in Section 3.3.

[14] Appendix A.3 compares this persistence with that of the unsmoothed index and of the bilateral scores.



(a) Persistence of Geopolitical Alignment

(b) GDP per Capita

FIGURE 6. Dynamic Local-Projection Estimates

Notes: Panel (a) reports coefficients from projecting the geopolitical alignment index at $t + h$ on its current value, conditional on the baseline controls. The coefficient at $h = 0$ equals one by construction. Panel (b) reports coefficients from projecting log GDP per capita (×100) at $t + h$ on current geopolitical alignment, per one-standard-deviation change in the index. The solid blue line uses the long-coverage sample of 148 countries with at least 40 years of GDP data. The dashed orange line uses the full unbalanced panel. Both panels include four lags of GDP and geopolitical alignment, country fixed effects, and region-year fixed effects. Panel (a) spans $h = 0, \ldots, 20$. Panel (b) spans $h = -10, \ldots, 25$, with horizons $h = -10, \ldots, -5$ providing timing diagnostics. Coefficients at $h = -4, \ldots, -1$ are zero by construction because the corresponding lags of GDP enter the controls. Blue shaded areas show 95% confidence intervals based on Driscoll–Kraay standard errors (Driscoll and Kraay 1998).

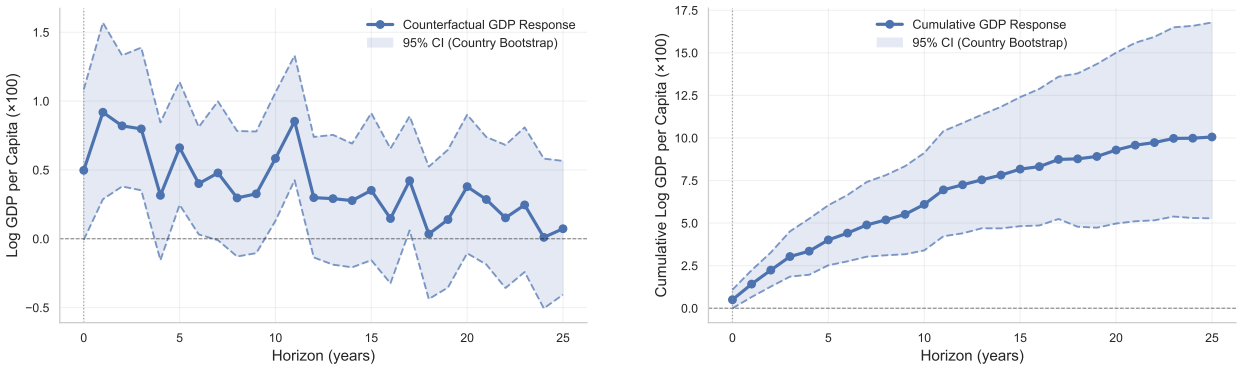
From $h = 0$ onward, the estimated GDP response in panel (b) is positive and hump-shaped. The coefficient rises from 0.5 log points at $h = 0$ to 3.3 log points in year 11, then declines to 1.4 log points by year 20. Estimates from the 148-country long-coverage sample and the full unbalanced panel closely overlap. Subsequent specifications use the long-coverage sample unless otherwise noted. The response does not depend on shifts in partner GDP shares: fixing partner weights over time yields similar estimates, as do regime-fixed weights, equal weights, and weights based on military capability across the 24 major nations (Appendix Figure B.1).[15]

Before impact, the coefficients are small and not individually distinguishable from zero, which provides one timing diagnostic. A reverse projection likewise shows no systematic relationship between current GDP and subsequent geopolitical alignment: no coefficient after impact is distinguishable from zero (Appendix Figure B.2, panel (b)).

Transitory and Sustained Alignment Paths. The conditional responses in Figure 6 combine a current change in geopolitical alignment with the subsequent dynamics of the index. To isolate the response associated with a purely transitory change, we follow Sims (1986) and Bilal and Känzig (2026) and construct a counterfactual path that raises the index by one standard deviation on impact and returns the change to zero immediately thereafter. Cumulating the transitory coefficients gives the corresponding mapping for an increase maintained at one standard deviation at all subsequent

[15] Appendix B.3 also reports bootstrap-based inference and robustness to alternative lag specifications, and Appendix Figure B.4 shows that the binned residual means are approximately linear over the observed range.

horizons.[16]



(a) Transitory Alignment Path

(b) Sustained Alignment Path

FIGURE 7. GDP Responses Associated with Transitory and Sustained Alignment Paths

Notes: Panel (a) reports the response associated with a one-standard-deviation change in alignment on impact that returns to zero thereafter. Panel (b) reports the cumulative response associated with maintaining a one-standard-deviation change at each subsequent horizon. The outcome is log GDP per capita (×100). Baseline specification as in Figure 6. Shaded areas represent 95% confidence intervals from 500 country-block bootstrap iterations.

Figure 7 maps the estimated conditional responses into transitory and sustained counterfactual paths. Under the transitory path in panel (a), the GDP coefficient is 0.5 log points on impact and 0.9 log points after one year. It remains positive but fluctuates over subsequent horizons. The wide confidence intervals indicate substantial uncertainty about the medium- and long-run path.

Panel (b) shows the cumulative response associated with the sustained path: the implied GDP coefficient rises steadily, reaching about 10 log points after 25 years. The trajectory continues to rise after year 20, though at a diminishing rate. Long-horizon local-projection estimates are imprecise. Section 5.1 presents a complementary analysis using dynamic panel methods that impose an autoregressive structure to trace the mapping beyond the local-projection horizon.

Economic Magnitude. The index has a standard deviation of 0.141 and ranges from −0.43 (hostile relations) to 0.55 (strong cooperation), with a median of 0.26. A sustained one-standard-deviation improvement corresponds to about 10 log points after 25 years (95 percent bootstrap interval: 5.3 to 16.8).[17] An interquartile improvement, from 0.154 to 0.340 (1.32 standard deviations), implies about 13 log points. Table 4 compares these magnitudes with observed realignments: Russia's deterioration through the Crimean crisis (2005–2015) is about one standard deviation and Vietnam's normalization with the United States (1990–2000) is 1.4.

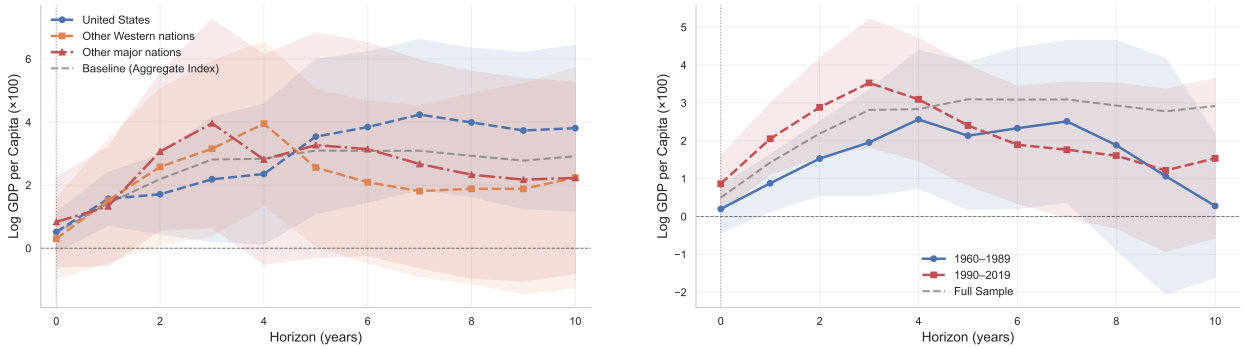
[16]The mapping assumes linearity and that the response to a sequence of unanticipated innovations equals the response to the same alignment path anticipated at time zero. It adds no identifying variation beyond the baseline local projections. The constructed paths retain the same conditional interpretation. Appendix B.5 details the construction and assumptions.

[17]A delta-method interval clustered by country and by year gives 4.1 to 16.0 log points. A split-panel jackknife correction for fixed-effects bias (Dhaene and Jochmans 2015) gives a 25-year coefficient of 10.2 log points.

3.3. Robustness and Diagnostics

3.3.1. Partner Groups and Historical Periods

To interpret what the index captures and to assess whether the growth relationship is concentrated in alignment with a single bloc, we decompose the index into relations with the United States, other Western nations, and other major nations. We then enter the three components jointly in the baseline local-projection specification.[18] We also estimate the baseline specification separately for 1960–1989 and 1990–2019 to assess whether the relationship appears across two distinct international systems.



(a) Partner Groups (b) Historical Periods

FIGURE 8. Growth Responses by Partner Group and Historical Period

Notes: Panel (a) jointly estimates the three additive partner-group components: the United States, other Western nations (Australia, Belgium, Canada, Denmark, France, Germany, Italy, the Netherlands, Poland, Spain, Switzerland, and the United Kingdom), and other major nations (Argentina, Brazil, China, India, Indonesia, Japan, Mexico, the Republic of Korea, Russia, Saudi Arabia, and Turkey). Group membership is fixed over the sample period. The gray dashed line is the baseline response to the aggregate index, shown only as a reference. It is not the sum of the three component-specific coefficients. Panel (b) estimates the baseline specification separately for 1960–1989 and 1990–2019 and shows the full-sample estimate for reference. All series show conditional local-projection coefficients for log GDP per capita (×100) over horizons $h = 0, \ldots, 10$, per one standard deviation of the aggregate index. Shaded bands are 95% confidence intervals based on Driscoll–Kraay standard errors.

Panel (a) of Figure 8 shows broadly similar positive, hump-shaped point-estimate paths across the three partner groups. The component estimates are less precise than the aggregate response and their confidence bands overlap, but all three coefficients peak at roughly 4 log points per standard deviation of the aggregate index within the first decade, with the US path peaking later than the other two. Each component is the group's GDP-weighted contribution to the aggregate index, so similar paths support the index's construction, in which relations matter in proportion to partner GDP. After partialling out the baseline fixed effects and lagged controls, the US and other-major-nations components have a conditional correlation of 0.04 (shared $R^2 = 0.002$), so the

[18] Let p_{ct}^{US}, p_{ct}^{OW}, and p_{ct}^{OM} denote the GDP-weighted contributions from the United States, other Western nations, and other major nations. By construction, $p_{ct} = p_{ct}^{\mathrm{US}} + p_{ct}^{\mathrm{OW}} + p_{ct}^{\mathrm{OM}}$. At each horizon, we estimate

$$y_{c,t+h} = \alpha_h^{\mathrm{US}} p_{ct}^{\mathrm{US}} + \alpha_h^{\mathrm{OW}} p_{ct}^{\mathrm{OW}} + \alpha_h^{\mathrm{OM}} p_{ct}^{\mathrm{OM}} + \boldsymbol{\gamma}_h' \mathbf{x}_{ct} + \eta_c + \eta_{r(c)t} + \varepsilon_{c,t+h},$$

where $\mathbf{x}_{ct}$ contains four lags of GDP per capita and four lags of each component, and η_c and $\eta_{r(c)t}$ are country and region-year fixed effects. We report $h = 0, \ldots, 10$.

two components vary largely independently. Residualizing each component on the other two, which removes their common movement, leaves positive paths of similar size (Appendix Figure B.5). The growth relationship is therefore neither driven only by a common improvement in all relations nor confined to one hegemon or ideological bloc.

Panel (b) shows positive short- and medium-horizon coefficients in both historical periods. The two paths rise with the full-sample estimate over the first few years and peak between 2.6 and 3.5 log points. They attenuate thereafter, while precision declines as longer horizons progressively reduce the usable observations within each 30-year subperiod. The relationship therefore appears in two structurally different international systems, even though geopolitical improvement took different political forms in each.[19]

3.3.2. Decomposing the Index: Event Domains and Formal Acts

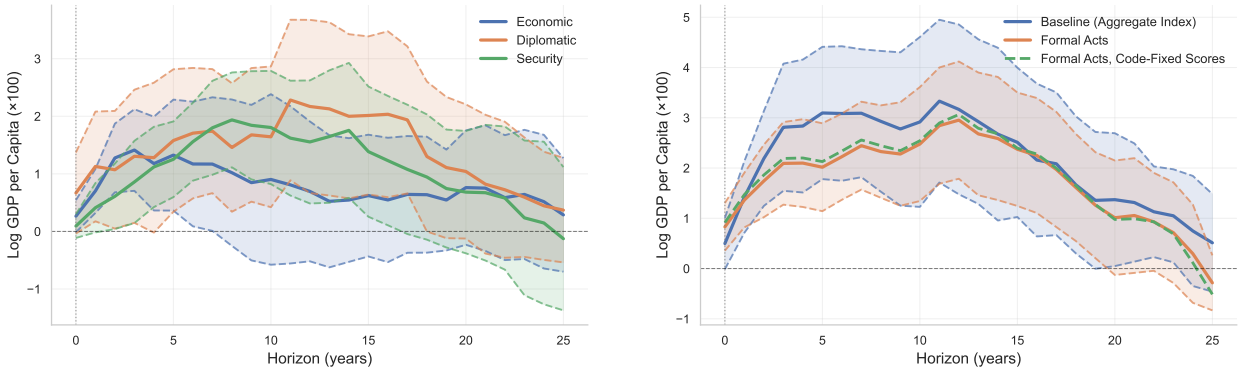
A natural concern is that the aggregate growth relationship may be concentrated in events with direct economic content, including trade agreements, sanctions, and investment treaties, which may themselves respond to growth conditions. Using the six event categories defined ex ante in the LLM prompt and held fixed across the sample (Section 2.1), we construct three components of the geopolitical alignment index: *economic relations* (category A), *diplomatic and political relations* (category B), and *security, defense, and territorial relations* (categories C and D).[20] We aggregate each component using the same GDP-weighted procedure as the baseline index in equation (2). Their correlations with the aggregate index are 0.73, 0.78, and 0.62, while their pairwise cross-correlations range from 0.30 to 0.48. The components therefore share meaningful but incomplete variation.

The aggregate growth relationship is not confined to events classified as economic. When all three components enter jointly, each with four lags, in a multivariate extension of equation (3) (panel (a) of Figure 9), the coefficients on all three components are positive and statistically distinguishable from zero over portions of the horizon. The economic coefficient peaks early, at approximately 1.4 log points within the first few years, then attenuates. The diplomatic coefficient is larger and more persistent, reaching 1.5–2.3 log points between horizons 5 and 12, while the security and territorial coefficient remains near 1.8 log points around horizon 10.

A second concern is look-ahead bias. The model and the human coders work from the current historical record, so they could record or score events with knowledge of what followed, making countries that later grew look better aligned in hindsight. Such bias would operate mainly through discretionary judgments, such as whether a consultation, statement, or cooperative

[19] Both periods contain similar variation in the index. In the long-coverage estimation sample at $h = 5$, the index has residualized standard deviations of 0.045 in 1960–1991 and 0.037 in 1992–2024 after the baseline fixed effects and lagged controls. Panel (b) uses 1960–1989 and 1990–2019 to accommodate forward horizons. Raw within-country variation is also similar under that split.

[20] Category A covers trade policy, financial relations, economic coercion, strategic sectors, and economic integration. Category B covers formal diplomacy, high-level interactions, public diplomacy, and cultural exchanges. Categories C and D cover military cooperation, security incidents, intelligence, legal actions, territorial and maritime disputes, and the movement of people. The remaining categories, multilateral governance (E) and other events (F), account for 20 percent of events and are omitted. Including them as a fourth component does not affect the results. See Table A.1 for subcategory details.



(a) Event Domains, Joint Specification

(b) Formal Acts

FIGURE 9. Decomposing Geopolitical Alignment: Event Domains and Formal Acts

Notes: Panel (a) estimates a multivariate extension of equation (3) with the three event-domain components entered simultaneously, each with four lags. Components: Economic (category A), Diplomatic (category B), Security (categories C and D). Panel (b) compares the baseline with an index built only from formal acts (solid) and with the same acts scored at the median Goldstein value of their CAMEO code (dashed). Both panels report conditional local-projection coefficients for log GDP per capita (×100) on the long-coverage sample, with baseline fixed effects and Driscoll–Kraay standard errors as in Figure 6. Shaded areas represent 95% confidence intervals. Coefficients are per one standard deviation of the aggregate index.

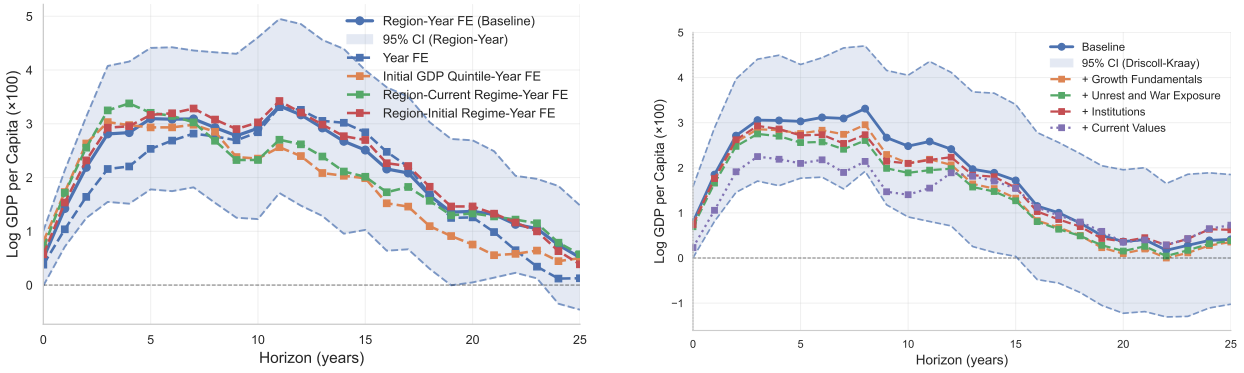
step meets the prompt's threshold and how intense it is. We therefore rebuild the index from formal acts only: agreements and diplomatic recognition, sanctions and their removal, diplomatic breaks and expulsions, uses of force, and state visits.[21] These acts are dated public decisions that contemporaneous sources record regardless of later outcomes. They account for 35 percent of events. Apart from the event set, the index is constructed as the baseline, and its correlation with the baseline is 0.81.

Panel (b) of Figure 9 shows that the formal-acts index reproduces the baseline path. Its coefficient lies within the baseline 95 percent confidence interval at every horizon from 0 to 15 and peaks at 3.0 log points in year 12, compared with 3.3 log points in year 11 for the baseline. The model also assigns each event an intensity score, which could carry hindsight even when the event itself is a formal act. For the dashed line, we score each act at the median Goldstein value of its CAMEO code, so that its score no longer depends on the model's judgment of intensity. Doing so changes the path by at most 0.2 log points. Together, the two paths weigh against look-ahead bias in which events are recorded or in their scores as the source of the relationship. As Section 2.3 shows, whether the database records externally documented events also does not depend on the subsequent growth of the countries involved.

[21]The formal-acts set comprises CAMEO codes 042–043 (visits), 054 (diplomatic recognition), 057 (formal agreements), 081 and 085 (easing of sanctions), 161–163 and 166 (breaking relations, halting aid, imposing sanctions, and expulsions or withdrawals), 172 and 174 (administrative sanctions and expulsions of individuals), and 190–204 (military force and mass violence).

3.3.3. Additional Controls and Fixed Effects

Alternative Fixed Effects Specifications. Panel (a) of Figure 10 shows similar coefficient paths under four alternative fixed-effects structures: year fixed effects only, initial GDP quintile-year effects,[22] region-initial regime-year effects (where regime indicates a country's democracy status), and region-current regime-year effects. Across specifications, the paths closely track the baseline and peak between about 3.0 and 3.4 log points. The modest attenuation under region-current regime-year effects, which allow region-year shocks to differ by current regime status, is consistent with some overlap between regime status and the variation summarized by the index. Section 4.2 examines this relationship.



(a) Alternative Fixed Effects

(b) Additional Controls

FIGURE 10. Robustness to Alternative Specifications

Notes: Panel (a) compares conditional coefficient paths under alternative fixed-effects specifications. The baseline (solid blue with 95% Driscoll–Kraay confidence interval) uses region-year fixed effects. Dashed overlays show year fixed effects only, initial GDP quintile-year, region-initial regime-year, and region-current regime-year fixed effects (regime = democracy status). Panel (b) cumulatively adds four lags of growth fundamentals (trade openness, log population, age-structure shares, investment share, log capital stock, and log human capital index), domestic unrest and five casualty-based war exposure measures, and five Varieties of Democracy (V-Dem) institutional quality indices. The final overlay adds the current values of all these controls. All paths in panel (b) use identical observations from 98 countries. Both panels use the baseline lag structure as in Figure 6. Coefficients are for log GDP per capita (×100) per one-standard-deviation change in the index. In panel (a), the baseline and year fixed effects use the 148-country long-coverage sample, and the other three paths use 122, 144, and 144 countries. In panel (b), the shaded area is the 95% Driscoll–Kraay confidence interval for the baseline.

Time-Varying Economic and Political Controls. Time-varying factors, including growth fundamentals, domestic instability, wars, and institutional change, may confound the estimated relationship if they correlate with geopolitical changes. We probe this concern by progressively adding controls, which enter as lags because contemporaneous controls may themselves respond to alignment. To limit compositional differences, all estimates use identical observations from the 98 countries with complete data on all controls.

[22] Following Acemoglu et al. (2019), we rank countries using Angus Maddison's GDP estimates for 1960, which are available for 145 countries in our panel, to maximize sample coverage.

Figure 10, panel (b), shows that the coefficient paths remain close as controls are cumulatively added, with peaks of about 3 log points in every lag-only specification. Adding four lags of standard growth fundamentals, including trade openness, investment, capital, and human capital, produces limited changes. Four lags of domestic unrest[23] and lagged casualty-based war exposure[24] likewise leave the path close to the common-sample baseline. Adding four lags of five V-Dem institutional quality indices (electoral, liberal, participatory, deliberative, and egalitarian democracy) (Coppedge et al. 2026), following Acemoglu et al. (2025), produces little further change.

The final specification adds the current values of all controls to their lags. The estimates attenuate to 2.1 log points at year 5 and 1.4 at year 10 but remain statistically distinguishable from zero at most horizons. Because current values may respond to alignment or lie on its transmission path, they are potentially bad controls, and we report this specification only as an overcontrol sensitivity. These results suggest that geopolitical alignment carries information about future growth beyond a broad set of observed economic and political conditions.

3.3.4. Placebo and Timing Diagnostics

We conduct two placebo exercises that ask whether the estimated GDP response depends on the realized country-specific assignment and timing of changes in geopolitical alignment. Both exercises re-estimate the full baseline local projection in each placebo draw, preserving the specification's dynamics and fixed-effects structure.

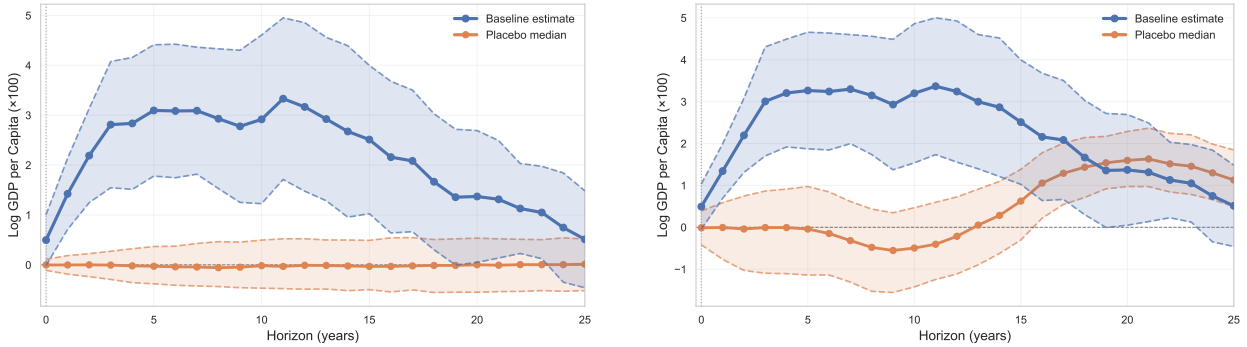
Within-Region-Year Reassignment. The first exercise randomly reassigns contemporaneous alignment values across countries within each region-year cell. The reassignment preserves the index's within-cell distribution while breaking its observed country assignment. Because the baseline already includes region-year fixed effects, the exercise does not test common regional dynamics. Instead, it asks whether the estimated response depends on the realized pairing between countries' GDP paths and alignment values within a region-year cell.

Panel (a) of Figure 11 shows that the placebo median remains flat near zero across all horizons and the placebo band stays tight, while the realized response lies well outside the placebo band throughout the main response window. This contrast shows that the realized within-cell pairing of alignment and GDP paths matters for the estimated response. It does not, however, distinguish the alignment relationship from time-varying omitted determinants that covary with actual alignment, because the reassignment breaks both relationships.

Future-Year Timing Placebo. The second exercise replaces each country's contemporaneous alignment value with its value 8 to 15 years in the future, with the lead drawn randomly for each country.

[23]The unrest indicator equals one if any domestic conflict event (assassinations, strikes, guerrilla warfare, government crises, purges, riots, revolutions, or anti-government demonstrations) is recorded in the Cross-National Time-Series Data Archive. See Appendix B.1 for details.

[24]We include four lags of five war variables from Federle et al. (2026): domestic war-site onset, domestic casualty intensity (scaled by prewar population), trade-weighted foreign war exposure, proximity-weighted foreign war exposure, and the count of active foreign war sites.



(a) Within-Region-Year Reassignment

(b) Future-Year Timing Placebo

FIGURE 11. Placebo Tests for the Conditional Dynamic GDP Response

Notes: Blue: baseline estimates and 95% Driscoll–Kraay CIs (in panel (b), estimated on 1960–2009 base years). Orange: placebo median and 5th–95th percentile range across 500 draws. Panel (a) reassigns alignment within region-year cells. Panel (b) replaces it with values 8–15 years ahead. Each draw re-estimates the baseline specification. The outcome is log GDP per capita (×100).

The shift keeps each country's own alignment series but breaks its contemporaneous timing with GDP. The exercise therefore asks whether the baseline path depends on the temporal match between geopolitical changes and subsequent GDP dynamics.

In panel (b) of Figure 11, the placebo median stays near zero through horizon 5, dips to about −0.6 log points around horizon 9, and then rises to the actual late-horizon path once the projection window reaches the years from which the future alignment values are drawn. This delayed emergence is a positive control: the design is sensitive to timing. The exercise does not distinguish geopolitical changes from omitted factors that move contemporaneously with them, and the long shift is less informative about confounding that operates over shorter intervals.

Averaged over horizons 0 to 10, the realized response exceeds all 500 placebo draws in both exercises (Appendix B.7).

3.4. Robustness to Bundled Domestic Transformations

Changes in a country's external geopolitical environment often coincide with major domestic transformations, including democratization, regime change, post-socialist transition, post-conflict reconstruction, and state collapse. These episodes may affect growth through channels beyond geopolitical alignment. We address this concern in two ways. The first absorbs transformation-period observations with separate country-year fixed effects. The second uses bilateral shifts around leadership transitions in major nations, anchoring the timing to events outside the focal country. Neither exercise eliminates all omitted-variable concerns, but together they reduce reliance on the variation most directly exposed to bundled domestic transformations.

3.4.1. Excluding Domestic Transformation Episodes

We identify domestic transformation periods using a historical-institutional codebook. The codebook evaluates candidate episodes according to whether they mark a recognized turning point, involve broad changes in domestic political or economic institutions, span multiple institutional margins, and plausibly affect the country's long-run economic trajectory. Geopolitical realignment itself is not a classification criterion. The resulting classification contains 329 episodes across 172 countries and designates 11.8 percent of country-years as transformation periods. Examples include China's reform and opening (1978–1984), Chile's regime change (1973–1976), and South Africa's democratization (1990–1994). Appendix B.8 describes the criteria, and Table B.4 reports summary statistics and additional examples.

We re-estimate the baseline local projection after adding a separate country-year fixed effect for every transformation-period observation, thereby removing these observations from the variation used in estimation. As an alternative classification check, we repeat the exercise using V-Dem's *Regimes of the World* measure, absorbing covered observations within three years before or after a change in regime type.[25]

Panel (a) of Figure 12 shows that both exclusion rules attenuate the estimated GDP path but preserve its positive medium-run profile. At year 5, the historical transformation exclusion yields 2.0 log points, compared with 3.1 in the 148-country baseline. The V-Dem exclusion gives 2.4 log points on its covered-observation sample.[26] Both exclusion estimates are distinguishable from zero and remain positive through year 15, although precision weakens at later horizons. Applying the same one-standard-deviation, 25-year sustained-path mapping within each estimation sample gives 6.3 log points under the historical exclusion and 5.9 under the V-Dem exclusion, compared with 10.1 in the baseline. The historical exclusion therefore preserves about three-fifths of the baseline magnitude, and the V-Dem mapping is similar on its covered sample.

3.4.2. Leadership Transitions in Major Nations

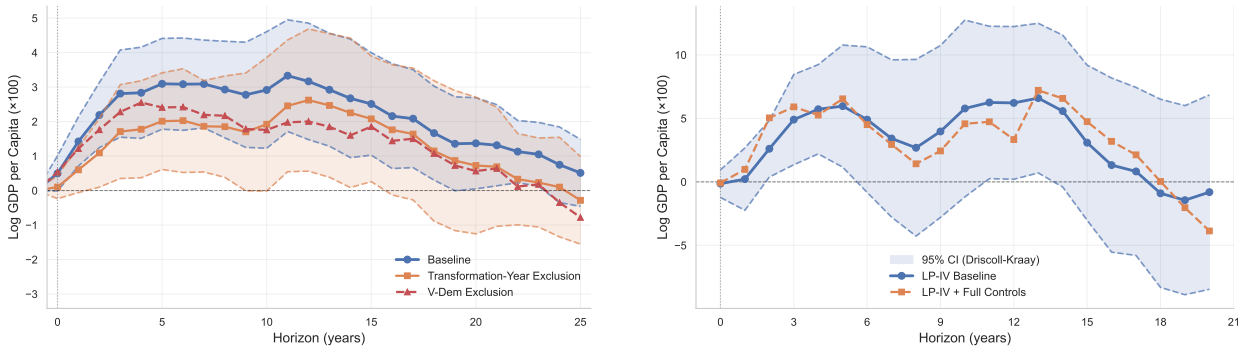
The second exercise uses bilateral alignment shifts around 67 leadership transitions across 21 major nations: 25 deaths in office from the Archigos database (Goemans, Gleditsch, and Chiozza 2009) and 42 close-election turnovers with vote margins below 10 percentage points (Marx, Pons, and Rollet 2025). For a transition in major nation j in year t, we construct

$$z_{ct}^{\mathrm{LC}} = \left(S_{cj,t+1} - S_{cj,t-1}\right) \times \text{GDP share}_{j,t-1},$$

summing across transitions when more than one occurs in the same year. We instrument the index with this GDP-weighted bilateral shift in the baseline local projection and add four lags of the

[25] Appendix B.8 explains why we absorb transformation-period observations rather than add fixed effects for all country-era cells.

[26] Because V-Dem coverage changes the estimation sample, Appendix B.8 reports a fixed 124-country comparison: 3.6 log points in the corresponding baseline and 2.5 under the V-Dem exclusion.



(a) Excluding Transformation Years

(b) Leadership Transitions in Major Nations

FIGURE 12. Robustness to Bundled Domestic Transformations

Notes: Panel (a) compares baseline conditional local-projection coefficients with estimates that absorb country-years classified as domestic transformations. Panel (b) reports LP-IV estimates instrumenting alignment with bilateral shifts around leadership changes in major nations. The baseline and transformation-year paths use 148 countries. The V-Dem path uses covered observations, and the full-controls IV uses a 109-country common sample. The V-Dem path absorbs observations within three years before or after a change in regime type in V-Dem's *Regimes of the World* measure. The full-controls IV adds four lags of trade openness, domestic unrest, war exposure, and V-Dem institutional indices. Shaded bands report 95% Driscoll–Kraay intervals for the baseline and transformation-year paths in panel (a) and the baseline IV in panel (b). Other paths are point estimates. The outcome is log GDP per capita (×100).

instrument.[27]

Panel (b) of Figure 12 shows positive paths that are less precise than the baseline. The baseline IV path reaches 4–6 log points at several horizons in the first decade and rises again around years 11–14, and its 95 percent intervals exclude zero in years 2–5 and 11–13. The full-controls specification displays a similar pattern. The wider confidence intervals reflect the limited number of underlying transitions. Because the instrument uses the realized bilateral change around each transition, not the transition alone, it may also capture other bilateral developments in the short event window. A narrower window, $S_{cj,t} - S_{cj,t-1}$, gives a stronger first stage ($F = 53.7$, against 16.9) and 3.1 log points at year 5, close to the baseline. Appendix B.9 reports the event-type split and alternative bilateral-shift windows.

3.5. Interpretation of the Growth Estimates

The estimates above are conditional dynamic responses. They admit a causal interpretation if, after conditioning on country and region-year fixed effects and on past GDP and alignment, countries whose alignment changes would otherwise have followed similar GDP paths. That is, current changes in alignment must be unrelated to other shocks that affect GDP now or later. Formally, the maintained condition is as follows.

[27] Following Jordà (2005) and Plagborg-Møller and Wolf (2021), at each horizon the estimator combines a reduced-form projection of $y_{c,t+h}$ on z_{ct}^{LC} with a first-stage projection of p_{ct} on z_{ct}^{LC}. Both use the baseline fixed effects and lag controls, augmented with four lags of the instrument. The LP-IV response is $\widehat{\alpha}_h^{\text{LP-IV}} = \widehat{\alpha}_h^{\mathrm{RF}} / \widehat{\alpha}^{\mathrm{FS}}$.

ASSUMPTION 1. *The error $\varepsilon_{c,t+h}$, the effect on GDP of all structural shocks other than the current change in alignment, satisfies $\mathbb{E}\left[\varepsilon_{c,t+h} \mid p_{ct}, \{\mathbf{x}_{c\tau}\}_{t_0 \le \tau \le t}\right] = 0$ for all $\{\mathbf{x}_{c\tau}\}_{t_0 \le \tau \le t}$, all countries c, and all $t \ge t_0$.*

Under Assumption 1 and the linear specification, the LP coefficient α_h^{LP} identifies the impulse response at horizon h, for any value p_{c0} of the index:

$$\alpha_h^{\mathrm{LP}} = \mathcal{R}_{p\to y}(h) \equiv \mathbb{E}\big[y_{c,t+h} \mid p_{ct} = p_{c0} + 1, \mathbf{x}_{ct}\big] - \mathbb{E}\big[y_{c,t+h} \mid p_{ct} = p_{c0}, \mathbf{x}_{ct}\big], \qquad h = 0, 1, \ldots, H. \tag{4}$$

The impulse response traces the dynamic effect on GDP per capita of a one-unit improvement in current geopolitical alignment together with the subsequent evolution of the index (Jordà 2005). The reported coefficients scale the response in equation (4) to a one-standard-deviation improvement, $0.141\,\mathcal{R}_{p\to y}(h)$.

What does this response capture? Because the index summarizes a country's relations with all major nations rather than a single policy, the causal object is broad: a change in the access and risk conditions described in Section 2.4. The accompanying changes in Section 4.1 fit both margins, with unrest falling and trade openness rising at once and investment, capital, and productivity following. For the same reason, the estimates cannot separate these margins or isolate the effect of any single treaty, sanction, or partner.

No single diagnostic verifies Assumption 1. The exercises in Sections 3.2–3.4 each target a distinct threat. Conditioning on past GDP and alignment addresses reverse causality from past growth: negative-horizon coefficients are not individually distinguishable from zero, and a reverse projection shows no systematic relationship between current GDP and subsequent alignment. The component and partner-group estimates show that the relationship is not confined to economic events or to relations with one bloc. Alternative fixed effects and lagged growth fundamentals, conflict, and institutional controls address observed pre-existing conditions, and adding their current values attenuates but does not remove the relationship. Placebo reassignment and future-year shifts address mechanical artifacts and timing, and the transformation exclusions and the shifts around major nations' leadership transitions address domestic reforms and regime changes.

The main remaining concern is an unobserved contemporaneous shock, at home or in a major partner, that changes both bilateral relations and future growth. A domestic shift that improves relations with all major nations at once, such as a broad opening or the end of a civil conflict, would move the three partner components together, yet the part of each component that does not move with the other two is still followed by GDP gains (Section 3.3.1). Subject to the limitation that these exercises cannot rule out every unobserved confounder, the evidence is consistent with a causal interpretation of the conditional dynamic response.

4. Channels and Implications

This section interprets the dynamic relationship between GDP and the external geopolitical environment. We first compare the relationship across output measures and examine accompanying changes in candidate growth fundamentals. We next study the interaction between democratic

institutions and geopolitical reorientation. We conclude with illustrative accounting exercises that map observed realignments into implied GDP differences under maintained assumptions.

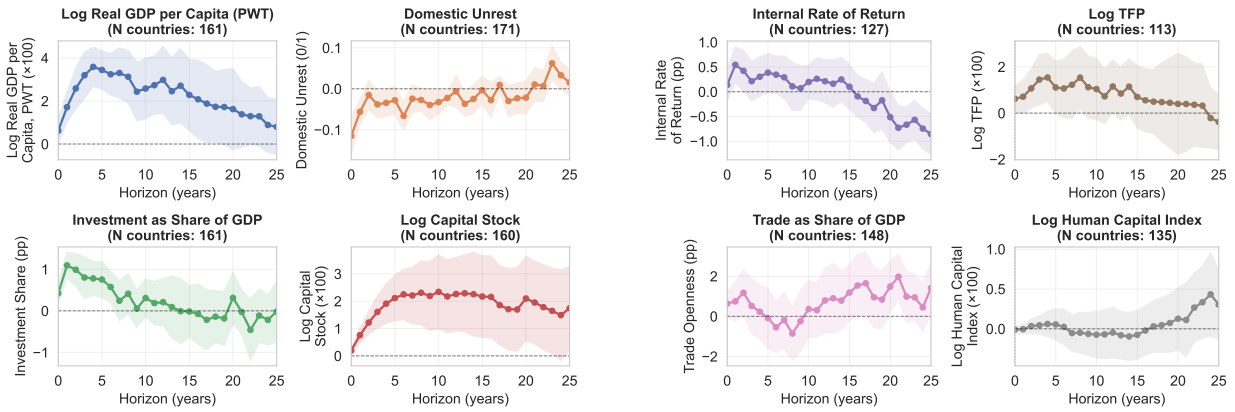
4.1. Output Measures and Growth Fundamentals

Alternative Output Measures. Our baseline uses GDP per capita from the World Development Indicators to maximize country and temporal coverage. As an alternative, panel (a) of Figure 13 uses real GDP per capita from the Penn World Table. The similar response peaks at about 3.6 log points around year 4 and remains above 2 log points through year 15.

Growth Fundamentals. To examine whether changes in the index are accompanied by changes in candidate growth fundamentals, we estimate the following reduced-form responses:

$$m_{c,t+h} = \alpha_h^m p_{ct} + \sum_{\ell=1}^{4} \beta_{h\ell} y_{c,t-\ell} + \sum_{\ell=1}^{4} \gamma_{h\ell} p_{c,t-\ell} + \sum_{\ell=1}^{4} \lambda_{h\ell} m_{c,t-\ell} + \eta_c + \eta_{r(c)t} + \varepsilon_{c,t+h}^m \tag{5}$$

where $m_{c,t+h}$ denotes the outcome at horizon h. Each specification adds four lags of the outcome to the baseline fixed effects and lagged controls.



(a) Output, Stability, and Capital Accumulation (b) Growth Fundamentals

FIGURE 13. Growth Correlates Following Changes in the Geopolitical Alignment Index

Notes: This figure reports horizon-specific coefficients on the country-level geopolitical alignment index from local projections of the listed outcomes. Panel (a) shows real GDP per capita from the Penn World Table, domestic unrest, investment share, and capital stock. Panel (b) shows internal rate of return, total factor productivity, trade openness, and human capital. All specifications follow equation (5) with baseline fixed effects and lag structure as in Figure 6, adding four lags of the dependent variable. Each outcome uses an outcome-specific country set, with counts shown in parentheses. Country counts are constant across horizons except for trade, for which 147 of 148 countries contribute at horizons 18–24. Coefficients are per one-standard-deviation change in the index. Log outcomes are multiplied by 100. Investment share, trade openness, and the internal rate of return are in percentage points. Domestic unrest is a binary indicator. Shaded areas represent 95% confidence intervals based on Driscoll–Kraay standard errors.

Figure 13 shows distinct timing across outcomes. Domestic unrest, our measure of domestic political instability, falls sharply on impact and then moves back toward zero. The fall in unrest speaks to the risk margin described in Section 2.4: political stability can sustain domestic production

and reduce the political and sovereign risks that deter investment and capital accumulation. Consistent with this interpretation, the internal rate of return rises during the first two post-impact years, investment rises by about 1 percentage point within the first few years, and the capital stock reaches about 2.2 log points around years 8–12. At long horizons, after the initial alignment improvement has dissipated, the internal rate of return turns negative.

Total factor productivity rises to about 1.5 log points by year 4 and remains above 1 log point through year 10. This response is consistent with the access margin described in Section 2.4 and with the roles of technology adoption and international R&D spillovers emphasized by Parente and Prescott (1994) and Coe and Helpman (1995). Trade openness, defined as exports plus imports of goods and services relative to GDP, rises by about 0.6 percentage points on impact and, after imprecise intermediate estimates, reaches 1.4–2.0 percentage points at several horizons after year 15.[28] Human-capital estimates remain near zero until late in the horizon, and no post-impact coefficient is individually distinguishable from zero.

Appendix B.10 reports positive point-estimate paths for the tax-to-GDP ratio and primary enrollment, and GDP-like paths for consumption and domestic absorption per capita. It finds no systematic response in market reform, secondary enrollment, employment, or the labor share.

The sequence of lower domestic political instability and higher investment and productivity, followed by capital accumulation, helps interpret the aggregate GDP response. Trade openness rises at short and long horizons. Because the outcomes are estimated separately, their causal ordering and roles in mediating the GDP response remain unresolved.

4.2. Democracy and Geopolitical Alignment

Domestic political institutions and the external political environment are distinct dimensions of development, but they need not evolve independently. Institutions are a central determinant of development (Acemoglu, Johnson, and Robinson 2001), and Acemoglu et al. (2019) document substantial long-run GDP gains from democratization. Internationally, democratic institutions are associated with preferential foreign aid (Alesina and Dollar 2000) and more stable relations among democracies (Maoz and Russett 1993). We first examine how democratic status is associated with bilateral reorientation across major nations, then compare the GDP responses to democracy and geopolitical alignment when both enter jointly.

4.2.1. Democracy and Bilateral Geopolitical Reorientation

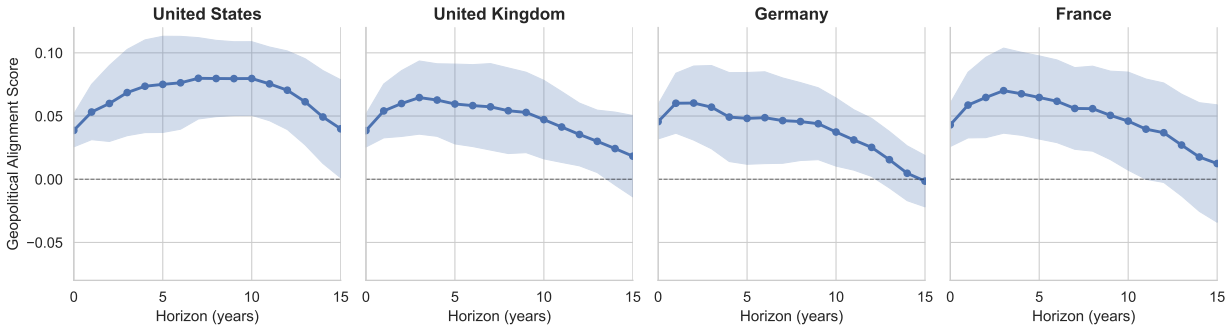
Using the binary democracy indicator from Acemoglu et al. (2019) (henceforth ANRR), we estimate

$$S_{cj,t+h} = \beta_h^j D_{ct}^{\text{ANRR}} + \sum_{\ell=1}^{4} \gamma_{h\ell}^j S_{cj,t-\ell} + \eta_c + \eta_t + \varepsilon_{cj,t+h} \tag{6}$$

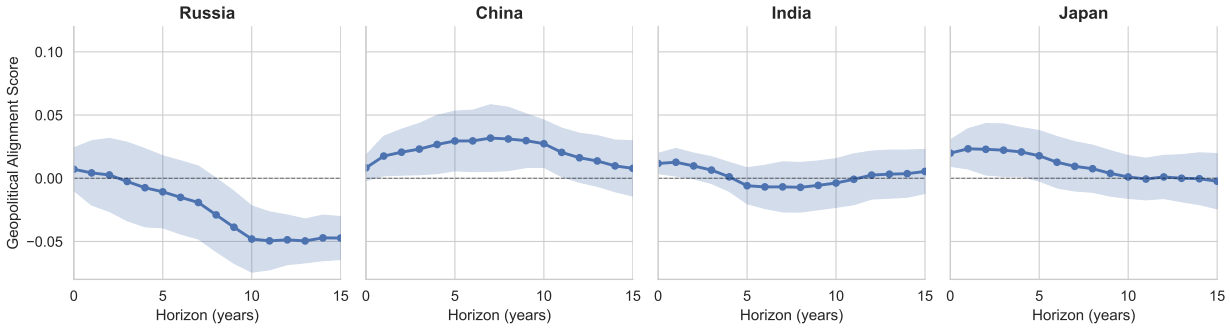
[28] Fan, Wo, and Xiang (2025) study the dynamic relationship between trade flows and bilateral alignment in a dyad-level gravity framework and find that bilateral trade rises in the same year as an improvement in alignment.

where $S_{cj,t}$ is country c's bilateral alignment score with major nation j, and D_{ct}^{ANRR} is its democratic status. The specification exploits within-country changes in democratic status after conditioning on four lags of the bilateral score, country fixed effects, and year fixed effects. Because D_{ct}^{ANRR} records democratic status rather than a separately coded transition, β_h^j traces its dynamic conditional association with bilateral relations at each horizon.

Panel (a) of Figure 14 shows positive coefficients for all four Western democracies. The US coefficient rises from about 0.04 on impact to 0.08 around years 7–10 and remains positive through year 15. The coefficients for the United Kingdom, Germany, and France peak between 0.06 and 0.07 within the first five years before attenuating. Their common direction is consistent with closer bilateral ties among countries that share democratic institutions.



(a) Selected Western Democracies



(b) Other Major Nations

FIGURE 14. Democracy and Bilateral Geopolitical Reorientation

Notes: This figure reports horizon-specific coefficients on the ANRR democracy indicator from equation (6) over horizons 0–15. Panel (a) covers the United States, United Kingdom, Germany, and France. Panel (b) covers Russia, China, India, and Japan. The dependent variable is the bilateral geopolitical alignment score with the named major nation. The sample covers the ANRR democracy data (182 countries, 1960–2010). All specifications include four lags of the bilateral score, country fixed effects, and year fixed effects. Shaded areas are pointwise 95% confidence intervals based on Driscoll–Kraay standard errors.

Panel (b) shows heterogeneous associations with the other major nations. The Russian coefficient is initially near zero and declines to approximately −0.05 after year 10. The Chinese coefficient is positive at medium horizons, reaching about 0.03, while the Indian and Japanese coefficients are smaller and less persistent. The panels therefore show a geopolitical reorientation toward Western

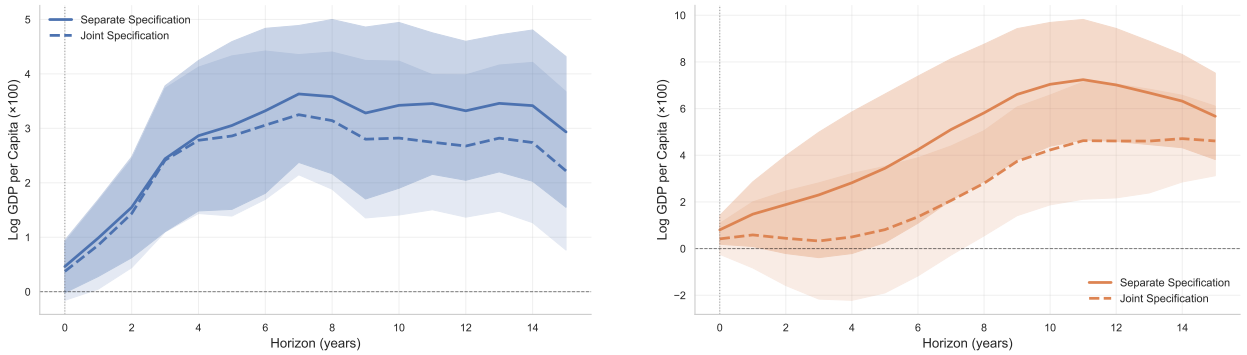
democracies rather than a uniform improvement across all bilateral relations.

4.2.2. Democracy, Geopolitical Alignment, and GDP

We next ask whether democratic status and the index contain the same information about subsequent GDP by estimating

$$y_{c,t+h} = \alpha_h^{\text{Geo}} p_{ct} + \alpha_h^{\text{Dem}} D_{ct}^{\text{ANRR}} + \boldsymbol{\gamma}_h' \mathbf{x}_{ct} + \eta_c + \eta_t + \varepsilon_{c,t+h}$$

where both the geopolitical alignment index (p_{ct}) and democratic status enter jointly, and $\mathbf{x}_{ct}$ contains four lags of GDP per capita and geopolitical alignment. Following ANRR, we use year rather than region-year fixed effects, retaining the variation associated with regional democratization waves.[29]



(a) Geopolitical Alignment (b) Democracy

FIGURE 15. Democracy, Geopolitical Alignment, and GDP

Notes: This figure compares horizon-specific GDP coefficient paths from separate and joint local projections using the original ANRR GDP series. All specifications include four lags of GDP, country fixed effects, and year fixed effects. In panel (a), both specifications include four lags of geopolitical alignment, and the joint specification adds democratic status. In panel (b), the separate specification follows ANRR, while the joint specification adds current geopolitical alignment and its four lags. The outcome is log GDP per capita (×100). Alignment coefficients correspond to a one-standard-deviation index change, while democracy coefficients correspond to a zero-to-one difference. Shaded areas are pointwise 95% confidence intervals based on Driscoll–Kraay standard errors.

Panel (a) of Figure 15 shows that the index retains a positive, hump-shaped GDP path when democratic status is added. The separate and joint coefficients peak at 3.6 and 3.3 log points at year 7. The joint path is moderately lower thereafter but remains positive through year 15. The external geopolitical environment therefore contains growth-related variation not subsumed by political regime type.

Panel (b) shows that adding alignment lowers the democracy coefficient most at short and medium horizons. The separate democracy coefficient rises from 0.8 log points on impact to 7.2 log

[29] Regional democratization waves, including Latin America in the 1980s, Eastern Europe in the 1990s, and the Arab Spring in the 2010s, induce strong regional correlation in democratic transitions. Region-year fixed effects absorb this variation and may therefore attenuate the democracy coefficients. Under our baseline country and region-year fixed effects, however, adding democratic status leaves the GDP coefficient path of the index nearly unchanged (Appendix Figure B.9, panel (b)).

points at year 11, then declines to 5.7 log points by year 15. Adding current geopolitical alignment and its four lags lowers the coefficient throughout: the joint path reaches 4.7 log points at year 14 and is 4.6 log points at year 15.[30]

Together with the bilateral reorientation in Figure 14, these estimates show that the early democracy–GDP relationship overlaps with changes in geopolitical alignment, while a distinct democracy relationship emerges more gradually, consistent with slower-moving domestic institutional forces. The coefficient changes measure conditional overlap rather than a mediation share.

4.3. Geopolitical Growth Accounting

We use the estimated response paths to map observed changes in geopolitical alignment into implied GDP contributions. The first exercise applies the paths to within-country realignments. The second maps cross-country alignment gaps into GDP differences under a homogeneity assumption and is illustrative rather than separately identified.

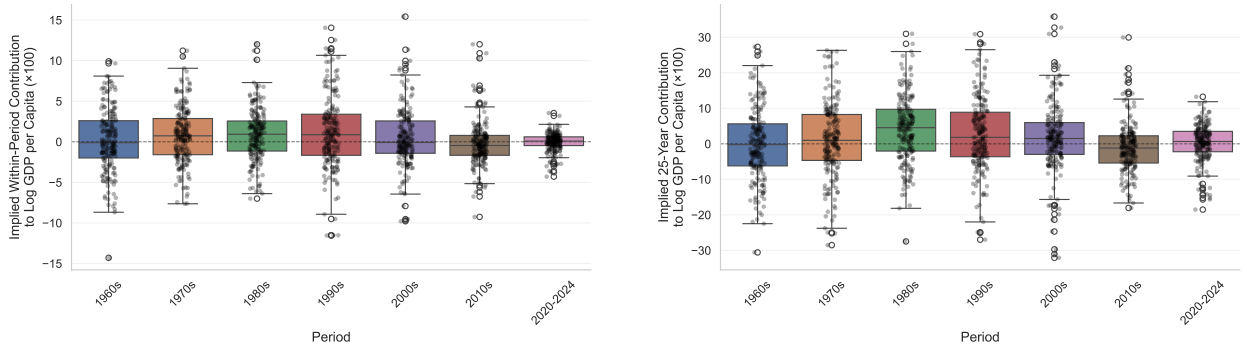
4.3.1. Decadal Realignments and Implied GDP Contributions

For each country and calendar period, we compute implied within-period and 25-year GDP contributions from the transitory and sustained response paths.[31]

Figure 16 shows positive median accounting contributions in the 1970s through the 1990s. The within-period median is 0.7–0.9 log points across those decades, while the 25-year median reaches 4.6 log points in the 1980s and remains 1.5–1.8 log points in the 1990s and 2000s. The within-period median is near zero in the 2000s, and both medians turn negative in the 2010s. This reversal gives economic scale to the distributional shift documented in Section 2.4: post-Cold War convergence maps into positive median contributions, while fragmentation after 2010 maps into a slightly negative median. The 2020–2024 period shows a partial median recovery but continued wide dispersion across countries.

[30] Appendix Figure B.8 maps the democracy coefficients into transitory and sustained paths. The separate sustained democracy path reaches 19.1 log points after two decades, compared with 13.1 log points when alignment is included. The separate estimate closely matches the approximately 20 percent long-run GDP effect estimated by Acemoglu et al. (2019).

[31] Specifically, for within-period contributions, we calculate $\sum_{t=0}^{9} \alpha_{9-t}^{\text{transitory}} (p_{c,\tau+t} - p_{c,\tau})/\sigma_p$, where $\sigma_p = 0.141$ is the standard deviation of the index, so that index changes enter in the standard-deviation units of the reported coefficients, $\alpha_h^{\text{transitory}}$ is the transitory-path coefficient at horizon h, $p_{c,\tau+t} - p_{c,\tau}$ is the cumulative change in the geopolitical alignment index from the start of the decade, and τ denotes the first year of the decade. For 25-year contributions, we use $\alpha_{25}^{\text{sustained}} \times (p_{c,\tau+9} - p_{c,\tau})/\sigma_p$, where $\alpha_{25}^{\text{sustained}}$ is the sustained-path coefficient at the 25-year horizon. For 2020–2024, the analogous calculation uses the available five-year interval.



(a) Within-Period Contribution

(b) 25-Year Contribution

FIGURE 16. Implied GDP Contributions of Geopolitical Realignments by Period

Notes: This figure reports the distribution across countries of implied GDP contributions associated with observed within-period changes in geopolitical alignment. Panel (a) applies the transitory response path to changes realized during each period. Panel (b) applies the 25-year sustained-path coefficient to the period-end change. Values are log GDP per capita multiplied by 100. Boxes show interquartile ranges and whiskers extend to 1.5 times the interquartile range. Gray dots show individual countries, and open circles mark values beyond the whiskers. The dashed line marks zero. Periods cover calendar decades from the 1960s through the 2010s and 2020–2024, the last spanning only five years. Each period includes all 193 countries in the panel, since the implied contributions use only the alignment index.

4.3.2. Illustrative Cross-Country GDP Contributions

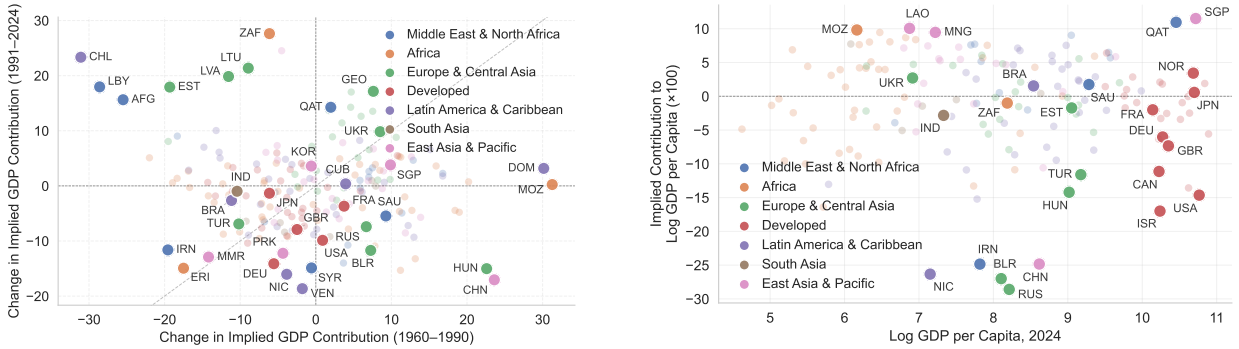
To provide cross-country scale, we apply the estimated within-country response paths to each country's deviation in geopolitical alignment from the annual cross-country median.[32]

Panel (a) of Figure 17 compares changes in implied contributions over 1960–1990 and 1991–2024. Values span approximately −31 to +31 log points and are dispersed across all four quadrants. Chile shows the largest reversal, from about −31 log points in the first period to +23 in the second, and South Africa's contribution rises by about 28 log points in the second period. China and Hungary move in the opposite direction, and Venezuela and Nicaragua also move sharply toward more negative contributions in the second period. The limited concentration around the 45-degree line indicates that relative accounting contributions differ substantially across periods.

Panel (b) reports the 2024 cross-country mapping. Implied contributions range from approximately −29 to +12 log points, with a longer negative than positive tail. Russia, Belarus, Nicaragua, Iran, and China lie at the lower end. Israel and Canada also have sizable negative contributions. Levels for the 24 major nations are mechanically lower because their index omits their own share of world GDP. Singapore lies at the upper end, followed by Qatar, Laos, and Mozambique.

There is no evident monotonic relationship between GDP per capita and the implied contribution. The range shows the scale implied by cross-country differences in the external geopolitical

[32]Formally, the geopolitical contribution to country c's GDP in year t is: $\Delta y^{\text{geo}}_{c,t} = \sum_{s=t-25}^{t} \alpha^{\text{transitory}}_{t-s} \times (p_{c,s} - p^{\text{median}}_{s})/\sigma_p$, where $\sigma_p = 0.141$ is the standard deviation of the index, $\alpha^{\text{transitory}}_{h}$ is the transitory-path coefficient at horizon h and p^{median}_{s} is the cross-country median of the geopolitical alignment index in year s. Deviations before 1960 are set to their 1960 values. Panel (a) plots the within-period change $\Delta y^{\text{geo}}_{c,\text{end}} - \Delta y^{\text{geo}}_{c,\text{start}}$ for each period. Panel (b) applies the 25-year sustained-path coefficient to cross-country differences in 2024: $(p_{c,2024} - p^{\text{median}}_{2024})/\sigma_p \times \alpha^{\text{sustained}}_{25}$.



(a) Changes in Implied GDP Contributions

(b) Implied GDP Contributions, 2024

FIGURE 17. Illustrative Cross-Country Accounting for Geopolitical Alignment

Notes: This figure reports illustrative GDP accounting contributions obtained by applying the estimated response paths to countries' deviations in geopolitical alignment from the annual cross-country median. Panel (a) plots the change in each country's implied contribution during 1960–1990 on the horizontal axis and during 1991–2024 on the vertical axis. Panel (b) plots each country's implied contribution in 2024 against log GDP per capita, among countries with 2024 GDP data, with deviations measured from the 2024 median among these countries. Accounting contributions are measured as log GDP per capita multiplied by 100. Colors indicate regions. Horizontal dashed lines mark zero in both panels. Panel (a) also includes a vertical zero line and a diagonal 45-degree line.

environment under the maintained homogeneity assumption. It is not a causal decomposition of cross-country income differences.

5. Additional Robustness and Alternative Measures

Dynamic panel estimates and an unsmoothed index give 25-year sustained-path coefficients of 10.5 and 10.7 log points, close to the baseline 10.1, while indices built from machine-coded event databases or UN voting do not reproduce the GDP relationship. Sanctions and formal alliances leave the alignment path nearly unchanged.

5.1. Dynamic and Construction Robustness

Dynamic Panel Estimates. We complement the local-projection estimates with a dynamic panel model. Following the preferred specification in Acemoglu et al. (2019), we regress log GDP per capita on current geopolitical alignment, four lags of GDP and alignment, and the baseline country and region-year fixed effects (Appendix B.12). Under stable finite-order linear dynamics, this model and the local projections recover the same population response paths (Plagborg-Møller and Wolf 2021), and the dynamic panel can improve precision by imposing the autoregressive structure (Montiel Olea et al. 2026). The response to a transitory alignment change rises from 0.5 log points on impact to 0.9 after one year and then declines while remaining positive. Cumulated into a sustained one-standard-deviation path, it reaches 10.5 log points at year 25, with a 95 percent confidence interval of 5.4 to 16.7, close to the local-projection estimate of 10.1 and its interval of 5.3 to 16.8 (Appendix Figure B.10). The dynamic panel mainly extends the horizon by imposing parametric

dynamics.

Unsmoothed Index. Our baseline index smooths bilateral event scores to emphasize persistent relations. To test whether smoothing itself accounts for the GDP dynamics, we replace the index with the unsmoothed GDP-weighted average of bilateral event scores $\tilde{S}_{ct}$, equivalent to setting $\delta = 1$ in equation (1).[33]

Appendix Figure B.11 shows that the GDP coefficient remains positive when we use the unsmoothed index but is smaller than for the baseline index: it peaks at 1.2 log points in year 11, compared with 3.3 log points for the smoothed index. This attenuation is consistent with the rapid mean reversion of annual event scores: 58 percent of an initial change dissipates within one year. When the event-score path is sustained, however, the year-25 coefficient reaches 10.7 log points, close to the baseline estimate of 10.1. Smoothing therefore primarily filters transitory event variation. Appendix B.13 reports the supporting persistence and transitory-path results.

5.2. Alternative Data Sources and Bilateral Benchmarks

Machine-coded event databases provide alternative event streams, UNGA ideal points summarize multilateral voting positions, and sanctions and formal alliances capture narrower bilateral margins.

Machine-Coded Event Databases. We reconstruct country-level indices from ICEWS, GDELT, and Phoenix using the same aggregation and dynamic recursion as for the baseline index, then standardize each index within the relevant overlap window. Appendix B.14 shows that our geopolitical alignment index produces a positive, hump-shaped GDP coefficient path in all three windows. The ICEWS and GDELT paths are statistically indistinguishable from zero, while the Phoenix path is small and fades after the first few years. Because the indices are standardized within each window, pure differences in scale cannot account for the contrast. Together with the measurement comparison in Section 2.3, this pattern is consistent with the contrast arising from compiling consequential geopolitical events rather than averaging over news reports.

UN General Assembly Voting. UNGA voting similarity is widely used to measure foreign-policy affinity (Signorino and Ritter 1999). It locates countries in a common multilateral voting space, a different object from bilateral relationship intensity. We use the negative Ideal Point Distance (IPD) from Bailey, Strezhnev, and Voeten (2017), so higher values indicate closer voting positions.

Panel (a) of Figure B.14 in Appendix B.15 reports our primary comparison, negative IPD to the United States. Under the baseline country and region-year fixed effects, the GDP coefficient path is statistically indistinguishable from zero throughout the horizon. Panel (b) replaces the United States reference with a GDP-weighted average of negative IPD to the 24 major nations. Because

[33] We define $\tilde{S}_{ct} = \sum_{j \in \mathcal{N} \setminus \{c\}} \tilde{S}_{cj,t} \times \text{GDP share}_{jt}$, with $\tilde{S}_{cj,t} = 0$ in dyad-years without events, and substitute it for p_{ct} in the baseline local projection. The specification includes four lags of GDP and $\tilde{S}_{ct}$, country fixed effects, and region-year fixed effects.

ideal-point distances locate countries in a common multilateral space, their GDP-weighted average is not a direct counterpart to our aggregation of bilateral relations. We nonetheless report it as a broader check that the US comparison is not driving the result. Its coefficient path is also statistically indistinguishable from zero.

These results and the case comparisons in Section 2.2 indicate that IPD captures broad geopolitical affinity but not the same time-varying bilateral information as the geopolitical alignment index.

Sanctions and Formal Alliances. Sanctions capture a salient but narrow margin of bilateral coercion. Using the Global Sanctions Database (Felbermayr et al. 2020; Yalcin et al. 2025), we aggregate active sanctions imposed by the 24 major nations, weighted by their GDP shares, and enter this exposure jointly with geopolitical alignment. Figure B.15 in Appendix B.16 shows that adding sanctions leaves the alignment coefficient path nearly unchanged. The sanctions coefficient is negative when entered separately and reaches its trough in year 3. At that horizon, it attenuates from −8.2 to −5.6 log points in the joint specification, a reduction of 31 percent, and no joint sanctions coefficient is statistically distinguishable from zero. This attenuation reflects the different timing of the two measures: the alignment index records the diplomatic deterioration surrounding a sanction and its announcement or lifting, while the sanctions measure tracks active enforcement.

Formal alliances capture a narrow, slow-moving margin of institutionalized security cooperation. Using the Alliance Treaty Obligations and Provisions dataset (Leeds et al. 2002), we aggregate active alliances with the same major nations and enter the resulting exposure jointly with geopolitical alignment. Figure B.16 in Appendix B.17 again shows little change in the alignment coefficient path. The alliance coefficients are positive and build through the first decade. In year 9, the estimate declines from 17.8 to 15.3 log points when alignment is included, a 14 percent attenuation, and the joint estimates are statistically indistinguishable from zero. Here the alignment index records diplomatic cooperation surrounding alliance formation and maintenance, whereas the alliance variable is a persistent treaty stock. The stability of the alignment path in both exercises shows that neither narrow bilateral margin accounts for the broader alignment–GDP relationship.

6. Conclusion

This paper introduces and validates an event-based measure of bilateral geopolitical alignment and uses it to document a large, persistent relationship between improvements in a country's external geopolitical environment and its subsequent economic growth. Our results carry a broader message for the growth literature: the traditional separation between "domestic" and "international" determinants of growth obscures important interactions between domestic factors and the external political environment. Improvements in geopolitical alignment are accompanied by lower domestic unrest and higher investment, capital, productivity, and trade openness, each with its own timing.

Two limitations remain. First, because the estimates exploit within-country changes, they speak to changes in geopolitical relations, not to the level consequences of sustained cooperation or isolation, and their causal reading requires that these changes be unrelated to other shocks to

future growth. Second, the measure relies on LLM-compiled event records. Documented-event benchmarks and the human audit support its conflict–cooperation signal, but the set of recorded events remains an important source of measurement error.

The implications go beyond the general case for openness. Our accounting exercises suggest that the gains from cooperative relations are neither automatic nor permanent: fragmentation after 2010 turned the median implied contribution slightly negative, after positive contributions in the preceding decades. Because GDP gains follow improved relations with Western and other major nations alike, the relationship is not tied to alignment with any single bloc. As the post-Cold War order gives way to renewed strategic competition, the external political environment is an increasingly important dimension of economic development.

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Online Appendix A. Event-based Measure of Geopolitical Alignment

Appendix A documents the event database and the alignment measures in the order of Section 2: the reference set (A.1), event compilation and statistics (A.2), dynamics of the scores (A.3), case studies and maps (A.4–A.5), the validation exercises (A.6–A.10), and index statistics and the comparison with the Geopolitical Risk Index (A.11–A.12).

A.1. Major Nations

The reference set consists of 24 major nations: the G20 member countries other than South Africa, together with Belgium, Denmark, the Netherlands, Poland, Spain, and Switzerland. Twenty-three of them rank among the 24 largest economies by average share of world GDP over 1960–2024, and together they account for 82–90 percent of world GDP over the sample period. Figure A.1 presents the time series of aggregate and individual GDP shares for these 24 major nations.[34]

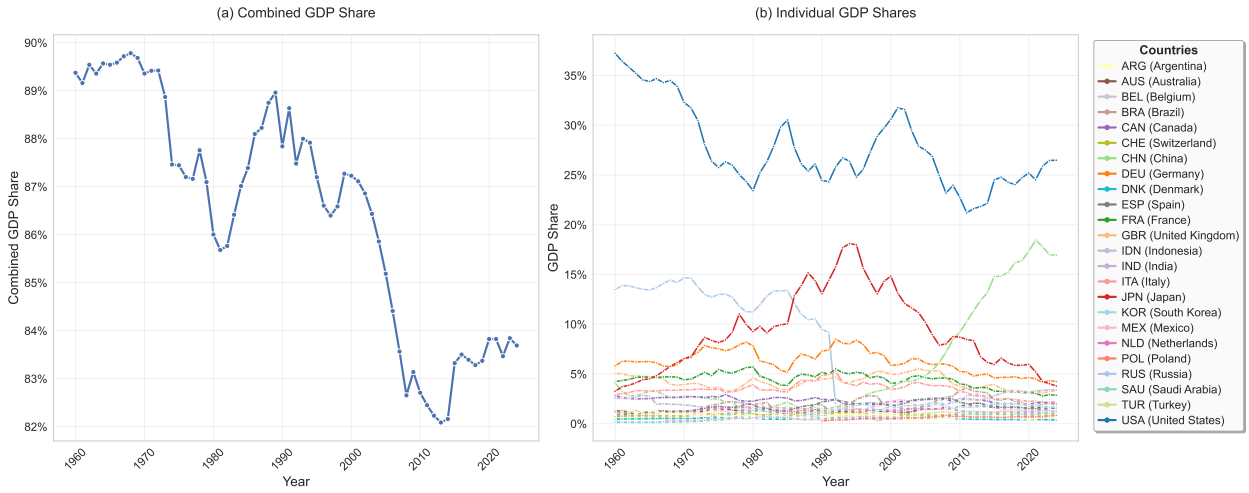


FIGURE A.1. GDP Shares of Major Nations

Notes: Panel (a) shows the combined GDP share of 24 major nations (Argentina, Australia, Belgium, Brazil, Canada, Switzerland, People's Republic of China, Germany, Denmark, Spain, France, United Kingdom, Indonesia, India, Italy, Japan, Republic of Korea, Mexico, Netherlands, Poland, Russian Federation, Saudi Arabia, Turkey, and United States) relative to global GDP over time. Panel (b) shows the individual GDP share of each nation. Data are sourced from the World Development Indicators, with Soviet Union GDP before 1992 equal to US nominal GDP times the Soviet–US real GDP ratio at purchasing power parity from the Maddison Project (Bolt and Van Zanden 2025).

A.2. Geopolitical Event Data: Methodology and Descriptive Statistics

The all-pairs event compilation contains 833,485 events with complete CAMEO and Goldstein coding over 1950–2024. Unless otherwise noted, the descriptive statistics below concern the 373,020 major-involving events used in this paper over 1960–2024. Observations from the 1950s initialize

[34] GDP data are nominal GDP in current USD and sourced from the World Development Indicators. For the Soviet Union before 1992, GDP is US nominal GDP multiplied by the ratio of Soviet to US real GDP at purchasing power parity from the Maddison Project, so the Soviet weight is not valued at market exchange rates.

the dynamic bilateral scores. Appendix C summarizes the prompt, and the full production prompt is released with the replication package.

A.2.1. Event Compilation and Analysis

This subsection provides additional detail on the event compilation and analysis procedure described in Section 2.1.

Entity Verification and Historical Accuracy. Our framework begins by verifying the political entities corresponding to each country pair for the target year. This step handles state succession, such as the dissolution of the Soviet Union and emergence of the Russian Federation on December 26, 1991. The prompt directs the model to official, international-organization, news-archive, and academic sources. For years when specified entities did not exist, the framework identifies the primary political entities controlling the relevant territories or populations and applies them consistently throughout the analysis. For example, the Baltic states' relations before 1991 are recorded through the Soviet Union, and Germany before 1990 combines the two German states.

Event Identification. Table A.1 lists the six dimensions and 25 subcategories, fixed in the prompt, that define which geopolitical events the LLM records, with representative examples for each. Within these subcategories, the prompt asks for events that affect or indicate the state of the bilateral relationship and defines in advance what clears this threshold, including regular high-level meetings, formal agreements, trade actions, and diplomatic communications. The prompt instructs the model to check events against reliable sources and to exclude unverified ones. Appendix C.2 details the criteria.

CAMEO Classification and Goldstein Scoring. Each identified event is classified using the Conflict and Mediation Event Observations (CAMEO) framework, which categorizes international political actions along two dimensions: cooperation versus conflict and verbal versus material. The two dimensions define four quadrant classes. Within them, two-digit root codes identify general categories and three-digit event codes identify specific actions. The prompt supplies definitions, coding conventions, and anchored examples for this classification. After selecting a CAMEO code, the LLM assigns a Goldstein scale score ranging from −10.0 (maximum conflict) to +10.0 (maximum cooperation), using intensity bands and action-based anchors while allowing the event's specific bilateral context and severity to inform the final value.

Economic events are those in subcategories A1–A6 of Table A.1. Appendix C.3 summarizes the CAMEO classification and Goldstein scoring rules.

Tables 1 and A.2 illustrate the range of scores: 2022 US–Russia events are dominated by conflict scores (−4.5 to −8.0), while the 1972 détente period features exclusively cooperative events (+5.0 to +9.0).

TABLE A.1. Categories of Bilateral Geopolitical Events

Subcategory	Representative Events	Subcategory	Representative Events
A. Economic Relations		*D. Legal, Territorial & Movement*	
A1. Trade Policy	Tariff imposition/removal; non-tariff barriers; FTA negotiations, signing, ratification, withdrawal	D1. Legal Actions	ICJ/WTO disputes; international arbitration; extradition; law enforcement cooperation
A2. Financial Relations	Asset freezes; SWIFT exclusions; capital market denial; currency swaps; FDI restrictions; investment treaties	D2. Territorial & Maritime	EEZ/continental shelf disputes; freedom of navigation operations; border demarcation; sovereignty claims
A3. Economic Coercion	Trade embargoes; sectoral sanctions; Entity List designations; export controls; foreign aid; debt relief	D3. Movement of People	Visa regime changes; travel bans; guest worker programs; refugee/asylum policies
A4. Strategic Sectors	Energy supply agreements; pipeline projects; nuclear cooperation; 5G bans; rare earth controls		
A5. Integration	BRI/B3W infrastructure; port concessions; friend-shoring; supply chain arrangements; customs unions	*E. Multilateral & Global Governance*	
A6. Other Economic	Regulatory harassment; boycott campaigns; government procurement restrictions; gray zone pressures	E1. Int'l Organizations	UNSC confrontations; GA coalition building; regional bloc membership changes
		E2. Global Issues	Climate cooperation; pandemic coordination; vaccine diplomacy; human rights
B. Diplomatic & Political Relations		*F. Other Significant Events*	
B1. Formal Diplomacy	Embassy/consulate openings and closures; ambassador recalls; staff expulsions; démarches	F1. Historical & Symbolic	Apologies for historical wrongs; memorial visits; monument disputes
B2. High-Level Interactions	Presidential visits; bilateral summits; summit boycotts; ministerial meetings; strategic dialogues	F2. Humanitarian	Disaster aid offers/rejections; joint rescue operations; evacuation cooperation
B3. Public Diplomacy	Policy speeches; parliamentary resolutions; propaganda campaigns	F3. Sports & Events	Olympic boycotts; joint hosting; World Expo participation
B4. Cultural & Educational	Cultural agreements/boycotts; scholarship programs; university partnerships; student visa policies	F4. Technology & Space	Joint space missions; tech theft accusations; research terminations
		F5. Environment	Transboundary river/pollution disputes; conservation conflicts
C. Security & Defense		F6. Communications	Journalist expulsions; broadcasting restrictions; information warfare
C1. Military Cooperation	Alliances; defense pacts; base arrangements; arms sales/embargoes; joint exercises; military aid	F7. Other	Emerging forms of bilateral interaction
C2. Security Incidents	Border skirmishes; airspace/maritime violations; naval encounters; hybrid warfare		
C3. Intelligence & Cyber	Espionage revelations; intel officer expulsions; intel-sharing pacts; state-sponsored cyber attacks		

Notes: The production prompt assigns each event one primary subcategory (`event_category`). The table lists all 25 subcategories with representative events. Appendix C.2 describes the main ones.

TABLE A.2. US–Soviet Union Bilateral Geopolitical Events in 1972: LLM Analysis Results

Event Name	Event Description	CAMEO Class.	Category	Goldstein Score
Declaration of Basic Principles	Joint declaration of twelve principles guiding bilateral relations, including peaceful coexistence and renouncing unilateral advantage	Verbal Coop. (01-019)	Diplomatic & Political	+5.0
US-Soviet Grain Deal	Soviet purchase of 19 million metric tons of American grain, including nearly a quarter of the US wheat harvest	Material Coop. (06-061)	Economic	+6.0
Moscow Summit	President Nixon's historic visit to Moscow, the first official US presidential visit to the USSR	Verbal Coop. (04-042)	Diplomatic & Political	+6.0
Environmental Protection Agreement	Bilateral agreement establishing cooperation on 11 areas including air and water pollution and nature preservation	Verbal Coop. (05-057)	Other Significant Events	+6.5
Biological Weapons Convention	US and USSR signed the BWC prohibiting development and stockpiling of biological weapons	Verbal Coop. (05-057)	Multilateral	+7.0
US-Soviet Trade Agreement	Comprehensive agreement providing reciprocal MFN tariff treatment, official trade offices, and government credits	Verbal Coop. (05-057)	Economic	+7.0
Apollo-Soyuz Agreement	Agreement for joint US-Soviet space mission, leading to the 1975 orbital rendezvous of American astronauts and Soviet cosmonauts	Verbal Coop. (05-057)	Other Significant Events	+7.5
Incidents at Sea Agreement	Protocols to prevent accidents between US and Soviet navies, including rules against collisions and simulated attacks	Verbal Coop. (05-057)	Security & Defense	+8.0
SALT I Accords	ABM Treaty limiting defensive systems and Interim Agreement freezing ICBM/SLBM launchers	Verbal Coop. (05-057)	Security & Defense	+9.0

Notes: The table lists all nine events the model records for the US–Soviet Union pair in 1972, ordered by Goldstein score. For years before 1992 the model analyzes the Soviet Union in place of Russia (Appendix C.1). Event names and descriptions are abridged from the model output. Columns are defined as in Table 1.

A.2.2. Statistics of Geopolitical Events

Table A.3 and Figure A.2 show three patterns. First, recorded events are mostly cooperative: 83 percent fall in the verbal or material cooperation classes. Second, total event volume more than tripled from 29,289 in the 1960s to 96,175 in the 2010s, driven primarily by expanding cooperative interactions. Third, mean Goldstein scores rose from Cold War levels (2.86–3.28) to a post-Cold War peak (3.98–4.09) before declining modestly from 3.88 in the 2010s to 3.73 in the 2020s, consistent with the shifts in the distribution of alignment in Section 2.4. Across the six event categories, diplomatic and political interactions (category B) account for 39 percent of all events, followed by economic relations (A, 27 percent) and multilateral governance (E, 15 percent). Security and defense events

TABLE A.3. Summary Statistics of Geopolitical Events by Decade, 1960–2024

	1960s	1970s	1980s	1990s	2000s	2010s	2020s	Total
CAMEO Event Classification								
Verbal Cooperation	14,092	17,523	18,393	24,878	36,418	53,698	31,555	196,557
Material Cooperation	7,919	9,349	11,264	15,659	22,812	29,180	16,990	113,173
Verbal Conflict	4,729	4,643	6,012	4,816	6,480	8,587	5,235	40,502
Material Conflict	2,549	2,710	3,378	3,173	3,169	4,710	3,099	22,788
Goldstein Scale Statistics								
Mean	2.86	3.28	2.92	3.98	4.09	3.88	3.73	3.67
Std. Dev.	4.87	4.62	4.74	4.27	3.87	3.78	3.88	4.18
Median	5.00	5.00	4.50	6.00	5.00	4.50	4.50	5.00
Event Category Classification								
Economic Relations (A)	6,296	9,045	10,288	14,067	20,364	26,366	14,847	101,273
Diplomatic & Political (B)	11,559	12,515	14,082	16,058	25,591	41,086	23,529	144,420
Security & Defense (C)	3,635	3,205	3,879	4,361	5,507	8,343	5,175	34,105
Legal & Territorial (D)	1,588	1,889	1,614	2,226	3,565	4,736	2,443	18,061
Multilateral & Governance (E)	5,239	6,212	6,624	9,765	10,780	11,268	7,459	57,347
Other Significant Events (F)	972	1,359	2,560	2,049	3,072	4,376	3,426	17,814
Summary								
Total Events	29,289	34,225	39,047	48,526	68,879	96,175	56,879	373,020

Notes: CAMEO classifications follow the Conflict and Mediation Event Observations framework. The Goldstein scale ranges from −10 (most conflictual) to +10 (most cooperative). Event categories (A)–(F) and their subcategories are those of the LLM prompt (Table A.1). All entries represent event counts except Goldstein scale statistics. Each event is assigned to exactly one category, so category totals sum to the overall total. The table covers the 373,020 events of 1960–2024 in country pairs involving at least one of the 24 major nations; the 2020s column covers 2020–2024.

(C) constitute 9 percent, while legal/territorial (D) and other events (F) account for 5 percent each.

Coverage. The event record becomes denser over time and is uneven across dyads. The share of major-involving dyad-years with at least one recorded event rises from 32 percent in the 1950s to 80 percent in the 2020s, reflecting both the intensification of bilateral interactions and the greater availability of digitized source material for recent decades in LLM training corpora. Overall, 46 percent of major-involving dyad-years contain no recorded event, density is lowest for the poorest partners, and the most active fifth of dyads accounts for 42 percent of all events. Because the annual event scores average event-level Goldstein scores within each dyad-year, they capture the average intensity of recorded events rather than raw event volume, although the dynamic score moves toward neutral in years without recorded events. Appendix A.6 benchmarks the event record and intensity against externally documented events by decade and by partner income.

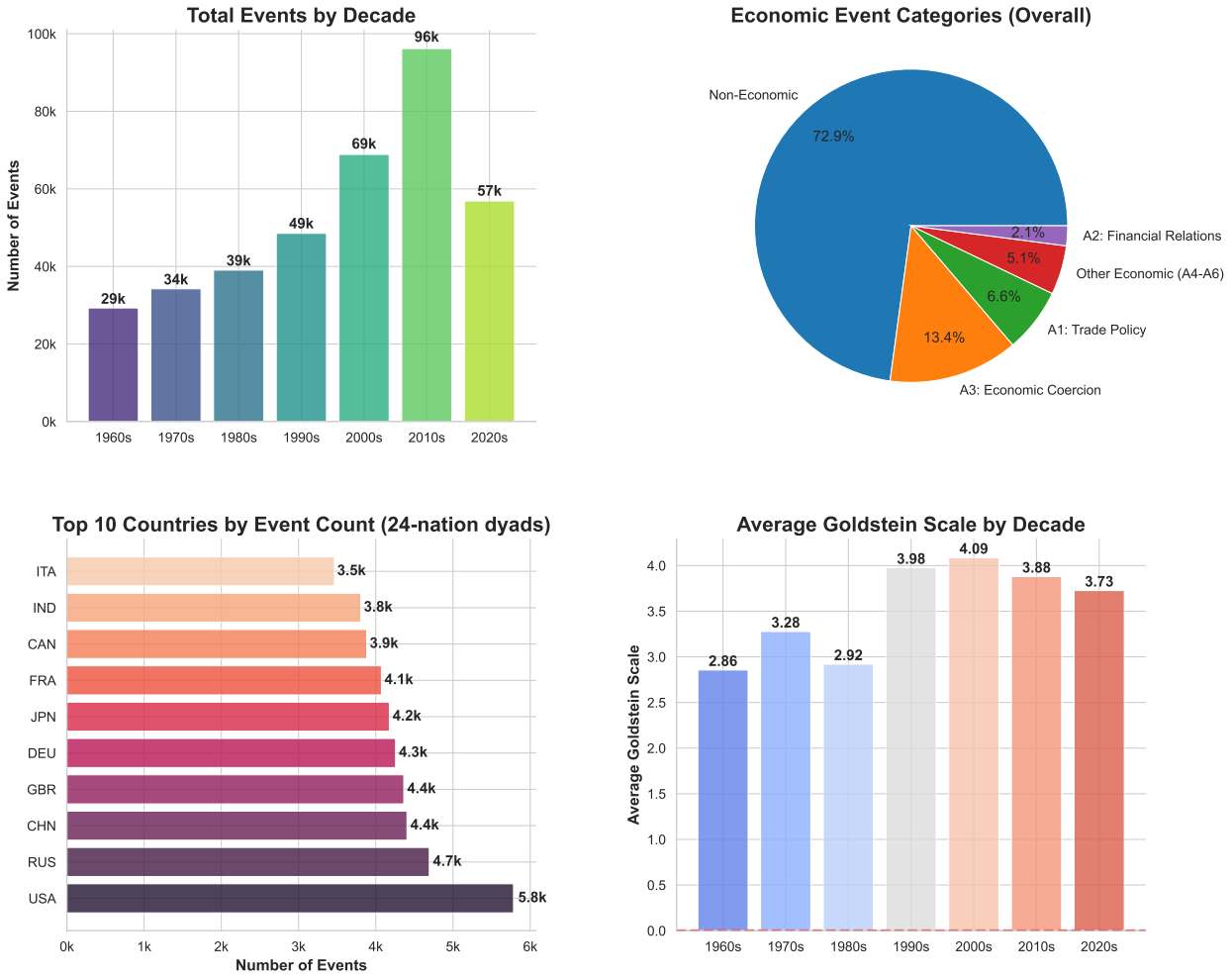


FIGURE A.2. Geopolitical Events Summary (1960–2024)

Notes: The upper panels show total events by decade (the 2020s cover 2020–2024) and the shares of events by economic category, where categories A1–A6 of Table A.1 are economic and all other events non-economic. The lower-left panel counts each country's events in dyads between two of the 24 major nations. The lower-right panel shows the average Goldstein score by decade.

A.3. Dynamics of Geopolitical Alignment

This section estimates the persistence of the geopolitical alignment index and of the bilateral geopolitical alignment scores, comparing each with its unsmoothed counterpart. Persistence matters because the GDP coefficients in Section 3.2 reflect both the initial change in alignment and the index's subsequent dynamics.

Methodology. For the geopolitical alignment index, we estimate:

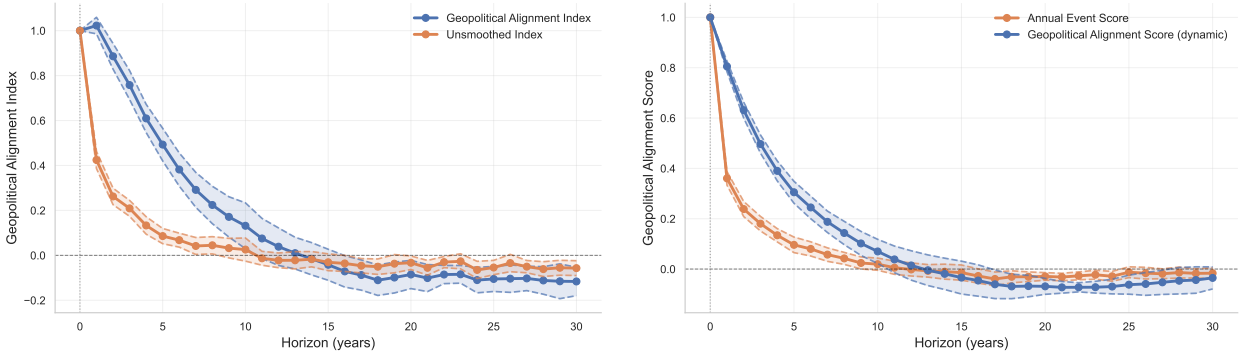
$$p_{c,t+h} = \alpha_h^{\text{Geo. Align.}} p_{ct} + \boldsymbol{\gamma}_h' \mathbf{x}_{ct} + \varepsilon_{c,t+h}, \quad h = 0, 1, \ldots, 30$$

where p_{ct} denotes either our dynamic geopolitical alignment index or the unsmoothed index, $\mathbf{x}_{ct}$ includes four lags of GDP and the respective geopolitical alignment measure, and we control for country and region-year fixed effects.

For bilateral geopolitical alignment scores, we estimate:

$$S_{ij,t+h} = \alpha_h^{\text{Score}} S_{ij,t} + \boldsymbol{\gamma}_h' \mathbf{x}_{ij,t} + \varepsilon_{ij,t+h}, \quad h = 0, 1, \ldots, 30$$

where $S_{ij,t}$ stands for either the bilateral geopolitical alignment score or the annual event score $\tilde{S}_{ij,t}$, $\mathbf{x}_{ij,t}$ contains four lags of the score variable, and we include country-pair and year fixed effects with Driscoll–Kraay standard errors.



(a) Country-Level Geopolitical Alignment Index (b) Bilateral Geopolitical Alignment Scores

FIGURE A.3. Persistence of Geopolitical Alignment Measures

Notes: Coefficients from projecting each measure at $t + h$ on its current value. Panel (a): Geopolitical alignment index with dynamic measure (blue) and unsmoothed index (orange). Panel (b): Bilateral geopolitical alignment scores with dynamic score (blue) and annual event score (orange). Shaded areas are 95% confidence intervals based on Driscoll–Kraay standard errors. Panel (a) controls for four lags of GDP and of the respective measure, country fixed effects, and region-year fixed effects. Panel (b) controls for four lags of the respective score, country-pair fixed effects, and year fixed effects. Panel (a) uses the full unbalanced country panel. Panel (b) uses all country pairs involving at least one of the 24 major nations, 1960–2024.

Results. Panel (a) of Figure A.3 shows that the dynamic geopolitical alignment index is persistent, with a half-life of about five years, while the unsmoothed index mean-reverts much faster (see also Figure B.12). The dynamic measure turns modestly negative after about year 15, as in panel (a) of Figure 6.

Panel (b) shows similar patterns at the bilateral level. The annual event score mean-reverts quickly, with a half-life under one year. In contrast, the dynamic score is more persistent, with coefficients remaining significant through year 10. This persistence comes from the decay parameter $\delta = 0.3$, which gives the weight on past events a half-life of about two years, consistent with a typical four-year electoral cycle (Section 2.2).

A.4. Additional Case Validation for Geopolitical Alignment Scores

This section provides additional validation of our bilateral geopolitical alignment scores beyond the cases presented in the main text (Figures 2 and 3). We examine eight bilateral relationships spanning different regions and relationship types: great-power rivalries, ideological revolutions, war-to-normalization transitions, sectarian conflicts, and diplomatic reversals.

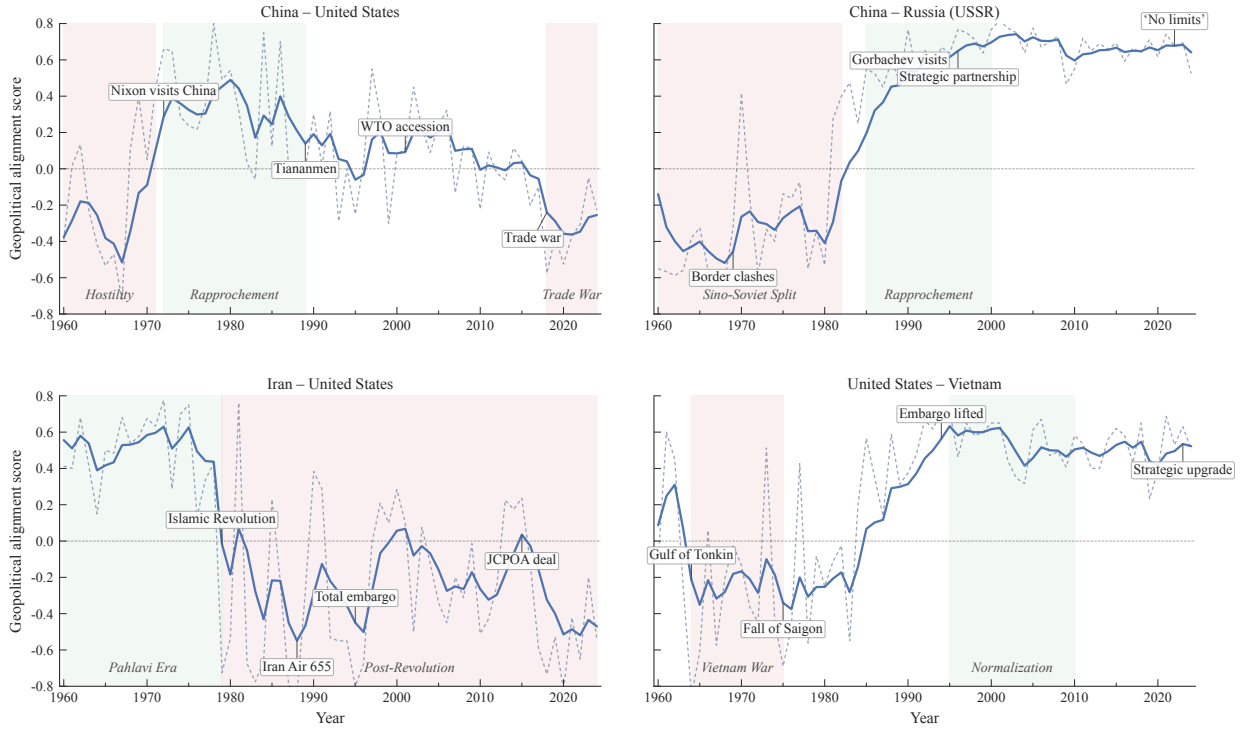


FIGURE A.4. Geopolitical Alignment Scores: Relationships Involving Major Nations

Notes: Dynamic geopolitical alignment scores for four bilateral relationships involving major nations, 1960–2024. In each panel, the blue solid line shows the dynamic alignment score and the gray dashed line the annual (unsmoothed) event score. Shaded regions mark key historical periods, and boxed labels mark major bilateral events. Panels (a)–(d) run left to right and top to bottom: China–United States, China–Russia, Iran–United States, and United States–Vietnam.

Figure A.4 presents four relationships involving major nations. The China–US trajectory (panel (a)) traces five distinct phases: Cold War-era hostility in the 1960s with scores near −0.4, Nixon's 1972 opening (scores rising to about +0.4), a post-Tiananmen deterioration around 1989, a period of WTO-era cooperation in the early 2000s, and the sharp trade-war decline after 2018 that returns scores to levels comparable to the 1960s. The China–Russia case (panel (b)) moves from alliance through the Sino-Soviet split and 1969 border clashes (scores near −0.5) to Gorbachev's 1989 visit, the strategic partnership of the late 1990s, and the contemporary "no limits" partnership with scores near +0.7. Iran–US relations (panel (c)) break sharply at the 1979 Islamic Revolution, with scores collapsing from approximately +0.5 during the Pahlavi era to sustained hostility. Scores fall below −0.4 in several periods, punctuated by the Iran Air 655 incident (1988), the total embargo (1995), and the brief JCPOA-related recovery around 2015. The US–Vietnam case (panel (d))

demonstrates a complete war-to-normalization transition: scores fall from moderate levels in the early 1960s through the Gulf of Tonkin escalation and the Fall of Saigon to deep hostility, before recovering steadily from the mid-1980s, reaching cooperative levels around the 1994 embargo removal and remaining there through the 2023 strategic upgrade. Before 1976, the database records US relations with South Vietnam through 1963 and mainly with North Vietnam from 1964, so the early decline also reflects this change in counterpart.

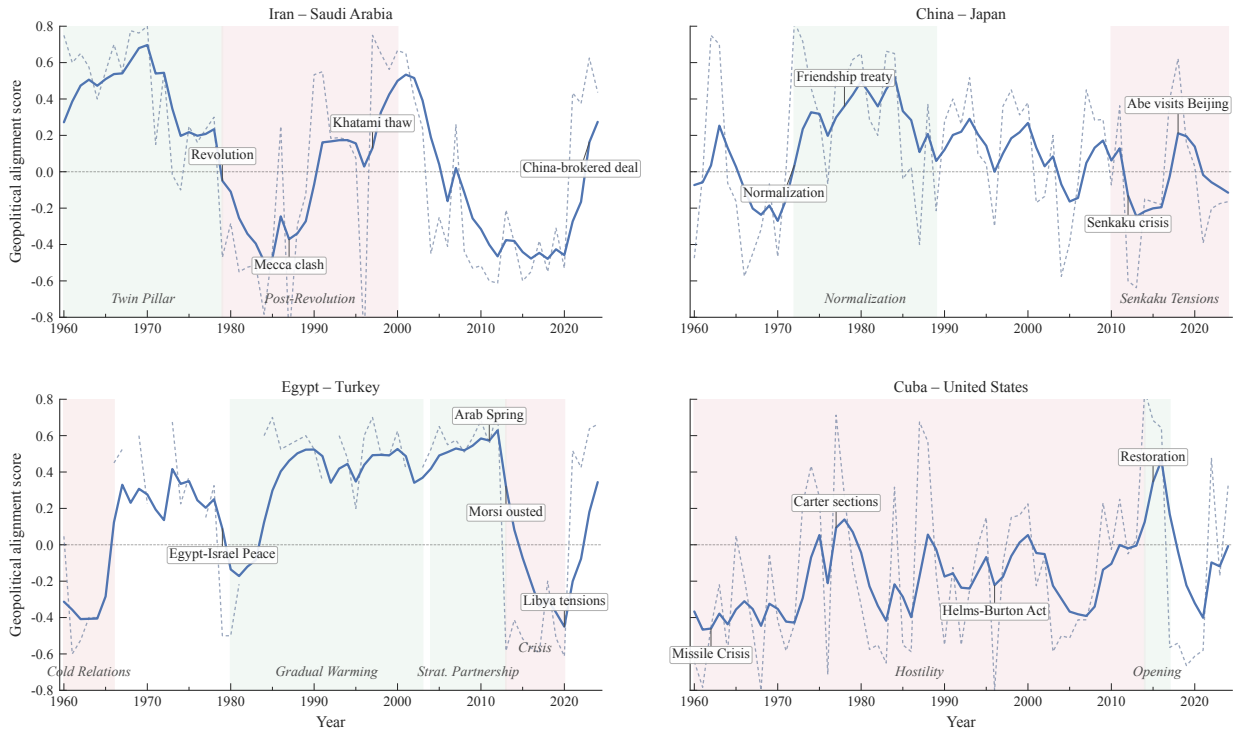


FIGURE A.5. Geopolitical Alignment Scores: Regional Rivalries and Bilateral Dynamics

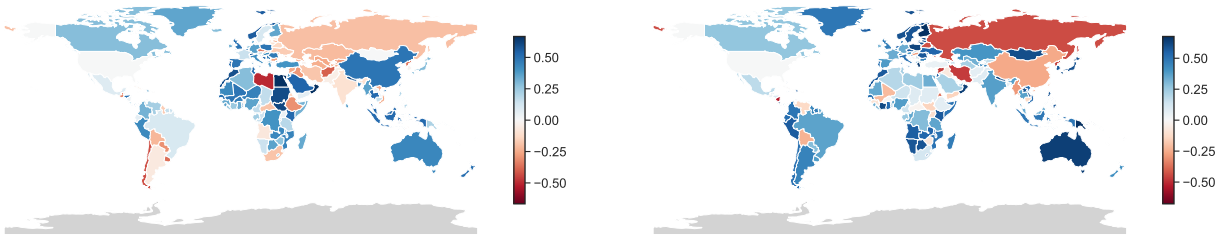
Notes: Dynamic geopolitical alignment scores for four regional bilateral relationships, 1960–2024. In each panel, the blue solid line shows the dynamic alignment score and the gray dashed line the annual (unsmoothed) event score. Shaded regions mark key historical periods, and boxed labels mark major bilateral events. Panels (a)–(d) run left to right and top to bottom: Iran–Saudi Arabia, China–Japan, Egypt–Turkey, and Cuba–United States.

Figure A.5 presents four regional relationships. The Iran–Saudi Arabia case (panel (a)) tracks the Twin Pillar era of cooperation under the Shah (scores peaking near +0.7 in 1970), the sharp post-revolutionary deterioration including the 1987 Mecca clash (scores dropping to approximately −0.4), the Khatami-era thaw that temporarily restored scores above +0.5, and the subsequent deterioration followed by the 2023 China-brokered rapprochement. China–Japan relations (panel (b)) show the post-war normalization around 1972 and the friendship treaty period of the late 1970s (scores reaching about +0.5 by the early 1980s), followed by gradual erosion and the sharp deterioration after the 2012 Senkaku/Diaoyu crisis, with a modest recovery around Abe's 2018 visit to Beijing. The Egypt–Turkey case (panel (c)) shows cold relations in the early 1960s (scores near −0.4), mostly positive scores from 1966 apart from a dip in the early 1980s, a strategic partnership peak during

the Arab Spring and Morsi presidency in 2011–2012 (scores reaching approximately +0.6), and the collapse after Morsi's ouster in 2013 with continuing tensions over Libya. Finally, Cuba–US relations (panel (d)) show persistent hostility following the Castro revolution and the 1962 Missile Crisis, interrupted briefly by the Carter-era interest sections in the late 1970s, tightened further by the 1996 Helms-Burton Act, and punctuated by the sharp Obama-era restoration around 2015 before subsequent reversal.

A.5. Maps of Bilateral Geopolitical Alignment with the US and Its Rivals

This section provides geographic visualization of geopolitical alignment, comparing Cold War patterns (1980) with the contemporary pattern (2024) for both the United States and its principal rivals.

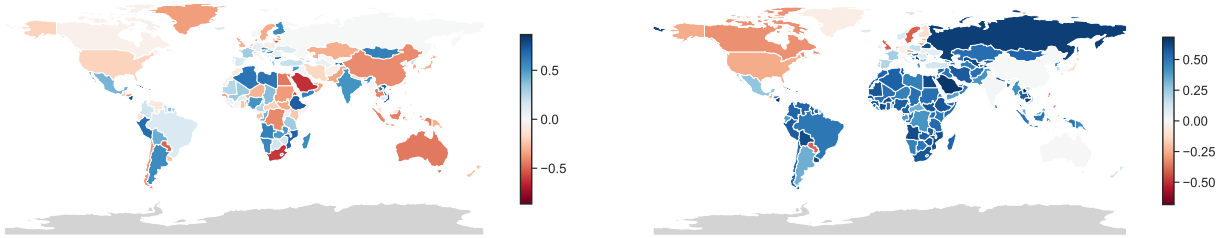


(a) United States, 1980 (b) United States, 2024

FIGURE A.6. Map of Geopolitical Alignment with the United States

Notes: World maps showing bilateral geopolitical alignment scores with the United States in 1980 and 2024. Blue indicates positive and red negative alignment.

Figure A.6 shows the shift from the Cold War pattern of 1980, with strong relations across NATO and Pacific allies and negative relations with the Soviet Union and parts of Eastern Europe, to a 2024 pattern in which core Western alliances remain intact, Russia appears in red and China in lighter red, and much of Latin America, Africa, and Southeast Asia has moderate or neutral relations.



(a) Soviet Union, 1980 (b) China, 2024

FIGURE A.7. Map of Geopolitical Alignment: Soviet Union (1980) vs. China (2024)

Notes: World maps showing bilateral geopolitical alignment scores with the principal US rival in 1980 (Soviet Union) and 2024 (China). Blue indicates positive and red negative alignment, as in Figure A.6.

Figure A.7 contrasts the geographic reach of America's principal rivals. The Soviet Union's

1980 positive relations were concentrated in Eastern Europe, Cuba, and select African and Middle Eastern allies, with China hostile (reflecting the Sino-Soviet split). China's 2024 map shows a different pattern: positive relations extend broadly across Sub-Saharan Africa, Central Asia, and Latin America, though China's relations with the United States, Canada, the Philippines, and several European countries are negative. The contrast between the opposing core blocs of the 1980 US and Soviet maps and the overlapping US and China maps of 2024, where most countries maintain positive relations with both, is consistent with the shift from Cold War bipolarity to the more complex distributional patterns documented in Figure 5.

A.6. Validation against Externally Documented Events

This section evaluates the event database against 4,670 bilateral developments independently dated by established external sources. The benchmark covers eight mutually exclusive conflict and cooperative event classes involving at least one of the 24 major nations between 1950 and 2023. Table A.4 summarizes the event definitions, sources, matching windows, and class sizes.

TABLE A.4. Construction of the Externally Documented Benchmark

Documented event class	Definition and date	External source	Matching window	Documented events
Conflict events				
War onset	First year of an interstate war	COW Dyadic MID 4.03	[−1, +1]	51
Use-of-force MID, excluding wars	First year of a nonwar militarized dispute involving force	COW Dyadic MID 4.03	[−1, +1]	619
Sanction imposition	First year of a unilateral sanction spell	Global Sanctions Database v4	[−1, +1]	1,326
Diplomatic severance	First census wave with no exchange following mutual resident representation	COW Diplomatic Exchange 2006.1	[−4, +1]	28
Cooperative events				
Sanction removal	First year after a unilateral sanction spell ends, *end* + 1	Global Sanctions Database v4	[−1, +1]	705
Diplomatic establishment	First census wave with mutual resident representation following no exchange	COW Diplomatic Exchange 2006.1	[−4, +1]	505
Trade-agreement entry	Year that a regional trade agreement enters into force	CEPII Gravity V202211	[−1, +1]	1,199
Alliance formation	Year that a bilateral alliance enters into force	ATOP 5.1	[−1, +1]	237
All classes				4,670

Notes: The sample contains mutually exclusive bilateral developments involving at least one of the 24 major nations between 1950 and 2023. COW = Correlates of War; MID = Militarized Interstate Disputes. Sources: Maoz et al. (2019), Felbermayr et al. (2020), Yalcin et al. (2025), Bayer (2006), Conte, Cotterlaz, and Mayer (2022), and Leeds et al. (2002). Zero in the matching-window column denotes the documented year. When a use-of-force dispute overlaps a war in the same dyad-year, only the war onset is retained: of 46 such disputes, 45 are represented by war onsets, and one, KOR–VNM in 1965, is excluded because common country codes cannot distinguish North from South Vietnam.

Detection and Annual-Score Differences. For each documented event, *any event recorded* indicates that the database contains an event for the dyad within the matching window. *Event in the expected direction* additionally requires conflict for the first four classes or cooperation for the final four. A *broad event-family match* further requires either the expected substantive category or the expected

TABLE A.5. Detection and Conditional Annual-Score Differences

	Detection rate			Conditional annual-score difference (SE)		
Documented event class	Any event recorded	Expected direction	Broad event-family match	Full sample	Before 1990	1990 onward
Conflict events						
War onset	1.00	1.00	1.00	−5.02 (0.62)	−5.75 (0.67)	−1.77 (0.97)
Use-of-force MID, excluding wars	0.99	0.92	0.78	−2.03 (0.22)	−2.21 (0.31)	−0.85 (0.27)
Sanction imposition	1.00	0.89	0.73	−2.15 (0.13)	−2.53 (0.33)	−1.76 (0.13)
Diplomatic severance	0.96	0.86	0.79	−1.15 (1.08)	−1.39 (1.77)	−1.15 (1.25)
Cooperative events						
Sanction removal	1.00	0.94	0.80	+0.39 (0.17)	+0.01 (0.43)	+0.66 (0.17)
Diplomatic establishment	0.93	0.92	0.88	+0.59 (0.18)	+0.68 (0.23)	+0.46 (0.26)
Trade-agreement entry	0.95	0.93	0.91	+0.58 (0.09)	+1.52 (0.30)	+0.36 (0.08)
Alliance formation	1.00	1.00	0.99	+1.82 (0.19)	+2.77 (0.33)	+0.85 (0.21)
All classes	0.98	0.92	0.83			

Notes: The detection columns use the matching windows in Table A.4. "Expected direction" requires conflict for the first four classes and cooperation for the final four. A broad event-family match further requires either the expected substantive category or the expected CAMEO root. The final three columns report coefficients from equation (A.1), estimated over the full panel and separately before 1990 and from 1990 onward. All eight documented-event indicators enter jointly in every regression. The three estimation samples contain 174,104, 70,158, and 103,946 dyad-years, respectively. Standard errors in parentheses are clustered by dyad.

CAMEO root. This criterion permits related actions within the same bilateral development. It does not require the recorded text to name the specific documented episode.

To examine the annual score, let $g_{ij,t}$ denote the mean Goldstein score across all events recorded for dyad ij in year t, before smoothing and expressed in raw Goldstein points. Equivalently, $g_{ij,t} = 10\tilde{S}_{ij,t}$, where $\tilde{S}_{ij,t}$ is the annual event score defined in Section 2.2. Let $B^k_{ij,t}$ equal one when a documented event of class k occurs in that dyad-year. We estimate

$$g_{ij,t} = \sum_{k=1}^{8} \beta_k B^k_{ij,t} + \mu_{ij} + \lambda_{i,t} + \lambda_{j,t} + \varepsilon_{ij,t}, \tag{A.1}$$

where all eight documented-event indicators enter jointly, μ_{ij} are dyad fixed effects, and $\lambda_{i,t}$ and $\lambda_{j,t}$ are country-by-year fixed effects for the two endpoints. Standard errors are clustered by dyad. Because $g_{ij,t}$ is defined only when the dyad records at least one event, the estimation sample consists of dyad-years with recorded events. The coefficient β_k is the descriptive conditional difference in the annual mean Goldstein score associated with a documented event of class k, not a causal effect.

Table A.5 reports the detection rates and annual-score differences for the full sample, before 1990, and from 1990 onward. Across the 4,670 documented events, the database records some event for 97.6 percent, an event in the expected direction for 92.1 percent, and a broad event-family match for 82.6 percent. All eight full-sample annual-score differences have the expected direction, and seven are statistically distinguishable from zero at the 5 percent level. The expected directions are

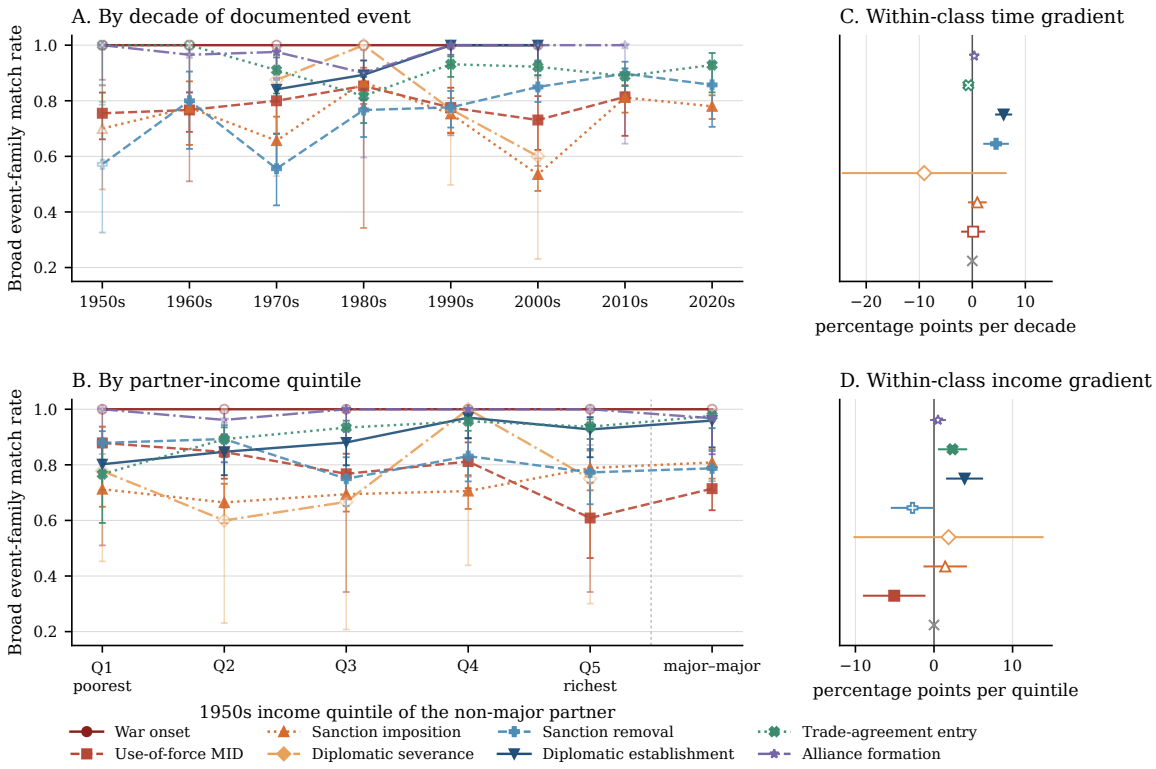


FIGURE A.8. Broad Event-Family Match Rates by Time and Partner Income

Notes: Panels A and B report class-specific broad event-family match rates with pointwise 95% Wilson confidence intervals. Panels C and D report the corresponding within-class slopes with 95% confidence intervals based on standard errors clustered by dyad. Cells based on fewer than 30 documented events are shown with open or faded markers. A gray cross denotes a class with a constant outcome, for which the slope is not estimable. Panels B and D exclude events without a mapped 1950s income quintile. Panel B shows major–major events separately.

preserved before and after 1990, although the absolute differences are smaller after 1990 in seven of the eight classes.

Variation in Matching over Time and Partner Income. We examine whether broad event-family matching varies systematically with calendar time or partner income. For documented event a, we estimate separate linear probability models,

$$D_a^F = \alpha_{k(a)} + \theta Z_a + u_a, \tag{A.2}$$

where D_a^F indicates a broad event-family match, $\alpha_{k(a)}$ are documented-event-class fixed effects, and Z_a is either decades since 1950 or the 1950s income quintile of the non-major partner. Standard errors are clustered by dyad. The income specification excludes major–major events and events without a mapped income quintile.

Figure A.8 reports the class-specific match rates and corresponding within-class gradients. The pooled income gradient is +0.39 percentage points per quintile, with a 95 percent confidence

interval of [−0.80, +1.58]. The estimate does not indicate systematically lower matching for poorer partners. The pooled match rate rises by 1.16 percentage points per decade, with a 95 percent confidence interval of [+0.36, +1.96]. This increase is concentrated in sanction removals and diplomatic establishments: excluding these two classes reduces the estimate to +0.24 percentage points per decade, with an interval of [−0.66, +1.14]. War and diplomatic severance are too sparse to establish class-specific stability across decades.

Variation in Matching over Subsequent Growth. We also test whether the database records documented developments selectively by the non-major country's subsequent growth. For documented event a involving non-major country c in year t, we estimate, separately for conflict and cooperative events,

$$D_a = \alpha_{k(a)} + \delta_{d(t)} + \theta\, G_{c,t} + \gamma X_{c,t} + \mu_c + u_a, \tag{A.3}$$

where D_a indicates an expected-direction or broad event-family match, $G_{c,t}$ is the growth of log GDP per capita over t to $t + 10$ relative to the region-year mean, $X_{c,t}$ is log GDP per capita in year t, $\delta_{d(t)}$ are decade fixed effects, and μ_c are country fixed effects. Table A.6 shows that θ is small and statistically indistinguishable from zero for both matching measures and both event types.

TABLE A.6. Documented-Event Matching and Subsequent Growth

	Conflict events	Cooperative events
Expected-direction match	−0.5 (1.5)	−0.4 (1.0)
Broad event-family match	−0.5 (2.4)	+1.8 (1.7)
Documented events	844	1,678
Countries	135	151

Notes: Entries are estimates of θ in equation (A.3), in percentage points per one standard deviation of the non-major country's growth in log GDP per capita over the following ten years relative to its region-year mean, with standard errors clustered by dyad in parentheses. Each regression includes documented-event-class, decade, and country fixed effects and log GDP per capita in the documented year. The sample contains documented events with exactly one major nation and observed growth over the following decade (events through 2014).

A.7. Comparison with Machine-Coded Event Databases

We next compare our annual event score with the corresponding scores from the Integrated Crisis Early Warning System (ICEWS; Boschee et al. 2015), the Global Database of Events, Language, and Tone (GDELT; Leetaru and Schrodt 2013), and the Cline Center Historical Phoenix database (Althaus et al. 2020). We harmonize actors to common country codes and construct undirected dyad-year measures under common aggregation rules. For each database, the annual measure is the mean Goldstein score in raw Goldstein points. None of the measures in this section is dynamically smoothed.

Event Volume and Detection. Table A.7 compares event volume and the two detection outcomes. GDELT records a median of 288.5 events in a documented-event dyad-year, compared with three in our database, yet the two databases record some event within the matching window at nearly identical rates, 98.0 and 98.2 percent respectively. GDELT records an event in the expected direction more often, 97.2 versus 91.7 percent. ICEWS and Phoenix also record substantially more events than our database but have lower pooled rates on both outcomes. Event volume therefore does not by itself determine whether a database records the benchmark dyad or its expected conflict or cooperation direction.

TABLE A.7. Event Volume and Detection across Event Databases, 1995–2018

	Documented events	Our database	ICEWS	GDELT	Phoenix
Panel A. Pooled volume and detection					
Median events in documented-event year		3	34	288.5	6
Mean events in documented-event year		3.0	371.7	4,083.0	61.3
Any event recorded within matching window	2,038	0.982	0.907	0.980	0.844
Event in expected direction within matching window	2,038	0.917	0.869	0.972	0.794
Panel B. Event in expected direction, by documented-event class					
War onset	7	1.000	1.000	1.000	1.000
Use-of-force MID, excluding wars	167	0.922	1.000	1.000	0.982
Sanction imposition	561	0.836	0.900	0.984	0.816
Diplomatic severance	13	0.769	0.615	0.769	0.385
Sanction removal	399	0.962	0.952	0.995	0.905
Diplomatic establishment	79	1.000	0.962	1.000	0.937
Trade-agreement entry	772	0.940	0.764	0.944	0.666
Alliance formation	40	1.000	0.975	1.000	0.875
All classes	2,038	0.917	0.869	0.972	0.794

Notes: The sample contains the 2,038 documented events over 1995–2018 that all four databases can support after actor harmonization. Volume is measured in the documented-event year; detection uses the class-specific matching window.

We estimate the class-adjusted linear probability model in equation (A.2) separately for each event database and detection outcome. Per decade, any-event detection changes by −0.19 [−0.96, +0.58] percentage points for our database, −4.45 [−6.59, −2.32] for ICEWS, +0.10 [−0.96, +1.17] for GDELT, and −9.06 [−11.83, −6.30] for Phoenix, where brackets report 95 percent confidence intervals. Expected-direction detection changes by +1.95 [+0.24, +3.66], −4.48 [−6.93, −2.03], −0.72 [−2.04, +0.60], and −10.23 [−13.36, −7.09], respectively. Across partner income, Figure A.9 shows that our database and GDELT have no statistically distinguishable gradients, whereas ICEWS and Phoenix detect richer-country dyads more often.

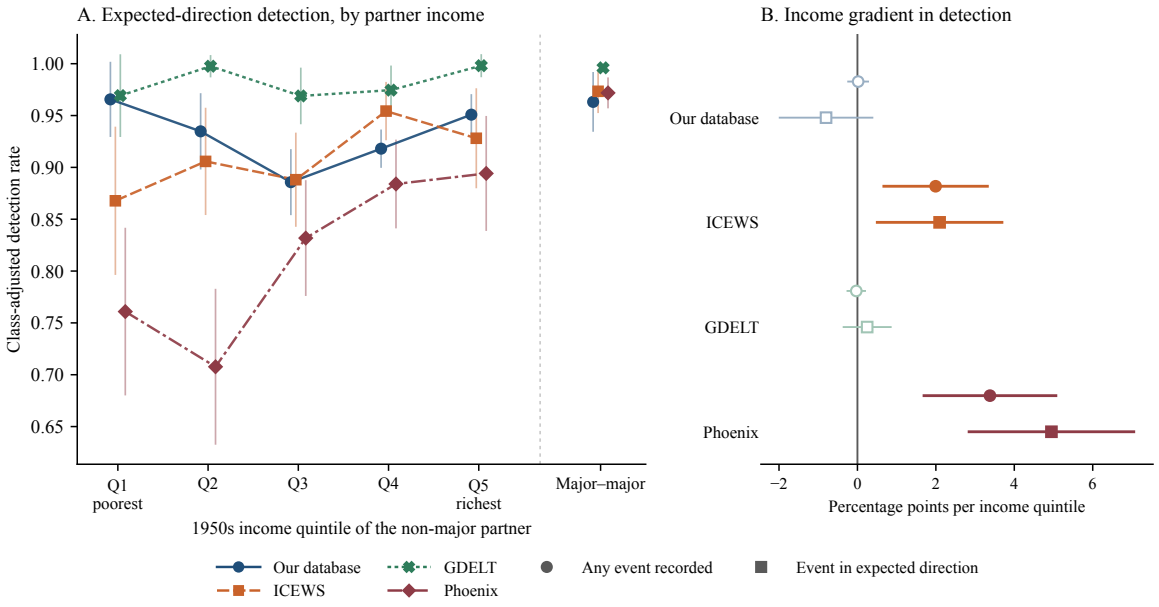


FIGURE A.9. Event Detection by Partner Income across Databases

Notes: Panel A reports expected-direction detection rates, adjusted for documented-event class, by the 1950s income quintile of the non-major partner. Major–major events are shown separately. Panel B reports percentage-point changes per income quintile from class-adjusted linear probability models for any-event and expected-direction detection. Filled markers indicate intervals that exclude zero. Bars in both panels are 95% confidence intervals based on standard errors clustered by dyad. Of the 2,038 documented events in the common sample, 1,405 enter the quintile sample, 238 are major–major events shown separately, and the remaining 395 lack a mapped partner-income quintile and are excluded.

Conditional Associations on Common Dyad-Year Samples. To compare the annual measures on a common specification, let $C_{ij,t}$ indicate that at least one of the four documented conflict classes occurs in dyad ij in year t, and let $K_{ij,t}$ indicate that at least one of the four documented cooperative classes occurs. For each annual measure m, we estimate

$$z^m_{ij,t} = \beta^m_C C_{ij,t} + \beta^m_K K_{ij,t} + \mu^m_{ij} + \lambda^m_{i,t} + \lambda^m_{j,t} + \varepsilon^m_{ij,t}, \tag{A.4}$$

where $z^m_{ij,t}$ is measure m for dyad ij in year t and both documented-event indicators enter jointly. The specification absorbs dyad fixed effects and country-by-year fixed effects for both endpoints, and standard errors are clustered by dyad. The coefficients report conditional differences in measure m during documented conflict and cooperation years. They are contemporaneous descriptive associations, not impulse responses or causal effects.

Panel A of Table A.8 uses the 34,412 dyad-years observed by all four databases that remain after fixed-effect singletons are dropped. Documented conflict is associated with annual differences of −1.467 Goldstein points in our database, −0.475 in ICEWS, −0.120 in GDELT, and −0.295 in Phoenix. Documented cooperation is associated with differences of +0.475, +0.194, +0.033, and −0.133, respectively. The cooperation coefficients for our database and ICEWS are statistically distinguishable from zero. Scaled by each database's standard deviation of annual scores in this

TABLE A.8. Conditional Associations of Annual Event Scores with Documented Conflict and Cooperation

	Our database	ICEWS	GDELT	Phoenix
Panel A. Common observed dyad-years				
Documented conflict	−1.467 (0.179)	−0.475 (0.078)	−0.120 (0.054)	−0.295 (0.084)
Documented cooperation	+0.475 (0.112)	+0.194 (0.060)	+0.033 (0.051)	−0.133 (0.096)
Dyad-years	34,412	34,412	34,412	34,412
Documented conflict events	627	627	627	627
Documented cooperative events	789	789	789	789
Panel B. Each database's observed dyad-years				
Documented conflict	−1.388 (0.155)	−0.561 (0.085)	−0.144 (0.060)	−0.260 (0.090)
Documented cooperation	+0.403 (0.085)	+0.116 (0.061)	+0.064 (0.057)	−0.086 (0.096)
Dyad-years	71,538	59,036	81,141	43,009
Documented conflict events	726	691	722	643
Documented cooperative events	1,181	1,019	1,204	848

Notes: Each column reports one regression of the annual event score in levels on jointly entered indicators for any documented conflict event and any documented cooperative event, as specified in equation (A.4). All regressions absorb dyad fixed effects and country-by-year fixed effects for both endpoints; standard errors in parentheses are clustered by dyad. Panel A uses identical observations for all four databases. Panel B uses each database's complete observed panel over 1995–2018. All scores are in raw Goldstein points on the $[-10, 10]$ scale.

sample, the conflict differences are −0.45, −0.26, −0.08, and −0.12 and the cooperation differences are 0.15, 0.11, 0.02, and −0.05, so the ordering does not reflect differences in score dispersion alone. Panel B uses each database's complete observed panel and preserves the relative magnitudes, although the ICEWS cooperation coefficient is not statistically distinguishable from zero at the 5 percent level.

The annual-score regressions do not establish that event selection causes the differences across databases. Appendix B.14 compares the smoothed indices built from each database and their relationships with economic growth.

A.8. Model Robustness and Replicability

LLM-based measures may depend on the specific model or vary across independent runs. We evaluate both dimensions using four models from three families: Gemini, GPT, and Claude. Cross-model comparisons use Gemini 2.5 Pro as the production baseline and GPT 5.4, Claude Opus 4.6, and Claude Sonnet 4.6 as alternatives. Repeated-run tests use all four models.

Figure A.10 reports three complementary exercises. Panel (a) compares scores from alternative LLMs with the Gemini 2.5 Pro baseline for 300 randomly selected dyad-years. Scores cluster along the 45-degree line, with pairwise correlations exceeding 0.82 (pooled $\rho = 0.88$). Panel (b) tests within-model replicability. The pooled correlation is 0.93. Model-specific correlations are 0.96 for Opus 4.6, 0.94 for Sonnet 4.6, 0.93 for GPT 5.4, and 0.90 for Gemini 2.5 Pro. Because scores vary across runs and served models can change, we treat the released event database, rather than a model

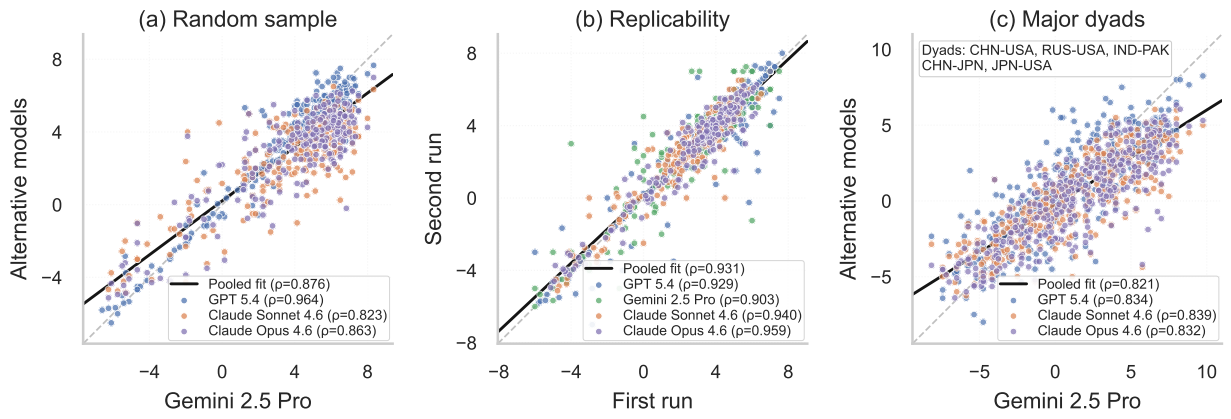


FIGURE A.10. Model Robustness and Replicability of Annual Event Scores

Notes: Each point represents the mean Goldstein score for a dyad-year observation. Panel (a) compares alternative-model scores with the Gemini 2.5 Pro production baseline for 300 randomly selected dyad-years. Panel (b) compares two runs on the same inputs within each model. The Gemini series uses Gemini 2.5 Pro. Panel (c) compares alternative models with the Gemini 2.5 Pro baseline for five high-salience bilateral relationships (CHN–USA, RUS–USA, IND–PAK, CHN–JPN, JPN–USA) across all years. The solid black line is the pooled OLS fit and the dashed gray line the 45-degree line. ρ is Pearson correlation. Exact model identifiers and provider-specific run settings are documented in the replication package.

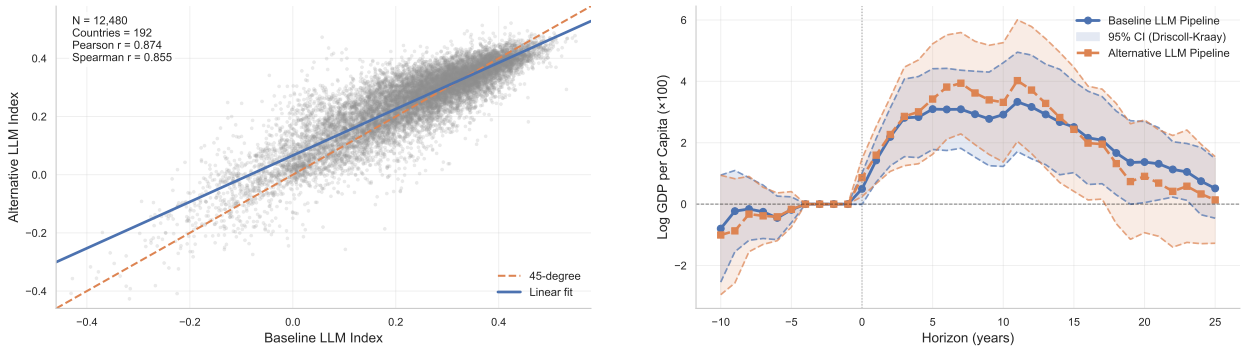
endpoint, as the reproducible artifact. Panel (c) compares scores for five high-salience bilateral relationships across all available years, a more demanding test because their dense and shifting event records require greater interpretive judgment. The pooled correlation is 0.82. Together, the exercises show that the common signal is large relative to model- and run-specific variation.

Full-Pipeline Robustness. The preceding exercises hold the prompt structure fixed and vary only the model on subsamples. We next vary both the LLM and the prompt, comparing the baseline index with an alternative LLM pipeline constructed primarily using Gemini 2.5 Flash (preview) and an earlier prompt.[35] To isolate measurement from aggregation differences, we reconstruct the alternative LLM country-year index using the same 24 major nations and time-varying GDP-share weights as the baseline.

Panel (a) of Figure A.11 shows that the two country-year indices are strongly correlated (Pearson $\rho = 0.874$, Spearman $\rho = 0.855$), with a fitted slope below one. In panel (b), the two pipelines produce similar coefficient paths: a hump-shaped response with its largest coefficient in year 11, followed by gradual decline. The alternative LLM pipeline's point estimates fall within the baseline 95 percent confidence interval at all 26 horizons from impact through year 25, and the correlation of point estimates across these horizons is 0.97.

Measurement Validity. The exercises above establish measurement *reliability*, not accuracy. Frontier LLMs draw on overlapping sources and may therefore share errors. The documented-event

[35]The alternative prompt pursues the same objective of identifying and scoring bilateral geopolitical events using the CAMEO/Goldstein framework but differs in scope, organization, and event taxonomy. Failed calls could fall back to Gemini 2.5 Pro (preview). The fallback share is not recoverable from the stored outputs.



(a) Country-Year Index Correlation

(b) GDP Coefficient Paths

FIGURE A.11. Full-Pipeline Robustness: Alternative LLM and Prompt Structure

Notes: Panel (a) plots the country-year geopolitical alignment index from the alternative LLM pipeline, constructed primarily with Gemini 2.5 Flash (preview) and an earlier prompt, against the production Gemini 2.5 Pro baseline for all 12,480 overlapping country-year observations across 192 countries. The blue solid line shows the linear fit. The orange dashed line shows the 45-degree line. Both indices use the same 24 major nations and time-varying GDP-share weights. Panel (b) overlays the coefficient paths of log GDP per capita (×100), per 0.141 index units (one standard deviation of the baseline index), from the baseline pipeline (blue) and the alternative pipeline (orange), both estimated on the 148-country long-coverage sample. Shaded areas are 95% confidence intervals based on Driscoll–Kraay standard errors.

benchmarks in Appendix A.6 and the staged human audit in Appendix A.9 address accuracy more directly by comparing the model record with independently compiled events, blind human reconstruction, and independent recoding.

A.9. Human Audit of the Event Data

The human audit covers 200 eventful dyad-years over 1960–2024 involving at least one of the 24 major nations. The stratified sample covers every decade, with 23–34 cases per decade. Based on the sign of the model's annual score, 44.5 percent of cases are conflictual and 55.5 percent cooperative. Six trained research assistants provide two retained assessments per case, giving 400 reads and 1,712 coder-event assessments of 856 model-generated events.

The audit follows a three-stage protocol. Before the audit, coders complete 15 training cases excluded from the analysis sample. In Stage 1, coders see only the country pair, year, and the codebook used by the model. They independently identify, classify, and score bilateral events. After this record is locked, Stage 2 reveals the model's event descriptions and dates while withholding its classifications and scores. Coders then assess validity and independently recode events. In Stage 3, the model's classifications and scores are revealed for diagnostic comparisons. Earlier assessments cannot be revised. Coders are prohibited from using generative AI and must document sources for each event. The research and coding session for each case is limited to 75 minutes.

Independent human reconstructions yield annual Goldstein scores that correlate at 0.80 with the model's scores (Figure A.12, panel (a)), compared with 0.76 between human coders. Human reconstructions contain 2.95 events per dyad-year on average, 31.1 percent fewer than the 4.28 events in the model record.

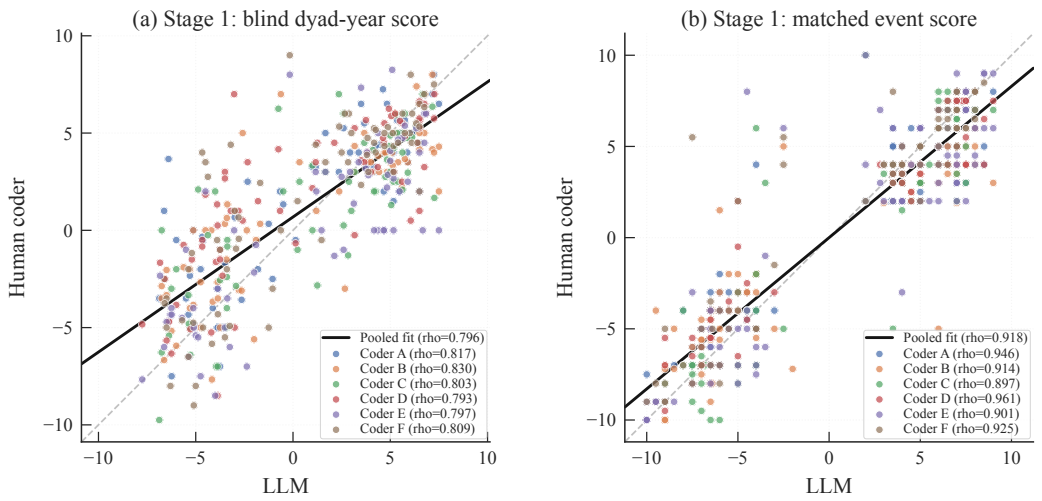


FIGURE A.12. Independent Human Reconstruction of Bilateral Events

Notes: Panel (a) compares blind human and model annual scores (n = 400 reads). Each is the unweighted mean of its event-level Goldstein scores. For 28 reconstructions with no events, the human score is the coder's holistic rating of the dyad-year. Panel (b) compares events with exact or near matches (n = 491 links). Colors denote coders. Dashed and solid lines show equality and the pooled OLS fit. Both panels use 200 dyad-years and uncorrected coder scores. All scores are in Goldstein units on $[-10, 10]$. ρ is Pearson correlation.

For independently reconstructed events with exact or near matches in the model record, Goldstein scores correlate at 0.92 (panel (b)). Coders flag these correspondences in Stage 2, but the human scores are locked in Stage 1. Strong agreement on matched events suggests that the set of recorded events is an important source of differences between the annual measures.

After event review, human and model scores correlate at 0.92 at the event level and 0.89 at the annual level (Figure A.13, panels (a) and (b)). The audit identifies three fabricated records among the 856 model-generated events, less than 0.5 percent of the sample.[36]

In Stage 2, coders also assign each event to one of the six broad categories of Table A.1 before the model's classification is revealed. Human and model categories agree in 79.0 percent of cases, compared with 75.6 percent between two coders who assess the same event. Most differences arise when coders classify as diplomatic and political relations an event that the model places in security and defense or in multilateral and global governance. After the reveal in Stage 3, coders who disagreed with the model prefer the model's category, prefer their own, or regard both as defensible in roughly equal shares.

To gauge what error of this size does to the growth estimates, we treat the model's annual score and the two blind human scores of each dyad-year as three measures of the same quantity with independent errors. Their covariances imply a standard deviation of 1.85 Goldstein points for the error in the model's annual score. Because coders reconstruct their own event sets, this error

[36]These records were classified as fabricated during adjudication: a Norwegian royal state visit to the Netherlands in 2017, a visit by US Secretary of State Condoleezza Rice to Angola in 2006, and a British condemnation attributing the 1994 AMIA bombing to Iran.

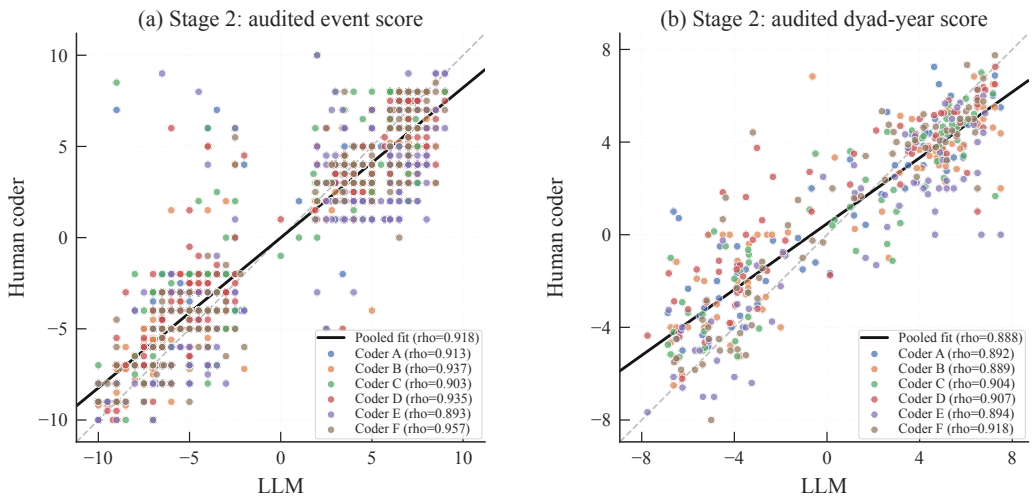


FIGURE A.13. Human Review of Model-Generated Events

Notes: Panel (a) uses all submitted human event recodes (n = 1,357), including those for flagged records. Unscored assessments are excluded. Panel (b) compares annual scores (n = 400 reads). The human score is the unweighted mean over the final event set after exclusions, corrections, merges, splits, and inclusion of unmatched Stage 1 events, with a holistic fallback for an empty set. The model score averages its original events. Both panels use 200 dyad-years and uncorrected coder scores. Colors denote coders. Dashed and solid lines show equality and the pooled OLS fit. All scores are in Goldstein units on $[-10, 10]$. ρ is Pearson correlation.

reflects differences in which events are recorded as well as in how they are scored. We add normally distributed error of this size to the annual score of every eventful dyad-year involving a major nation, independently across dyad-years and with event counts and empty years unchanged. We then rebuild the index and re-estimate the baseline local projection and the sustained path. Across 500 simulations, the year-11 coefficient falls by 14.3 percent on average (Monte Carlo standard error 0.2) and the 25-year sustained coefficient by 5.8 percent (0.3).

The audit thus shows agreement in annual scores and event coding within this stratified sample of eventful dyad-years. Because the sample excludes dyad-years without events and human searches are time-limited, the audit says less about the accuracy of the full event database.

A.10. Bilateral Outcomes

Table A.9 reports gravity regressions of four bilateral outcomes on the bilateral geopolitical alignment score $S_{ij,t}$. The sample comprises directed country pairs involving at least one of the 24 major nations over 1960–2024, the pairs that enter the alignment index. Columns (1)–(3) include distance controls and origin-year and destination-year fixed effects for each sub-period. Column (4) adds dyad fixed effects, so identification comes from changes within pairs over time. Bilateral alignment is strongly associated with trade, sanctions, migration, and foreign aid. With dyad fixed effects, improvements in alignment are associated with higher trade, a lower sanction probability, and larger aid flows, consistent with the score capturing economically relevant bilateral ties.

The second row of each panel removes outcome-linked events before rebuilding the score: all

TABLE A.9. Geopolitical Alignment and Bilateral Outcomes: Gravity Regressions

	Cold War 1963–1990 (1)	Globalization 1990–2010 (2)	Fragmentation 2010–2024 (3)	All Years (Dyad FE) (4)
Panel A. Dependent Variable: Log Bilateral Trade				
Baseline score	0.995***	0.800***	0.861***	0.292***
	(0.073)	(0.066)	(0.079)	(0.027)
Excluding Economic Relations events	0.741***	0.627***	0.718***	0.225***
	(0.077)	(0.069)	(0.080)	(0.026)
Observations	142,356	158,306	123,784	414,391
Panel B. Dependent Variable: Sanction Indicator (LPM)				
Baseline score	−0.028***	−0.052***	−0.146***	−0.044***
	(0.004)	(0.005)	(0.013)	(0.003)
Excluding sanction impositions and removals	−0.025***	−0.046***	−0.124***	−0.037***
	(0.005)	(0.005)	(0.013)	(0.003)
Sanction Rate	0.037	0.070	0.090	0.058
Observations	242,592	181,944	121,296	554,496
Panel C. Dependent Variable: Log Bilateral Migration				
Baseline score	0.447***	0.547***		
	(0.091)	(0.099)		
Excluding Movement of People events	0.416***	0.554***		
	(0.092)	(0.099)		
Observations	18,464	13,495		
Panel D. Dependent Variable: Log Bilateral ODA				
Baseline score	2.012***	1.784***	2.046***	0.902***
	(0.136)	(0.116)	(0.173)	(0.059)
Excluding Economic Relations events	1.421***	1.099***	1.442***	0.529***
	(0.140)	(0.111)	(0.166)	(0.056)
Observations	27,884	36,323	28,973	90,641
Distance Controls	Yes	Yes	Yes	—
Origin × Year, Dest. × Year FE	Yes	Yes	Yes	Yes
Dyad FE	—	—	—	Yes

Notes: Gravity regressions of bilateral outcomes on the geopolitical alignment score $S_{ij,t}$. The sample comprises directed country pairs involving at least one of the 24 major nations over 1960–2024. In each panel, the first row uses the baseline score, and the second row uses a score rebuilt after removing potentially outcome-linked events before dyad-year aggregation and the dynamic recursion, on an identical sample. Trade and ODA exclude all category A (Economic Relations) events, sanctions exclude sanction impositions and removals (CAMEO codes 081, 085, 163, and 172), and migration excludes category D3 (Movement of People) events. Columns (1)–(3): origin×year and destination×year fixed effects with distance controls. Column (4): dyad, origin×year, and destination×year fixed effects (distance controls absorbed by dyad FE). Period endpoints are inclusive, so 1990 enters columns (1) and (2), and 2010 enters columns (2) and (3). Panel A: log bilateral trade from COMTRADE (1962–2024). Panel B: linear probability model for bilateral sanction indicator from the Global Sanctions Database v4 (1960–2023), and Sanction Rate reports the unconditional probability. Panel C: log bilateral migration stocks from the World Bank in five decennial waves from 1960 through 2000; the reported period columns begin in 1963, so the 1960 wave is not used, and the later-period and sparse all-years dyad fixed-effects cells are omitted. Panel D: log gross ODA disbursements from the OECD DAC (1960–2024, constant prices); the sample comprises 47 donor and 156 recipient countries. Reported observation counts are before singleton fixed-effect removal. Distance controls include normalized geodesic distance, contiguity, and tree-weighted linguistic distance. Standard errors are clustered by dyad. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Economic Relations events for trade and official development assistance (ODA), sanction impositions and removals for sanctions, and Movement of People events for migration. Signs and significance are unchanged. With dyad fixed effects, the coefficients are 0.225 for trade, −0.037 for sanctions, and 0.529 for ODA, versus 0.292, −0.044, and 0.902 in the baseline, and the period-specific migration estimates barely change. Because the exclusions alter the score's dispersion and composition, the coefficient differences are not a decomposition. The second-row estimates show that the bilateral patterns are not generated solely by coding the outcome event itself.

A.11. Statistics of the Geopolitical Alignment Index

Table A.10 presents summary statistics for the geopolitical alignment index across seven decades. The index covers all 193 countries in every year: before independence, a country's relations are recorded through the state that controlled its territory, such as the Soviet Union for the Baltic states before 1991.

TABLE A.10. Summary Statistics of the Geopolitical Alignment Index by Decade, 1960–2024

	1960s	1970s	1980s	1990s	2000s	2010s	2020s	Overall
Summary Statistics								
Mean	0.173	0.192	0.182	0.260	0.287	0.296	0.281	0.235
Median	0.189	0.204	0.209	0.284	0.315	0.319	0.302	0.261
Std. Dev.	0.141	0.120	0.138	0.152	0.132	0.108	0.114	0.141
Min	−0.323	−0.315	−0.275	−0.430	−0.256	−0.335	−0.160	−0.430
Max	0.499	0.531	0.466	0.550	0.548	0.479	0.482	0.550
5th Pct.	−0.054	−0.013	−0.103	−0.027	0.023	0.082	0.059	−0.029
25th Pct.	0.060	0.108	0.108	0.188	0.226	0.251	0.223	0.154
75th Pct.	0.283	0.281	0.284	0.371	0.378	0.368	0.367	0.340
95th Pct.	0.378	0.368	0.359	0.451	0.446	0.425	0.420	0.422
N Country-Years	1,930	1,930	1,930	1,930	1,930	1,930	965	12,545
Countries with Lowest Geopolitical Alignment Index (Bottom 5)								
1.	S. Africa	S. Africa	Afghanistan	Libya	Myanmar	N. Korea	N. Korea	–
2.	China	Albania	Libya	Iraq	Belarus	Eritrea	Nicaragua	–
3.	Zimbabwe	Zimbabwe	S. Africa	Myanmar	Zimbabwe	Syria	China	–
4.	Albania	Timor-L.	Latvia	Serbia	Iraq	Venezuela	Russia	–
5.	Timor-L.	Chile	Lithuania	Montenegro	Gambia	Iran	Syria	–
Countries with Highest Geopolitical Alignment Index (Top 5)								
1.	Ethiopia	Romania	Mozambique	Hungary	Mozambique	Peru	Mozambique	–
2.	Italy	Finland	Egypt	Poland	Romania	Singapore	Qatar	–
3.	Nepal	Bangladesh	Bangladesh	Mozambique	Ghana	Sierra Leone	Singapore	–
4.	Afghanistan	Belgium	Senegal	Kuwait	Vietnam	Liberia	Senegal	–
5.	Colombia	Colombia	Mali	Mongolia	Albania	Senegal	Laos	–

Notes: The geopolitical alignment index is the GDP-weighted aggregate of each country's bilateral geopolitical alignment scores with the 24 major nations. Statistics are calculated using all available country-year observations within each decade. Countries are ranked by their decade-average geopolitical alignment index values.

The summary statistics show three patterns. First, the mean geopolitical alignment index improved from 0.173 in the 1960s to 0.296 in the 2010s—an increase of approximately 0.87 standard deviations—before declining to 0.281 in the 2020s. This trajectory matches the post-Cold War convergence and recent fragmentation documented in Section 2.4. Second, the standard deviation declined from 0.141 in the 1960s to 0.108 in the 2010s, indicating reduced heterogeneity as countries

converged toward more cooperative average relations, though it has since risen slightly to 0.114 in the 2020s. Third, the 5th percentile rose from −0.054 in the 1960s to 0.082 in the 2010s, so the lower tail of the distribution also improved, before falling to 0.059 in the 2020s.

The countries at the extremes provide a face-validity check on the measure. During the Cold War decades, apartheid-era South Africa, China, and Albania ranked among the most geopolitically isolated states, reflecting ideological divisions and international ostracism. The contemporary period's most isolated countries—North Korea, Syria, and Iran in the 2010s, joined by China, Russia, and Nicaragua in the 2020s—are all subjects of extensive international sanctions or geopolitical confrontation with Western powers. Conversely, the highest-scoring countries are often small states that maintain positive relations across geopolitical divides (Singapore, Senegal, Qatar) or beneficiaries of particular historical moments (Poland and Hungary during post-Cold War democratization, Vietnam and Albania following market reforms, African states like Mozambique and Ghana engaging multiple development partners).

A.12. Comparison with the Geopolitical Risk Index

The Geopolitical Risk Index (GPR) of Caldara and Iacoviello (2022) infers geopolitical risk from newspaper article counts discussing wars, terrorism, and military tensions—measuring media *attention* rather than the events themselves. Figure A.14 compares our GDP×GDP–weighted conflict intensity against the historical GPR for the global aggregate and three major nations. The two series co-move during traditional military confrontations (Vietnam War, Russia–Ukraine war). The correlations of the annual series are $\rho = 0.28$ (aggregate), 0.31 (United States), 0.50 (China), and 0.36 (Russia).

Two divergences are informative. During 2001–2004, the Iraq War produces the largest post-Cold War GPR spike, yet our measure barely responds, since Iraq's GDP was about 0.1 percent of world output. During 2017–2021 (orange shading), our measure rises with the US–China trade war and technology restrictions while the GPR stays flat, as its keywords target military threats rather than geoeconomic competition. Excluding 2017–2021 raises the aggregate correlation to $\rho = 0.41$ and the US correlation to 0.44. The GPR is also limited to countries with extensive English-language media coverage, whereas our event-based approach covers all 193 countries.

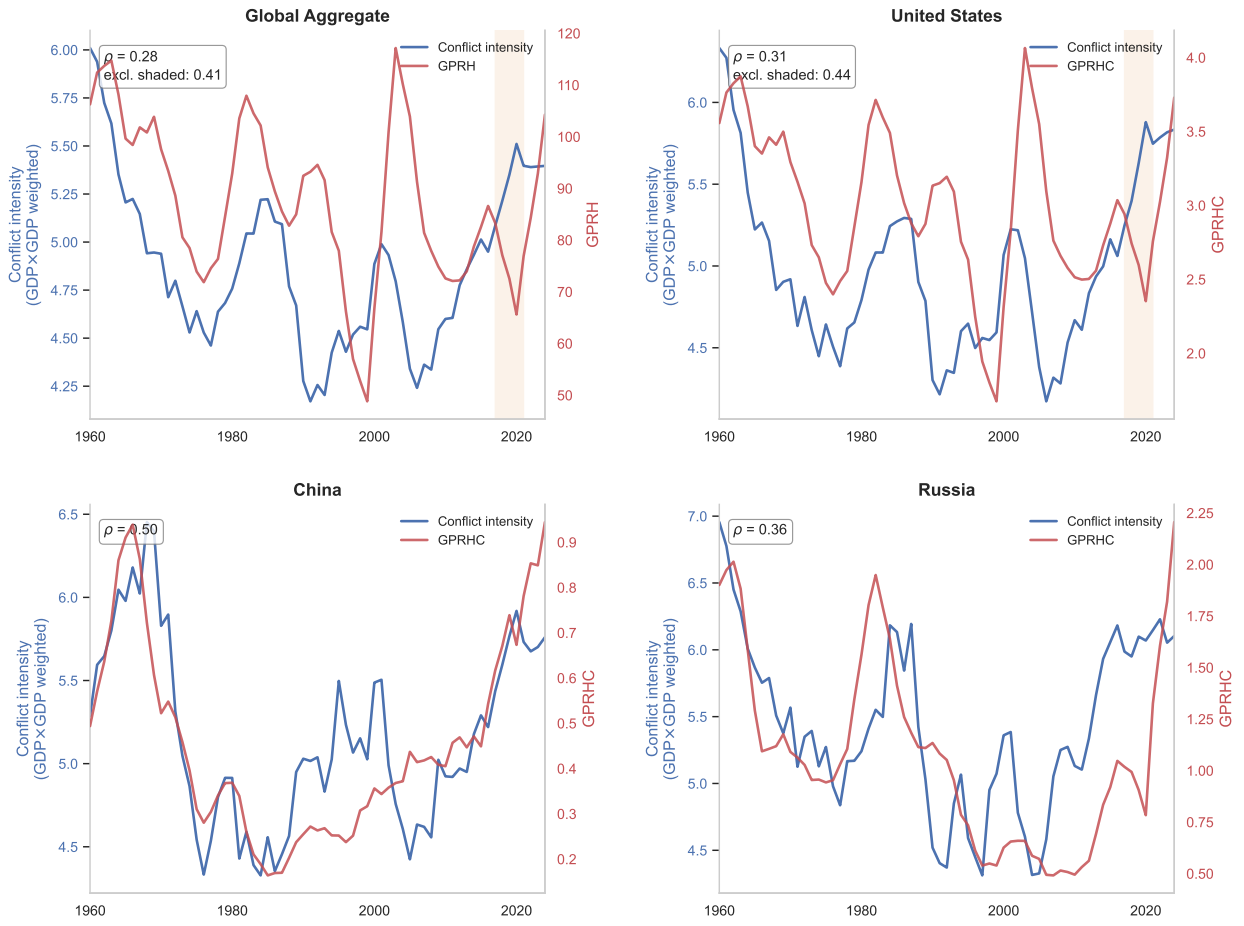


FIGURE A.14. GDP-Weighted Conflict Intensity versus Geopolitical Risk Index

Notes: Each panel plots our GDP×GDP–weighted conflict intensity (blue, left axis) against the historical Geopolitical Risk Index (GPRH/GPRHC, red, right axis) from Caldara and Iacoviello (2022). Conflict intensity is the weighted mean severity of recorded conflict events, $-\sum_n w_n g_n / \sum_n w_n$, where the sums run over all conflict events in the year, $g_n < 0$ is the Goldstein score of event n, and $w_n = \text{GDP}_i \times \text{GDP}_j$ for its country pair. Higher values indicate more severe conflict among economically significant country pairs. Both series are smoothed with a four-year centered moving average. Orange shading highlights the 2017–2021 geoeconomic divergence period. Boxed values are Pearson correlations of the unsmoothed annual series. For the aggregate and the United States, the second value excludes the shaded 2017–2021 period.

Online Appendix B. Additional Empirical Results

Appendix B follows Sections 3–5: data (B.1–B.2), the baseline estimates and their robustness (B.3–B.9), channels and democracy (B.10–B.11), and additional robustness and alternative measures (B.12–B.17). As in the main text, coefficients on the alignment index are per one standard deviation of the index (0.141).

B.1. Economic Data

Our dataset extends Acemoglu et al. (2019) to cover 193 countries over 1960–2024, incorporating updated democracy measures from Acemoglu et al. (2025) and economic indicators from the Penn World Table (Feenstra, Inklaar, and Timmer 2015).

Table B.1 organizes variables into five categories following the growth literature: enhanced Solow fundamentals (GDP, investment, capital, demographics), political institutions and governance (democracy, unrest, regime indicators), market institutions and reforms (liberalization indices, fiscal measures), human capital and labor markets (education, employment), and productivity and returns (TFP, internal rate of return, consumption, absorption). Coverage varies from 117 countries (TFP) to 192 countries (demographic variables). All monetary variables are in constant prices.

Domestic Unrest. Our domestic unrest indicator is a binary variable equal to one if any of eight domestic conflict components—assassinations, general strikes, guerrilla warfare, government crises, purges, riots, revolutions, and anti-government demonstrations—is strictly positive in a given country-year. We construct this variable from the 2024 edition of the Cross-National Time-Series Data Archive (CNTS), which extends coverage through 2023 (191 countries, 10,324 country-year observations). The original Acemoglu et al. (2019) unrest variable, derived from an earlier CNTS edition, covers 179 countries through 2010. Our updated indicator agrees with the original in 83 percent of country-years in the overlap period, with all discrepancies reflecting events added in the 2024 edition rather than removed.

B.2. Cross-Sectional Correlation with Economic Fundamentals

Table B.2 reports within-year cross-sectional correlations between economic fundamentals and (i) GDP per capita (Panel A) and (ii) the geopolitical alignment index (Panel B). GDP per capita correlates strongly with human capital (0.7–0.8), TFP (0.6), capital stock (0.5), and investment (0.4). Correlations with the geopolitical alignment index are generally modest and vary in sign across decades. Most are below 0.2 in absolute value, although capital stock reaches about 0.27 in the 1970s and 0.26 in the 1990s and human capital reaches 0.33 in the 1990s. The index is therefore distinct from, but not orthogonal to, standard measures of economic development.

TABLE B.1. Data Description and Coverage

Variable	Data Source	Country Coverage	Data Period
Enhanced Solow Fundamentals			
GDP per capita (Constant US Dollar)	WDI	184 countries	1960–2024
Real GDP per capita	PWT	172 countries	1960–2023
Trade as Share of GDP	WDI	176 countries	1960–2024
Investment as Share of GDP	PWT	172 countries	1960–2023
Capital Stock	PWT	169 countries	1960–2023
Population	PWT	172 countries	1960–2023
0–14 Population Share	WDI	192 countries	1960–2023
15–64 Population Share	WDI	192 countries	1960–2023
Political Institutions and Governance			
Democracy (Acemoglu et al. 2019)	Acemoglu et al. (2019)	183 countries	1960–2010
Democracy (Acemoglu et al. 2025)	Acemoglu et al. (2025)	183 countries	1960–2019
Unrest (ANRR)	Acemoglu et al. (2019)	179 countries	1960–2010
Unrest (CNTS 2024)	Cross-National Time-Series	191 countries	1960–2023
Soviet Union	Acemoglu et al. (2019)	183 countries	1960–2024
Region	Acemoglu et al. (2025)	192 countries	1960–2024
Market Institutions and Reforms			
Market Reform Index	Acemoglu et al. (2019)	152 countries	1960–2005
Tax-to-GDP	Acemoglu et al. (2019)	136 countries	1960–2005
Human Capital and Labor Markets			
Human Capital Index	PWT	142 countries	1960–2023
Employment	PWT	172 countries	1960–2023
Labor Share	PWT	132 countries	1960–2023
Primary Enrollment	Acemoglu et al. (2019)	176 countries	1970–2010
Secondary Enrollment	Acemoglu et al. (2019)	176 countries	1970–2010
Productivity and Returns			
TFP	PWT	117 countries	1960–2023
Internal Rate of Return	PWT	132 countries	1960–2023
Real Consumption	PWT	172 countries	1960–2023
Real Domestic Absorption	PWT	172 countries	1960–2023

Notes: Country coverage represents the number of countries with at least one non-missing observation for each variable. Data sources: WDI = World Development Indicators; PWT = Penn World Table; CNTS = Cross-National Time-Series Data Archive (2024 edition). Unrest (ANRR) is the original variable from Acemoglu et al. (2019), retained for comparability. Unrest (CNTS 2024) is our updated indicator, used throughout the analysis.

TABLE B.2. Cross-Sectional Correlation with Economic Fundamentals

	Overall	1960s	1970s	1980s	1990s	2000s	2010s	2020s
Panel A: Correlation with GDP per Capita								
Investment Share	0.390	0.404	0.312	0.376	0.433	0.446	0.408	0.441
Capital Stock	0.486	0.602	0.467	0.477	0.473	0.476	0.474	0.465
Internal Rate of Return	−0.196	−0.432	−0.303	−0.252	−0.098	−0.081	−0.130	−0.071
TFP	0.645	0.426	0.480	0.539	0.659	0.738	0.790	0.844
Trade Openness	0.313	0.094	0.235	0.295	0.282	0.343	0.387	0.411
Human Capital	0.738	0.764	0.628	0.732	0.752	0.766	0.791	0.775
Panel B: Correlation with Geopolitical Alignment Index								
GDP per Capita (PWT)	0.099	0.123	0.113	−0.166	0.241	0.127	0.133	0.007
Investment Share	0.106	−0.022	0.150	0.018	0.130	0.186	0.195	0.100
Capital Stock	0.137	0.141	0.266	−0.006	0.256	0.163	0.061	−0.153
Internal Rate of Return	−0.014	−0.005	−0.082	−0.010	0.001	−0.041	0.005	0.137
TFP	0.007	−0.025	−0.031	−0.128	0.167	−0.003	0.005	−0.072
Trade Openness	0.115	−0.047	0.088	0.024	0.181	0.160	0.146	0.119
Human Capital	0.086	0.029	0.067	−0.188	0.333	0.163	0.056	−0.094

Notes: Entries are pooled within-year cross-sectional correlations. Panel A reports correlations between GDP per capita and economic fundamentals. Panel B reports correlations between the geopolitical alignment index and the same variables. For each period, variables are first demeaned within year, and correlations are computed across country-year observations. Only years with at least 20 countries and nonzero cross-country variation are included. TFP is the PWT 11.0 level at current purchasing power parities (ctfp). The 2020s column covers 2020–2024.

B.3. Additional Results for Baseline Estimates

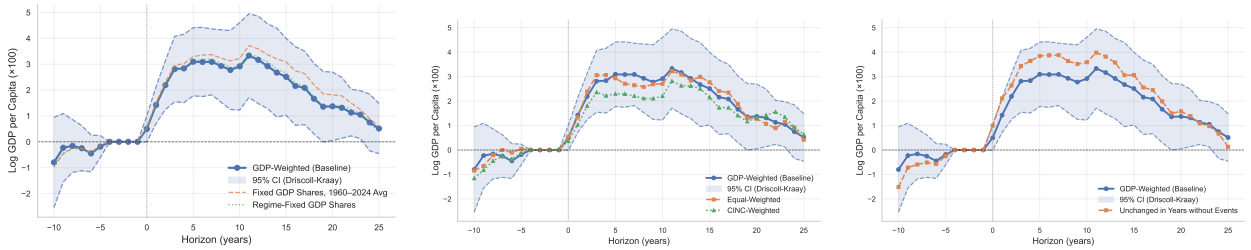
Table B.3 presents selected local-projection coefficients underlying Figure 6. The remaining results examine index weights, inference, reverse-direction timing, and lag specifications.

TABLE B.3. Local-Projection Coefficients for GDP per Capita

Horizon (years)	−15	−10	−5	+0	+5	+10	+15	+20	+25
Geopolitical Alignment Index	−1.758	−0.798	−0.185	0.497*	3.095***	2.917***	2.511***	1.371**	0.512
	(1.154)	(0.889)	(0.215)	(0.260)	(0.671)	(0.862)	(0.758)	(0.674)	(0.495)
Within-R^2	0.553	0.790	0.983	0.983	0.786	0.550	0.354	0.214	0.120
Observations	6,534	7,274	8,014	8,154	7,414	6,674	5,934	5,194	4,454
Countries	148	148	148	148	148	148	148	148	148

Notes: This table presents the local-projection coefficients from equation (3) for log GDP per capita (×100). Each column is a separate horizon regression; negative horizons provide timing diagnostics. Coefficients are per one standard deviation of the index. All specifications include country fixed effects, region-year fixed effects, and four lags of log GDP per capita and geopolitical alignment. Driscoll–Kraay standard errors are in parentheses. A Wald test that the coefficients at $h = -10, \ldots, -5$ are jointly zero, using the covariance of 500 country-block bootstrap draws, gives a p-value of 0.061. The underlying 148-country long-coverage sample spans 1960–2024. Estimation support varies by horizon because of the lags and horizon-specific outcome: $h = 0$ uses 1964–2024, while the displayed columns range from 1975–2024 at $h = -15$ to 1964–1999 at $h = 25$. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Alternative Index Weights and Construction. Because the baseline index weights partners by time-varying GDP shares, changes in partner GDP could move the index without any change in bilateral relations. Figure B.1 replaces these weights with fixed, regime-fixed, equal, and capability-based weights.



(a) Fixed GDP Weights (b) Equal and CINC Weights (c) Years without Recorded Events

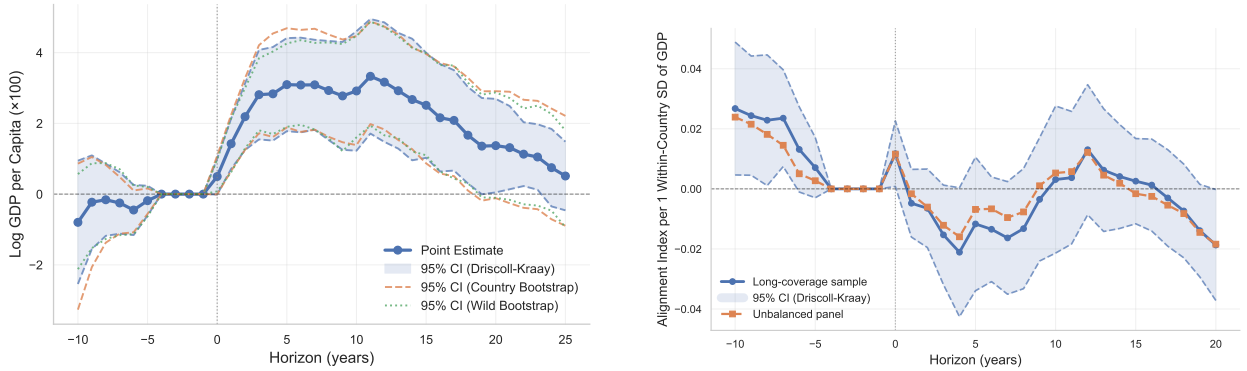
FIGURE B.1. Robustness to Alternative Index Weights and Construction

Notes: All panels report conditional local-projection coefficients for log GDP per capita (×100). Panel (a) compares the baseline using time-varying GDP shares (blue) with fixed GDP shares averaged over 1960–2024 (orange) and regime-fixed GDP shares using Cold War versus post-Cold War averages (green). Panel (b) compares the baseline GDP-weighted index (blue) with an equal-weighted index (orange) and a CINC-weighted index (green). Panel (c) compares the baseline (blue) with an index that holds bilateral scores unchanged in dyad-years without recorded events instead of moving them toward neutral (orange). All paths use the 148-country long-coverage sample. Each alternative index is rescaled to match the standard deviation of its corresponding baseline index before estimation. All specifications use the baseline controls. Shaded areas are 95% confidence intervals for the baseline, based on Driscoll–Kraay standard errors.

Panel (a) compares the baseline time-varying GDP shares with fixed shares averaged over 1960–2024 and regime-fixed shares for the Cold War and post-Cold War periods. Panel (b) instead

assigns equal weight to each of the 24 major nations. The resulting index has a correlation of 0.86 with the baseline. Panel (b) also weights partners by their share of the Composite Index of National Capability (Singer, Bremer, and Stuckey 1972). This index has a correlation of 0.89 with the baseline, and its coefficient path stays within the baseline confidence interval. Panel (c) holds bilateral scores unchanged in years without recorded events instead of moving them toward neutral. The resulting index has a correlation of 0.92 with the baseline, and its 25-year sustained-path coefficient is 10.0 log points. All five alternatives produce coefficient paths that closely track the baseline, weighing against changes in GDP weights, the aggregation scheme, or the treatment of years without events as the source of the result.

Inference and Reverse-Direction Diagnostics. Figure B.2 pairs alternative inference procedures with a reverse-direction timing diagnostic.



(a) Alternative Inference

(b) Reverse Direction: GDP to Alignment

FIGURE B.2. Inference and Reverse-Direction Diagnostics

Notes: Panel (a) displays the baseline GDP coefficient path with three types of confidence intervals: Driscoll–Kraay standard errors (blue shaded band), 500 country-block bootstrap iterations (orange dashed), and 500 wild bootstrap iterations using Rademacher weights (green dotted). Panel (b) reverses the projection, regressing fixed-weight geopolitical alignment at $t + h$ on log GDP per capita at t, conditional on four lags of both variables and the baseline fixed effects. Coefficients are per one within-country standard deviation of GDP. The blue line and shaded 95% Driscoll–Kraay confidence interval use the long-coverage sample. The dashed orange line uses the unbalanced sample. In panel (b), $h = -4, \ldots, -1$ are mechanically zero because the corresponding lagged alignment outcomes enter the controls. Horizons $h = -10, \ldots, -5$ relate past alignment to current GDP and do not test whether GDP precedes alignment.

Panel (a) compares the baseline Driscoll–Kraay bands with two resampling procedures. The country-block bootstrap resamples entire countries with replacement, preserving within-country dependence. The wild bootstrap applies observation-level Rademacher weights to residuals after absorbing the fixed effects, which allows for heteroskedasticity but not for serial or cross-country dependence.

The confidence intervals from 500 iterations of each bootstrap method are similar in width to the Driscoll–Kraay bands through year 20 and wider at longer horizons. The wild bootstrap intervals (green dotted) are slightly narrower than the country-block bootstrap intervals (orange dashed) at most horizons. Neither alternative changes the substantive inference.

Panel (b) asks whether current GDP predicts subsequent geopolitical alignment. Because the baseline index uses time-varying partner GDP shares, we hold each major nation's aggregation weight fixed at its sample mean, preventing changes in GDP weights from mechanically generating movement in the alignment index. In both the long-coverage and the unbalanced samples, current GDP shows no systematic or statistically meaningful association with subsequent alignment, so there is no evidence that rising GDP precedes geopolitical improvement.

Alternative Lag Specifications. Figure B.3 examines sensitivity to lag length selection.

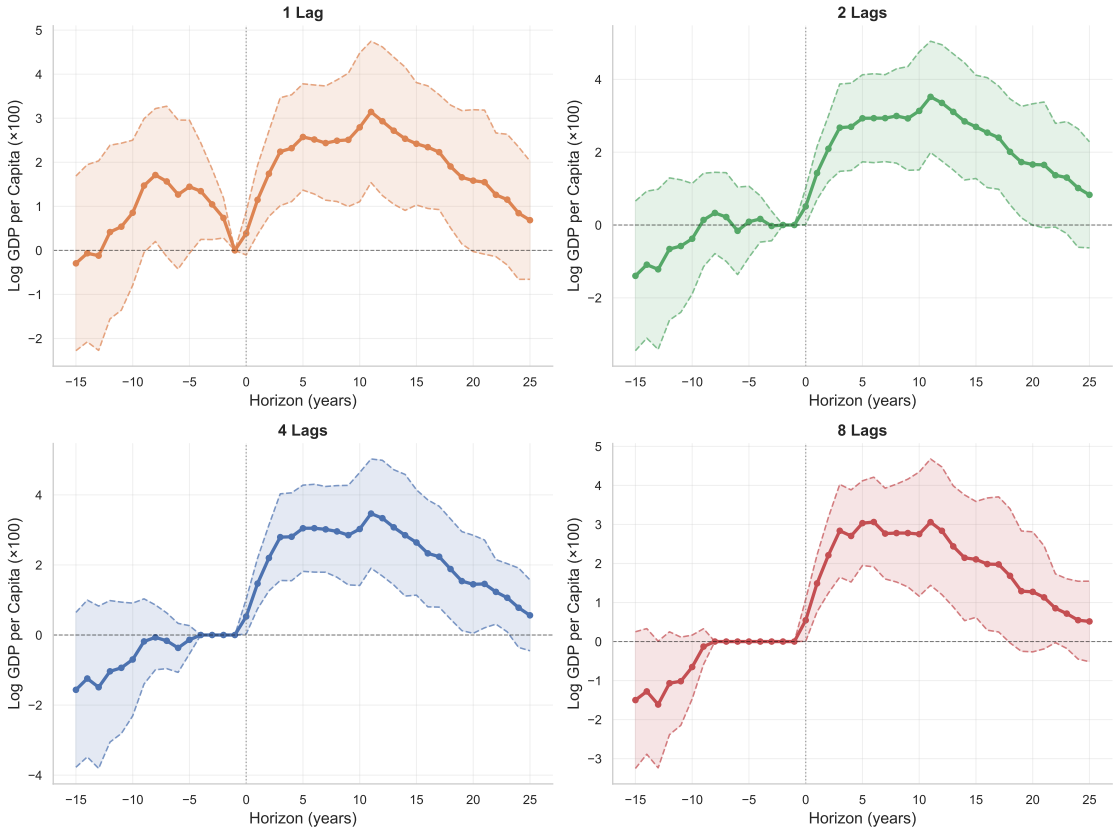


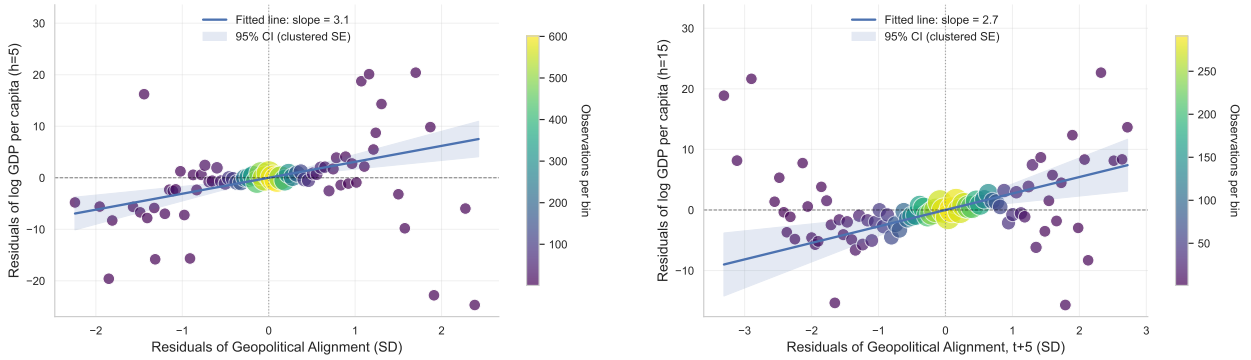
FIGURE B.3. Conditional GDP Coefficient Paths Under Alternative Lag Specifications

Notes: Each panel reports conditional local-projection coefficients for log GDP per capita (×100) with the indicated number of lags of GDP and geopolitical alignment. All specifications include country and region-year fixed effects. Shaded areas represent 95% confidence intervals based on Driscoll–Kraay standard errors.

The one-lag specification shows positive estimates at negative horizons and the two-lag specification negative ones at longer leads, indicating sensitivity to incomplete dynamic adjustment. With four and eight lags, the displayed negative-horizon coefficients are not individually distinguishable from zero and the post-zero paths are similar. We use four lags in the baseline to retain precision while allowing flexible dynamics.

B.4. Residualized Geopolitical Alignment and Subsequent GDP

To visualize the conditional relationship underlying the local projections, we residualize GDP and geopolitical alignment with respect to the baseline controls and fixed effects and plot their binned residual means.



(a) GDP at $t + 5$ vs. Geopolitical Alignment at t

(b) GDP at $t + 15$ vs. Geopolitical Alignment at $t + 5$

FIGURE B.4. Residualized Geopolitical Alignment and Subsequent GDP

Notes: Binned residual means of log GDP per capita against geopolitical alignment after partialling out the baseline controls and fixed effects used in Figure 6. Panel (a) relates GDP at $t + 5$ to geopolitical alignment at t. Panel (b) relates GDP at $t + 15$ to forward geopolitical alignment measured at $t + 5$. The plots use 100 equal-width bins. Marker size and color indicate the number of observations per bin. Shaded bands are 95% confidence intervals for the fitted lines, with standard errors clustered by country. GDP residuals are in log points (log GDP per capita ×100), and alignment residuals are divided by the standard deviation of the index (0.141).

The fitted slopes are 3.1 in panel (a) and 2.7 in panel (b). In both, the binned residual means follow an approximately linear positive pattern over the observed range, with no single segment driving the slope. This binscatter is a functional-form and outlier diagnostic, not evidence of causal symmetry or a basis for extrapolation.

B.5. Transitory and Sustained Alignment Paths

The coefficient paths in Section 3.2 combine the current association between geopolitical alignment and GDP with the subsequent dynamics of alignment. To isolate the component associated with a purely transitory change, we construct a counterfactual path in which the index increases by one unit on impact and returns to zero immediately thereafter. Following Sims (1986) and Bilal and Känzig (2026), we combine the projection coefficients for geopolitical alignment and GDP to construct this transitory path.

We begin by estimating the dynamics of the geopolitical alignment index using local projections:

$$p_{c,t+h} = \rho_h^p p_{ct} + \gamma_h' \mathbf{x}_{ct} + \varepsilon_{c,t+h}, \quad h = 0, 1, \ldots, H \tag{B.1}$$

where $\{\rho_h^p\}_{h=0,\ldots,H}$ are the horizon-specific projection coefficients on current geopolitical alignment, and $\mathbf{x}_{ct}$ contains the baseline lags and fixed effects. To construct the purely transitory path, we

introduce a series of auxiliary shocks $\{p_h^{\text{shock}}\}_{h=0}^{H}$ at each horizon that impose the desired response pattern $\tilde{\boldsymbol{\rho}}^p = (1, 0, \ldots, 0)'$. The required auxiliary series $\boldsymbol{p}^{\text{shock}}$ is obtained by solving:

$$\underbrace{\begin{pmatrix} p_0^{\text{shock}} \\ p_1^{\text{shock}} \\ \vdots \\ p_H^{\text{shock}} \end{pmatrix}}_{\boldsymbol{p}^{\text{shock}}} = \underbrace{\begin{pmatrix} 1 & 0 & \cdots & 0 \\ \rho_1^p & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \rho_H^p & \rho_{H-1}^p & \cdots & 1 \end{pmatrix}^{-1}}_{(\mathbf{R}^p)^{-1}} \underbrace{\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{\tilde{\boldsymbol{\rho}}^p} \tag{B.2}$$

Given this auxiliary series, the corresponding GDP coefficients $\tilde{\boldsymbol{\alpha}}$ under the purely transitory path are:

$$\underbrace{\begin{pmatrix} \tilde{\alpha}_0 \\ \tilde{\alpha}_1 \\ \vdots \\ \tilde{\alpha}_H \end{pmatrix}}_{\tilde{\boldsymbol{\alpha}}} = \underbrace{\begin{pmatrix} p_0^{\text{shock}} & 0 & \cdots & 0 \\ p_1^{\text{shock}} & p_0^{\text{shock}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ p_H^{\text{shock}} & p_{H-1}^{\text{shock}} & \cdots & p_0^{\text{shock}} \end{pmatrix}}_{\mathbf{P}^{\text{shock}}} \underbrace{\begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_H \end{pmatrix}}_{\boldsymbol{\alpha}} \tag{B.3}$$

The resulting coefficients $\tilde{\boldsymbol{\alpha}}$ trace the GDP path associated with a one-time, purely transitory improvement in geopolitical alignment. For the sustained path shown in Figure 7, we compute the cumulative coefficient $\sum_{s=0}^{h} \tilde{\alpha}_s$, which gives the GDP coefficient associated with maintaining geopolitical alignment one unit above baseline. The reported paths multiply these coefficients by 0.141, the standard deviation of the index. Confidence intervals come from a country-block bootstrap, which accounts for estimation uncertainty in both stages.

Methodological Caveat. This construction is an algebraic transformation of the conditional local-projection coefficients and does not add identifying variation beyond the baseline specification. A causal interpretation requires the conditions in Section 3.5, together with linearity and the assumption that the response to a sequence of unanticipated geopolitical innovations equals the response to the same alignment path anticipated at time zero. The latter assumption abstracts from forward-looking behavior that may differ with anticipation. We use the construction to compare persistence profiles and scale magnitudes, not as a structural policy experiment.

B.6. Residualized Partner Components

Figure B.5 regresses each partner component of Figure 8, panel (a), on the other two with country and year fixed effects,

$$p_{ct}^g = \sum_{k \neq g} \theta_k^g \, p_{ct}^k + \mu_c^g + \lambda_t^g + u_{ct}^g, \qquad g \in \{\text{US}, \text{OW}, \text{OM}\},$$

and enters the residual $\hat{u}_{ct}^{g}$ alone, with four lags of itself and of GDP, in the baseline local projection. The other two components explain 28, 33, and 9 percent of the variation in the US, other Western, and other major components that remains after country and year effects. The residualized paths peak at 3.2, 3.3, and 3.9 log points per standard deviation of the aggregate index, against roughly 4 in the joint specification.

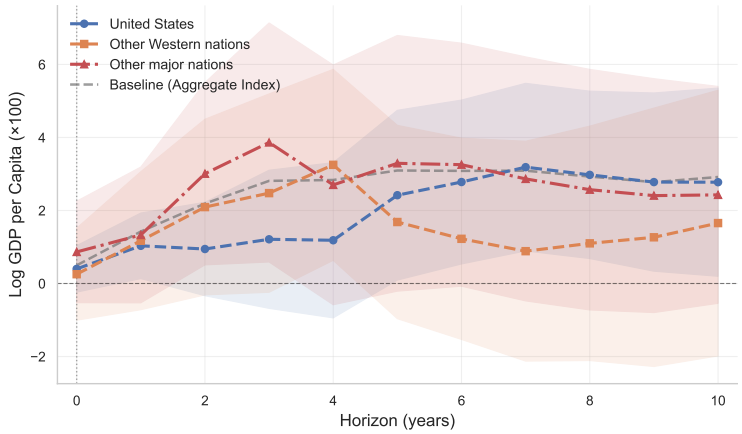


FIGURE B.5. Growth Responses by Partner Group: Residualized Components

Notes: Each partner component of Figure 8, panel (a), is regressed on the other two with country and year fixed effects. Its residual then enters the baseline local projection alone, with four lags of the residual and of GDP and with country and region-year fixed effects. Coefficients are for log GDP per capita (×100) per one standard deviation of the aggregate index over horizons $h = 0, \ldots, 10$. The gray dashed line is the baseline response to the aggregate index. Shaded bands are 95% confidence intervals based on Driscoll–Kraay standard errors.

B.7. Placebo Design

This section documents the placebo exercises summarized in Section 3.3.4.

Implementation. Both exercises use 500 independent placebo draws. In each draw, the full baseline local projection is re-estimated from scratch using the same specification: four lags of GDP and geopolitical alignment, country fixed effects, and region-year fixed effects, with Driscoll–Kraay standard errors. The summary statistic is the average estimated coefficient over horizons $h = 0, \ldots, 10$.

Placebo A: within-region-year reassignment. For each draw, we randomly permute realized alignment values across countries within each region-year cell. This preserves the region-year distribution of the index and the panel structure of outcomes and controls while breaking its country-specific assignment.

Placebo B: future-year timing. For each draw and each country, we replace current alignment with its value 8 to 15 years in the future, where the lead is drawn uniformly at random. To ensure that

all leads are well defined, we restrict base years to 1960–2009. This preserves each country's own alignment distribution, up to a time shift, while breaking the contemporaneous timing between alignment and subsequent GDP dynamics.

Results. Averaged over horizons 0 to 10, the realized coefficient is 2.51 under Placebo A and 2.67 under Placebo B, above all 500 placebo draws in each design, whose 95th percentiles are 0.29 and 0.42.

B.8. Domestic Transformation Periods

Definition and Boundaries. This section documents the classification of domestic transformation periods absorbed in Section 3.4. We evaluate candidate periods using four criteria: whether the period marks a recognized historical turning point, changes domestic political or economic institutions, spans multiple institutional margins (such as the constitutional order, state structure, property rights, or the economic-policy regime), and plausibly affects the country's long-run economic trajectory. A window begins when institutional flux becomes visible and ends when the new framework is operational. Transformation windows typically span two to seven years, but this is a guideline, not a hard restriction.

The classification excludes changes in geopolitical alignment, alliance membership, or bilateral relations unless they accompany a qualifying domestic transformation. It also excludes routine electoral alternation, gradual policy evolution, narrow sectoral reforms, macroeconomic crises, sanctions, and wars that do not transform the domestic institutional order. For example, Brexit is not classified because it changed external institutional membership without replacing the United Kingdom's domestic political-economic regime, and Japan's asset-price collapse is not classified because it was a macroeconomic shock within a continuous institutional framework. The rule is direction-neutral: democratizations and post-conflict reconstructions qualify alongside authoritarian consolidations and state collapses. Candidate episodes were reviewed for window placement, substantive scope, cross-country comparability, and coverage of well-established historical transformations, then normalized into a complete, nonoverlapping country-year classification.

Use in Estimation. The transformation-year specification adds a separate country-year fixed effect for every observation classified as part of a domestic transformation. This specification is algebraically equivalent to excluding these observations. We do not absorb fixed effects for all country-era cells, including stable-era means. Because geopolitical reorientation is persistent, full country-era fixed effects would remove the low-frequency component of alignment across long stable periods and change the estimand relative to the baseline.

As an alternative classification, we use V-Dem's *Regimes of the World* measure and absorb observations within three years before or after a change in regime type. V-Dem covers 8,385 of the 9,620 country-years in the long-coverage sample. It omits 16 countries, primarily microstates, and has shorter historical coverage for eight others. Observations without a V-Dem classification are

TABLE B.4. Domestic Transformation Periods: Coverage, Categories, and Examples

Panel A: Coverage and Duration		
Country-year grid	12,545 observations, 193 countries over 1960–2024	
Coded episodes	329 episodes across 172 countries; 21 countries with none	
Transformation years	1,476 country-years, or 11.8% of the grid	
Episode duration	Mean 4.5 years; median 4 years; range 1–13 years	
Duration distribution	79.6% last at most 5 years; 91.8% at most 7 years	
Panel B: Categories and Representative Examples		
Category	Episodes	Representative example
Regime change	91	Chile, military coup and economic restructuring, 1973–1976
Democratization	67	South Africa, end of apartheid, 1990–1994
Independence	57	Botswana, independence and state construction, 1966–1969
Authoritarian consolidation	30	Pakistan, Zia coup and Islamization, 1977–1980
Post-conflict transition	25	Rwanda, genocide and Rwandan Patriotic Front regime establishment, 1994–1999
Post-socialist transition	23	Poland, transition to market democracy, 1989–1992
Economic transition	18	China, reform and opening, 1978–1984
State collapse	14	Somalia, state collapse and civil war, 1988–1992
Reconstruction	4	Germany, reunification and institutional integration, 1990–1994

Notes: Panel A summarizes the complete country-year classification. Panel B reports the mutually exclusive category assigned to each episode and one representative case. The replication package contains the complete episode and country-year panels.

excluded from this exercise rather than treated as stable. Among covered country-years, both rules flag 7.9 percent, the domestic-transformation classification alone flags 4.7 percent, V-Dem alone flags 19.6 percent, and neither flags 67.8 percent. The domestic-transformation classification covers broader episode types, whereas V-Dem applies a mechanical seven-year window around each regime-type change. At year 5, the transformation-year and V-Dem exclusions yield coefficients of 2.0 and 2.4 log points, respectively. On the 124 countries with complete V-Dem coverage, the year-5 coefficient is 3.6 log points (95 percent confidence interval 2.3 to 4.9) in the corresponding baseline and 2.5 log points (0.8 to 4.1) under the V-Dem exclusion. The V-Dem sustained-path magnitude is slightly smaller than under the transformation-year exclusion despite its larger year-5 coefficient, partly because the mapping re-estimates alignment persistence within each exclusion sample. We therefore interpret V-Dem as an alternative exclusion rule, not as validation of individual historical episodes.

B.9. Leadership Transitions in Major Nations: Robustness

This section documents the leadership-change IV based on bilateral shifts measured around 67 leadership changes across 21 of 24 major nations: 25 deaths in office (from the Archigos database, 1950–2015) and 42 close election turnovers with vote margins below 10 percentage points (from Marx,

Pons, and Rollet (2025), 1960–2018). Switzerland is excluded due to its annually rotating presidency.[37] For each event in major nation j at time t, the instrument is $z_{ct}^{\text{LC}} = (S_{cj,t+1} - S_{cj,t-1}) \times \text{GDP share}_{j,t-1}$. Because the instrument uses the realized bilateral shift rather than the transition alone, it may also capture other bilateral developments in the event window.

Window Robustness. Table B.5 reports reduced-form and LP-IV estimates at horizon $h = 5$ under six bilateral-shift windows. The reduced-form point estimates range from 3.4 to 4.4 log points, while the IV estimates are larger for wider post-transition windows as the first stage weakens. The main-text window ($S_{t+1} - S_{t-1}$) is relatively short and has a first-stage F-statistic of 16.9.

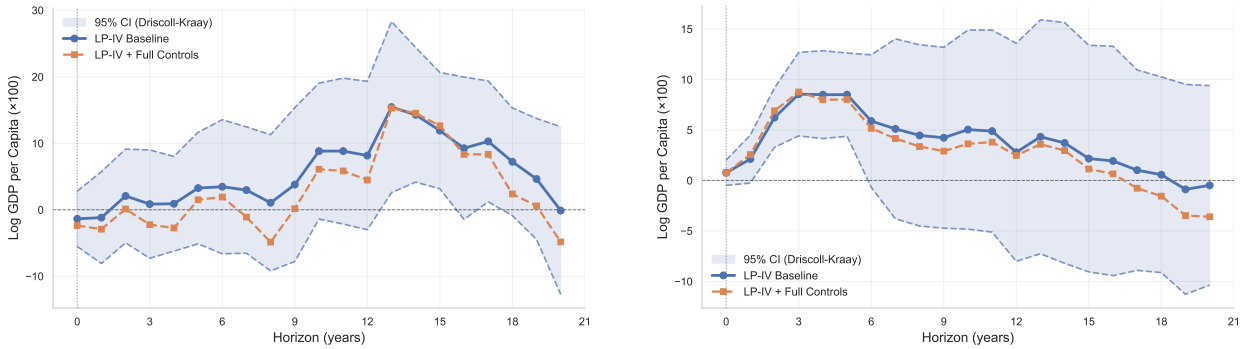
TABLE B.5. Leadership-Change IV: Window Robustness

Window	First-stage F	RF($h = 5$)	IV($h = 5$)
$S_t - S_{t-1}$	53.7	4.4 (2.3)	3.1 (1.4)
$S_{t+1} - S_{t-1}$	16.9	3.4 (2.4)	6.0 (2.5)
$S_{t+2} - S_{t-1}$	17.8	3.5 (2.8)	8.1 (3.7)
$S_t - S_{t-2}$	35.4	4.1 (2.6)	3.0 (1.8)
$S_{t+1} - S_{t-2}$	15.7	3.4 (2.8)	5.8 (3.1)
$S_{t+2} - S_{t-2}$	14.7	3.5 (3.2)	8.0 (4.4)

Notes: Each row uses a different bilateral-shift window to construct the leadership-change instrument. First-stage F: Wald F-statistic for the instrument in the first-stage projection of alignment. RF($h = 5$) and IV($h = 5$): reduced-form and LP-IV coefficients for log GDP per capita (×100) at $h = 5$ with Driscoll–Kraay standard errors in parentheses. Both are multiplied by 0.141, so the IV coefficient is per one standard deviation of the index and the reduced form is per 0.141 units of the instrument. All specifications use 67 events (25 deaths + 42 close elections), country and region-year fixed effects, and four lags of GDP and geopolitical alignment. Sample: 148-country long-coverage sample.

Deaths versus Close Elections. Figure B.6 splits the leadership-change IV by event type. The close-election component (panel (b)) yields a smoother hump-shaped path, with coefficients distinguishable from zero over horizons 2–5. The deaths-in-office component (panel (a)) is more volatile: its first-decade estimates are imprecise, and large positive coefficients appear at several horizons from 13 to 17. The death events provide more plausibly external timing, but their first stage is weak ($F = 7.9$, against 26.5 for close elections). Pooling the two event types does not remove the caveat created by using realized bilateral shifts.

[37] For France we use presidential elections only. We drop elections flagged in the source or lacking a reported margin, as well as elections within two years of a death in office in the same country. We also correct two Archigos codings. Deaths in office are Archigos exits by natural death, assassination, suicide, or irregular removal followed by death.



(a) Deaths in Office (25 events)

(b) Close Election Turnovers (42 events)

FIGURE B.6. Leadership-Change IV by Event Type

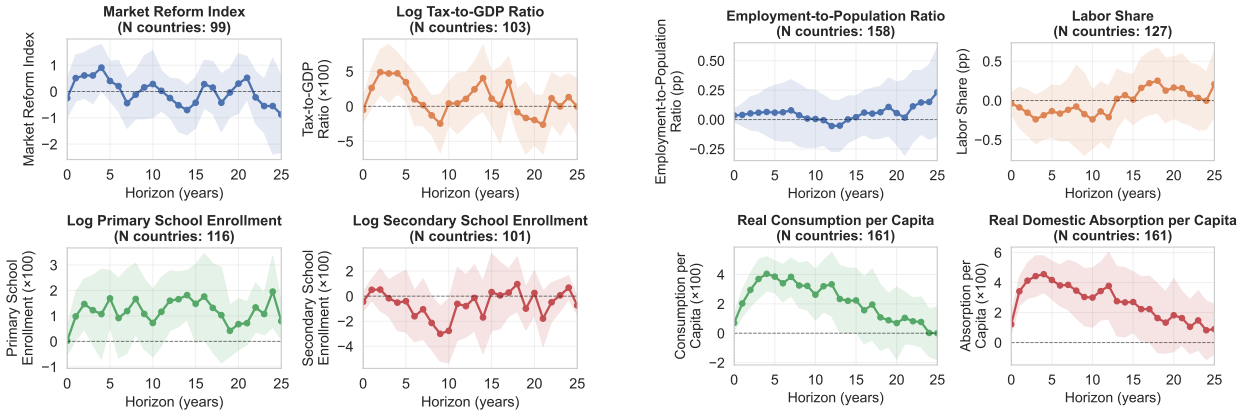
Notes: LP-IV coefficient paths for log GDP per capita (×100), estimated separately using deaths in office (panel (a)) and close election turnovers with margins below 10 percentage points (panel (b)). Blue solid: baseline specification. Orange dashed: adding four lags of trade openness, domestic unrest, war exposure, and V-Dem institutional indices (on the 109-country common sample, which the plotted baseline also uses). Shaded bands show 95% confidence intervals for the baseline, based on Driscoll–Kraay standard errors.

B.10. Other Correlates of Growth

Figure B.7 extends the growth-fundamentals analysis in Figure 13 to institutional, educational, labor-market, and expenditure outcomes.

Market Reforms and Human Capital Formation. Panel (a) shows a short-run positive response in the tax-to-GDP ratio and a more gradual increase in primary enrollment. Market reform and secondary enrollment show no systematic response.

Labor Markets and Domestic Absorption. Panel (b) shows positive, GDP-like paths for real consumption and domestic absorption per capita. The employment-to-population ratio and labor share show no systematic response. As in Figure 13, these outcomes are estimated separately and should be read as accompanying responses rather than identified mediating channels.



(a) Market Reforms and Education

(b) Employment and Expenditure

FIGURE B.7. Dynamic Responses of Additional Growth Correlates to Geopolitical Alignment

Notes: This figure reports horizon-specific coefficients on the country-level geopolitical alignment index from local projections of the listed outcomes. Panel (a) shows the market reform index, tax-to-GDP ratio, and primary and secondary enrollment. Panel (b) shows the employment-to-population ratio, labor share, real consumption, and domestic absorption per capita. Log outcomes are multiplied by 100. All specifications follow equation (5). Each outcome uses an outcome-specific country set, with counts shown in parentheses. Shaded areas represent 95% confidence intervals based on Driscoll–Kraay standard errors.

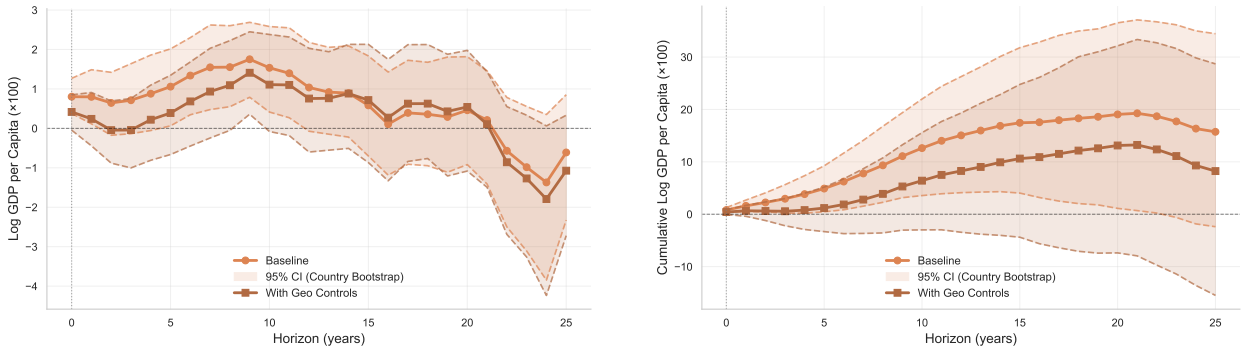
B.11. Additional Results for Democracy and Geopolitics

This section complements Section 4.2 by mapping the separate and joint democracy coefficients into transitory and sustained paths and by examining democratic status, aggregate alignment, and GDP under alternative fixed effects.

B.11.1. Transitory and Sustained Democracy Paths

We apply the construction in Appendix B.5 to the separate and joint specifications in Figure 15, extending them through year 25. For each specification, the estimated response of democratic status to itself determines the sequence of innovations that produces a one-period democracy path. Convolving this sequence with the GDP coefficients yields the transitory path, and its cumulative sum yields the sustained path.

Panel (a) shows small transitory paths that peak at 1.8 log points in the separate specification and 1.4 log points when alignment is included, before declining at longer horizons. Panel (b) shows the corresponding sustained paths. At year 20, the separate and joint estimates reach 19.1 and 13.1 log points. By year 25, they are 15.7 and 8.2 log points. These differences summarize conditional overlap between democracy and geopolitical alignment, not a mediation share.



(a) Transitory Democracy Path

(b) Sustained Democracy Path

FIGURE B.8. GDP Mappings for Transitory and Sustained Democracy Paths

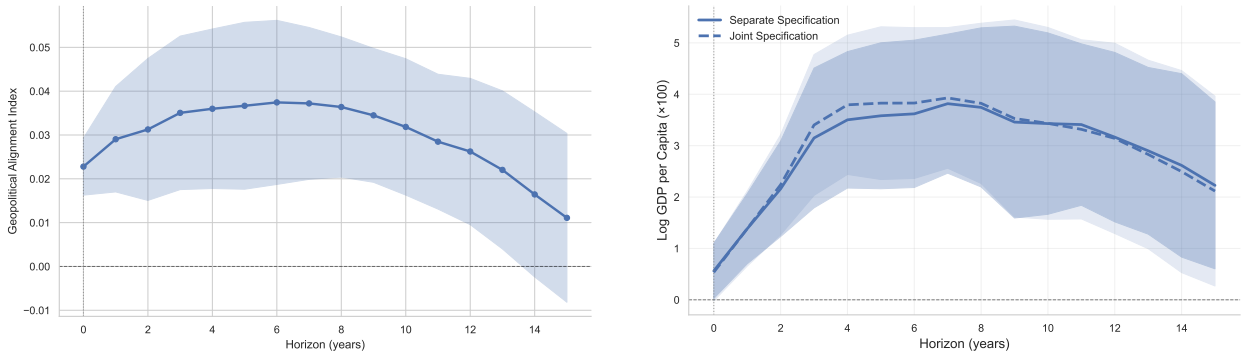
Notes: Panel (a) maps the democracy coefficients into the GDP path associated with a one-period increase in democratic status. Panel (b) cumulates that path to construct a sustained increase. The separate specification includes four GDP lags. The joint specification also includes current geopolitical alignment and its four lags. Both use the original ANRR GDP series and include country and year fixed effects. Shaded bands are pointwise 95% confidence intervals from 500 country-block bootstrap draws. The legends label the separate specification "Baseline" and the joint specification "With Geo Controls". The outcome is log GDP per capita (×100).

B.11.2. Aggregate Alignment and Region-Year Fixed Effects

Figure B.9 provides two additional checks. Panel (a) relates democratic status to aggregate geopolitical alignment, and panel (b) estimates the alignment–GDP relationship under the baseline region-year fixed effects.

Panel (a) shows a positive, persistent conditional relationship between democratic status and aggregate alignment. The coefficient is 0.023 on impact, peaks at 0.037 in year 6, and declines to 0.011 by year 15. Together with Figure 14, this pattern shows that democratic status is associated with both partner-specific geopolitical reorientation and a higher aggregate index.

Panel (b) shows that adding democratic status leaves the alignment–GDP coefficient path nearly unchanged under country and region-year fixed effects. Both paths peak near 3.8–3.9 log points, and their horizon-by-horizon correlation is 0.99.



(a) Democratic Status and Aggregate Alignment

(b) Alignment and GDP, Region-Year FE

FIGURE B.9. Democracy and Geopolitical Alignment: Additional Results

Notes: Panel (a) reports horizon-specific coefficients from local projections of aggregate geopolitical alignment on democratic status, including four alignment lags and country and year fixed effects. Panel (b) reports coefficients on geopolitical alignment from local projections of log GDP per capita (×100), with and without current democratic status, using the original ANRR GDP series. Both specifications in panel (b) include four GDP and alignment lags and country and region-year fixed effects. Panel (a) coefficients are in index units, and panel (b) coefficients are per one standard deviation of the index. Shaded bands report 95% confidence intervals based on Driscoll–Kraay standard errors.

B.12. Dynamic Panel Estimates

This section reports the dynamic panel estimates summarized in Section 5.1. Following the preferred specification in Acemoglu et al. (2019), we estimate

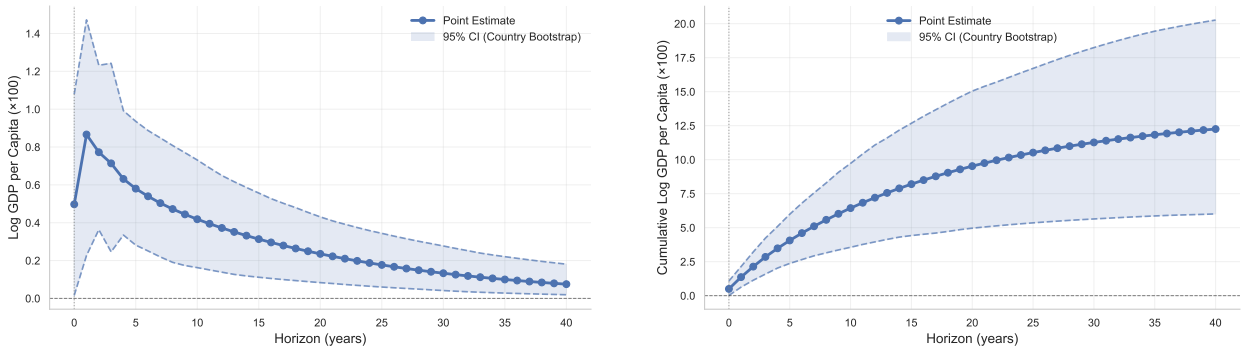
$$y_{ct} = \alpha p_{ct} + \sum_{\ell=1}^{J} \beta_\ell y_{c,t-\ell} + \sum_{\ell=1}^{J} \gamma_\ell p_{c,t-\ell} + \eta_c + \eta_{r(c)t} + \varepsilon_{ct}$$

where y_{ct} denotes log GDP per capita (×100), p_{ct} is the geopolitical alignment index, and η_c and $\eta_{r(c)t}$ denote country and region-year fixed effects. We set $J = 4$. A causal interpretation of the resulting responses rests on Assumption 1 (Section 3.5).[38]

Panel (a) of Figure B.10 shows the response to a one-period alignment change. It rises from 0.5 log points on impact to 0.9 after one year, then declines to 0.3 at year 15 and 0.2 at year 25. The point estimates remain positive, and their confidence intervals exclude zero throughout the displayed 40-year horizon.

In panel (b), the sustained-path coefficient reaches 4.1 log points at year 5, 8.2 at year 15, 10.5 at year 25, and 12.3 at year 40. At year 25, the 95 percent confidence interval is 5.4 to 16.7 log points, close to the local-projection estimate of 10.1 and its interval of 5.3 to 16.8.

[38]For a one-period change in alignment, the response satisfies $\psi_0 = \alpha$ and $\psi_k = \sum_{\ell=1}^{\min(k,J)} \beta_\ell \psi_{k-\ell} + \gamma_k \mathbf{1}[k \le J]$ for $k \ge 1$. The response to a sustained unit path at horizon h is $\Psi_h = \sum_{s=0}^{h} \psi_s$, with long-run multiplier $\Psi_\infty = (\alpha + \sum_{\ell=1}^{J} \gamma_\ell)/(1 - \sum_{\ell=1}^{J} \beta_\ell)$.



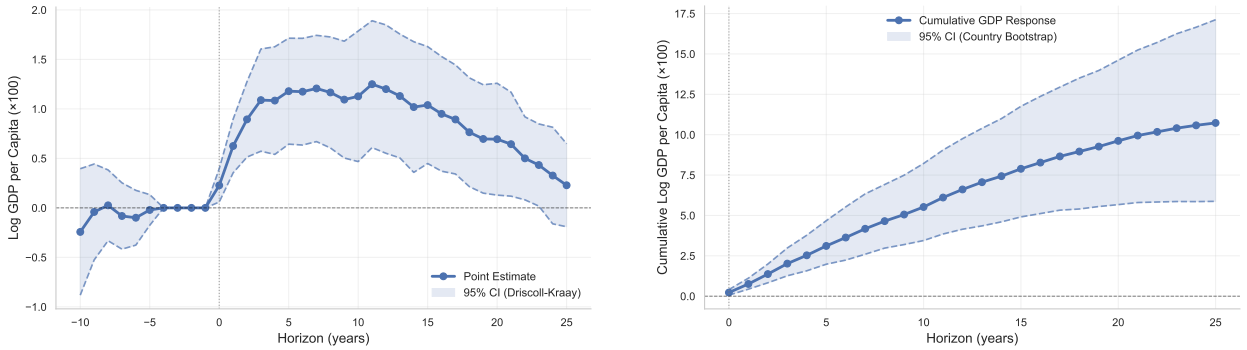
(a) Transitory Alignment Change (b) Sustained Alignment Path

FIGURE B.10. Dynamic Panel Estimates: GDP Responses to Transitory and Sustained Alignment Paths

Notes: Panel (a) shows the response of log GDP per capita (×100) to a one-period, one-standard-deviation change in geopolitical alignment. Panel (b) cumulates these responses to represent a sustained one-standard-deviation alignment path. Both panels include country and region-year fixed effects and four lags of GDP and alignment. Shaded areas report 95% confidence intervals from 500 country-block bootstrap iterations.

B.13. Dynamics of the Unsmoothed Index

This section reports persistence and transitory-path results for the unsmoothed index $\tilde{S}_{ct}$ examined in Section 5.1. Figure B.11 reports its GDP responses.



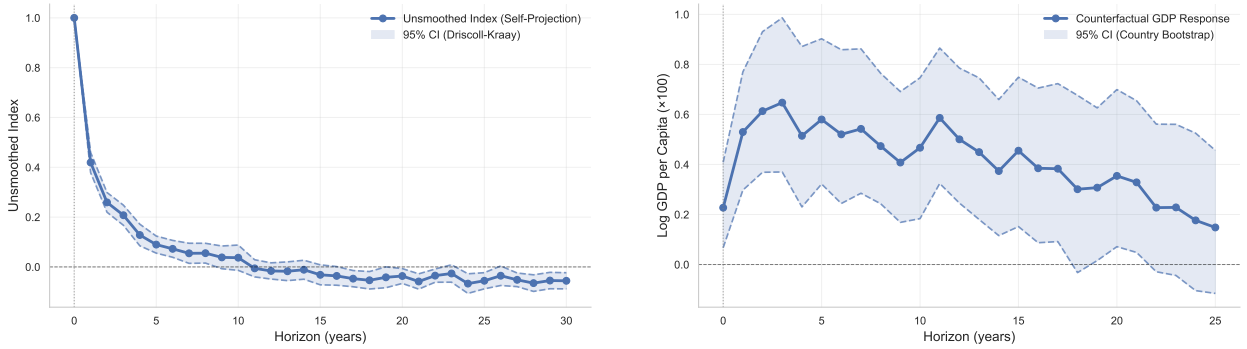
(a) Unsmoothed Index (b) Sustained Path, Unsmoothed Index

FIGURE B.11. GDP Responses to the Unsmoothed Index

Notes: Panel (a) reports conditional local-projection coefficients for log GDP per capita (×100) on the unsmoothed index, per 0.141 index units (one standard deviation of the baseline index). Panel (b) reports the sustained-path mapping obtained by cumulating the constructed transitory coefficients. Both panels include four lags of GDP and of the unsmoothed index, country fixed effects, and region-year fixed effects. Panel (a) reports 95% Driscoll–Kraay confidence intervals. Panel (b) reports 95% confidence intervals from 500 country-block bootstrap iterations.

Panel (a) of Figure B.12 shows that the unsmoothed index mean-reverts substantially faster than the baseline index (panel (a) of Figure 6). The coefficient falls from one on impact to 0.419 after one year, so 58.1 percent of the initial change dissipates. It is 0.055 at year 8, and its confidence interval first includes zero at year 9. Later point estimates are small and sometimes negative.

Panel (b) applies the construction in Appendix B.5 to remove this persistence. The constructed



(a) Persistence of the Unsmoothed Index

(b) Transitory Path, Unsmoothed Index

FIGURE B.12. Dynamics of the Unsmoothed Index

Notes: Panel (a) reports coefficients from projecting the unsmoothed index at $t + h$ on its current value, conditional on the baseline controls. Panel (b) reports the constructed GDP path associated with a 0.141 change in the unsmoothed index at $h = 0$ that returns to zero thereafter. Appendix B.5 details the construction. Panel (a) reports 95% Driscoll–Kraay confidence intervals. Panel (b) reports 95% confidence intervals from 500 country-block bootstrap iterations.

GDP coefficient peaks at 0.6 log points in year 3 and declines thereafter. The later estimates are less precise.

B.14. Machine-Coded Event Databases

This section reports the downstream GDP comparison for the machine-coded event databases introduced in Appendix A.7. Each panel of Figure B.13 estimates the baseline specification with the baseline index and with an index built from one alternative database, over their longest common window. Both indices use the same aggregation and dynamic recursion and are standardized by their in-sample standard deviations. The baseline coefficient path peaks at 1.88 log points after three years in the ICEWS window, 2.83 after three years in the GDELT window, and 3.31 after eleven years in the Phoenix window. The ICEWS and GDELT estimates remain statistically indistinguishable from zero. Phoenix yields a small positive response at horizons 0–2, peaking near 0.56 log points, but it fades by year 5. Standardization rules out pure scale differences as an explanation for the weaker and less persistent machine-coded paths. This exercise compares the persistence of the GDP responses. Appendix A.7 evaluates measurement accuracy and calibration directly.

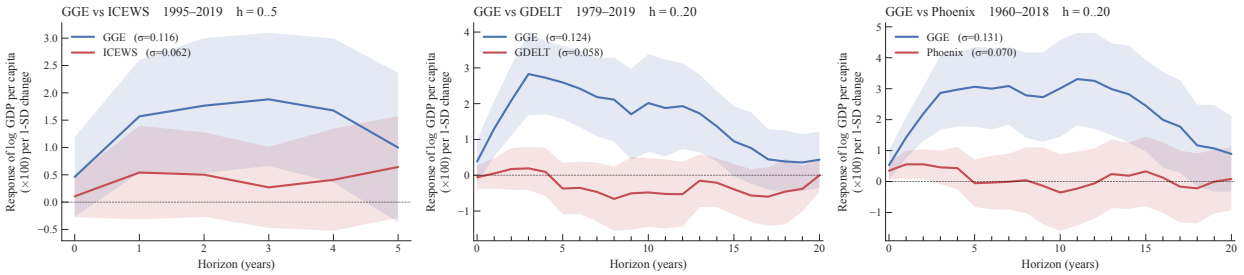
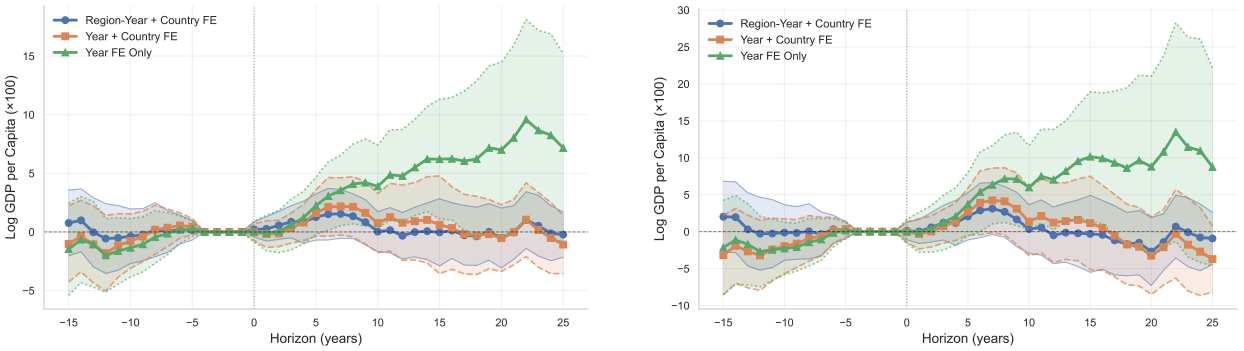


FIGURE B.13. Growth Responses Using Baseline and Machine-Coded Event Databases

Notes: Local-projection coefficients for log GDP per capita (×100) per one-standard-deviation improvement in geopolitical alignment. In the legends, GGE denotes the baseline index, and σ is each index's standard deviation in the regression sample, which scales the coefficients. Each panel compares the baseline index with one machine-coded database over the longest common estimation window: ICEWS, 1995–2019 and horizons 0–5; GDELT, 1979–2019 and horizons 0–20; Phoenix, 1960–2018 and horizons 0–20. All specifications include country and region-year fixed effects and four lags of GDP and geopolitical alignment. Shaded areas are 95% confidence intervals based on Driscoll–Kraay standard errors.

B.15. UNGA Voting and Economic Growth

UNGA ideal points locate countries in a common multilateral voting space. We use the negative Ideal Point Distance (IPD) from Bailey, Strezhnev, and Voeten (2017), which ranges from about −5.4 (complete disagreement) to 0 (perfect alignment), so higher values indicate closer voting positions. Panel (a) of Figure B.14 uses the United States as the reference point and is our primary comparison. Panel (b) instead takes the GDP-weighted average of negative IPD to the 24 major nations. This aggregation broadens the reference set, but it is not an exact analogue of the geopolitical alignment index because ideal-point distances describe relative positions in a common multilateral space rather than the intensity of bilateral relationships. We therefore interpret panel (b) as a check on whether the United States reference drives the comparison.



(a) Voting Proximity to United States

(b) GDP-Weighted Voting Proximity

FIGURE B.14. GDP Coefficient Paths Using UNGA Voting Proximity

Notes: Local-projection coefficients for log GDP per capita (×100) on UNGA voting proximity. Panel (a) uses negative IPD to the United States. Panel (b) uses the GDP-weighted average of negative IPD to the 24 major nations. The specifications include country and region-year fixed effects (blue, our baseline), country and year fixed effects (orange), or year fixed effects (green), together with four lags of GDP and the relevant IPD measure. Coefficients are per unit of negative IPD. Shaded areas report 95% confidence intervals based on Driscoll–Kraay standard errors.

Panel (a) shows that the coefficient path is statistically indistinguishable from zero under the baseline country and region-year fixed effects throughout the horizon. Replacing region-year fixed effects with year fixed effects produces a similar path. A positive coefficient emerges only when country fixed effects are removed, reaching approximately 7 log points after 20 years. This specification combines cross-country covariance between income and United States voting proximity with within-country changes, whereas the baseline design uses within-country variation.

Panel (b) reports the GDP-weighted check. With country fixed effects, the coefficient paths are statistically indistinguishable from zero. A positive path appears only with year fixed effects alone, at horizons 7 to 16. Broadening the reference set therefore does not alter the result from the United States comparison.

These results do not imply that UNGA voting is uninformative. The two measures describe different objects: multilateral voting positions can remain stable while bilateral relations change, as the historical cases in Section 2.2 illustrate. Within-country variation in IPD therefore does not reproduce the same bilateral dynamics in the growth application.

B.16. Sanctions and Geopolitical Alignment

This section asks whether active sanctions account for the alignment–GDP relationship. We use the Global Sanctions Database (GSDB) (Felbermayr et al. 2020; Yalcin et al. 2025) to identify active sanctions imposed by major nation j on country c from 1960 through 2023. We aggregate the dyadic indicators using the same 24 major nations as in the alignment index:

$$p_{ct}^{\text{Sanction}} = \sum_{j \in \mathcal{N} \setminus \{c\}} \mathbf{1}_{cjt} \times \text{GDP share}_{jt},$$

where $\mathbf{1}_{cjt}$ equals one if major nation j has an active sanction on country c in year t. All countries in the baseline panel are retained. Within the database's coverage, dyad-years with no active sanction are coded zero. Appendix A.6 separately benchmarks the event database against GSDB sanction onsets and removals.

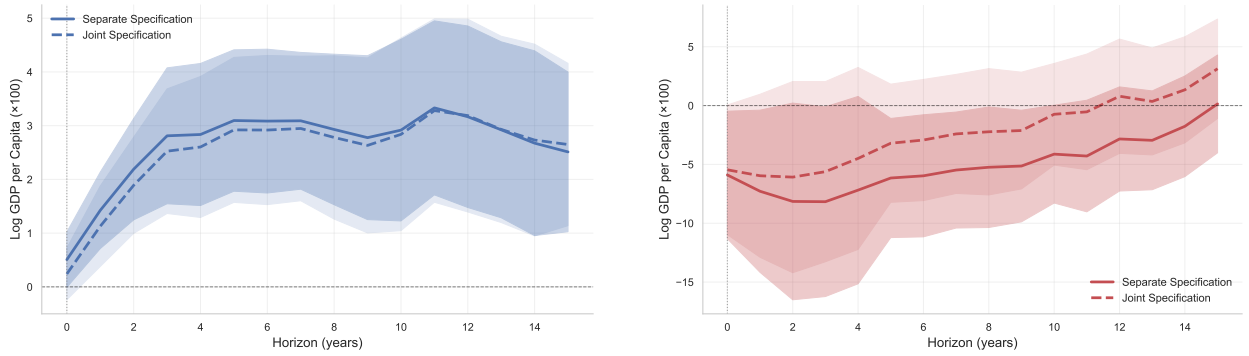
Empirical Specification. We estimate separate and joint specifications:

$$\text{Separate:} \quad y_{c,t+h} = \alpha_h^k p_{ct}^k + \sum_{\ell=1}^{4} \beta_{h\ell} y_{c,t-\ell} + \sum_{\ell=1}^{4} \gamma_{h\ell}^k p_{c,t-\ell}^k + \eta_c + \eta_{r(c)t} + \varepsilon_{c,t+h} \tag{B.4}$$

$$\begin{aligned} \text{Joint:} \quad y_{c,t+h} = {} & \alpha_h^{\text{Geo}} p_{ct} + \alpha_h^{\text{Sanc}} p_{ct}^{\text{Sanction}} + \sum_{\ell=1}^{4} \beta_{h\ell} y_{c,t-\ell} + \sum_{\ell=1}^{4} \gamma_{h\ell}^{\text{Geo}} p_{c,t-\ell} \\ & + \sum_{\ell=1}^{4} \gamma_{h\ell}^{\text{Sanc}} p_{c,t-\ell}^{\text{Sanction}} + \eta_c + \eta_{r(c)t} + \varepsilon_{c,t+h} \end{aligned} \tag{B.5}$$

where $k \in \{\text{Geo}, \text{Sanction}\}$ indexes the measure. The outcome and sample follow the baseline:

log GDP per capita, the 148-country long-coverage sample, country and region-year fixed effects, and four lags of GDP and each included geopolitical measure. At every horizon, the separate and joint specifications use identical country-year observations.



(a) Geopolitical Alignment Index (b) Sanctions

FIGURE B.15. Geopolitical Alignment and Sanctions

Notes: Local-projection coefficient paths for log GDP per capita (×100). Panel (a) compares separate and joint specifications for geopolitical alignment. Panel (b) does the same for sanctions exposure. All specifications use identical observations at each horizon and include four lags of GDP and the included geopolitical measures, country fixed effects, and region-year fixed effects. Shaded areas report 95% confidence intervals based on Driscoll–Kraay standard errors. Alignment coefficients are per one standard deviation of the index. Sanctions exposure is measured in raw GDP-share units, and the vertical scales differ across panels.

Results and Interpretation. In panel (a) of Figure B.15, the alignment coefficient paths track one another closely and peak in year 11 at 3.3 log points in both the separate and the joint specification. In panel (b), the separate sanctions coefficient reaches −8.2 log points in year 3. The corresponding joint estimate is −5.6, a 31 percent attenuation, and no joint sanctions coefficient is statistically distinguishable from zero. By year 10, the joint estimate is close to zero.

The two measures differ in timing. The alignment index records diplomatic deterioration and the announcement or lifting of sanctions, while the GSDB indicator records active enforcement. The stable alignment path shows that the narrower enforcement measure does not account for the broader alignment–GDP relationship.

B.17. Formal Alliances and Geopolitical Alignment

Data and Variable Construction. This section asks whether formal military alliances account for the alignment–GDP relationship. We use the Alliance Treaty Obligations and Provisions (ATOP) dataset, version 5.1 (Leeds et al. 2002), which codes formal military alliances through 2018.[39] For

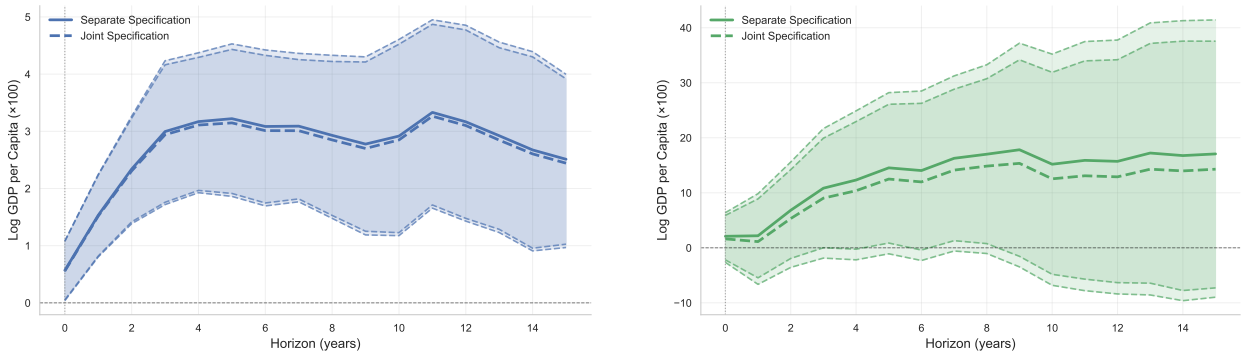
[39]The original ATOP dataset (Leeds et al. 2002) covered 1815–1944. We use version 5.1, which extends coverage through 2018.

each country c and year t, we construct a GDP-weighted alliance exposure measure:

$$p_{ct}^{\text{Alliance}} = \sum_{j \in \mathcal{N} \setminus \{c\}} \mathbf{1}_{cjt}^{A} \times \text{GDP share}_{jt},$$

where $\mathbf{1}_{cjt}^{A}$ equals one if country c maintains any active formal alliance (a defense pact, offensive pact, neutrality agreement, non-aggression pact, or consultation agreement) with major nation j in year t. We restrict the sample to years with observed ATOP coverage and retain all countries in the baseline panel. Within that period, dyad-years with no active alliance are coded zero. The dyadic indicators are binary, but the aggregate is continuous and can change with the major nations' GDP shares even when treaty status is unchanged. The measure ranges from zero to approximately 0.80.

Results. We estimate the same separate and joint local projections, on identical observations, as in equations (B.4)–(B.5), replacing p_{ct}^{Sanction} with p_{ct}^{Alliance}.



(a) Geopolitical Alignment Index

(b) Formal Alliances

FIGURE B.16. Geopolitical Alignment and Formal Alliances

Notes: Local-projection coefficient paths for log GDP per capita (×100). Panel (a) compares separate and joint specifications for geopolitical alignment. Panel (b) does the same for formal alliance exposure. All specifications use identical observations at each horizon and include four lags of GDP and the included geopolitical measures, country fixed effects, and region-year fixed effects. Shaded areas report 95% confidence intervals based on Driscoll–Kraay standard errors. Alignment coefficients are per one standard deviation of the index. Alliance exposure is measured in raw GDP-share units, and the vertical scales differ across panels.

In panel (a) of Figure B.16, the alignment coefficient paths are nearly identical and peak in year 11 at 3.3 log points in both the separate and the joint specification. In panel (b), the alliance coefficients are positive and build gradually through the first decade. In year 9, the estimate declines from 17.8 to 15.3 log points when alignment is included, a 14 percent attenuation. The joint alliance coefficients are statistically indistinguishable from zero throughout the horizon.

The two measures differ in timing: the alignment index records diplomatic cooperation surrounding alliance formation and maintenance, while the ATOP indicator records a persistent treaty stock. The alignment coefficient path remains stable after this stock is included.

Defense Pacts. As a robustness check, we restrict the alliance measure to defense pacts, the strongest form of alliance commitment. The alignment coefficient path remains nearly unchanged, while the defense-pact coefficients are imprecise in both the separate and joint specifications.

Online Appendix C. LLM Prompt: Event Category, CAMEO, and Goldstein Score

Appendix C describes the prompt structure (C.1), the event categories and inclusion threshold (C.2), and the CAMEO and Goldstein coding rules (C.3).

C.1. Prompt Structure

This section describes the LLM prompt structure for compiling and analyzing major political events shaping the bilateral relationship between two countries, {country_1} (code: {country_1_code}) and {country_2} (code: {country_2_code}), or their historical predecessors, during {target_year}. The analysis employs the Conflict and Mediation Event Observations (CAMEO) framework and the Goldstein scale, both detailed in Appendix C.3, to classify events and assess their intensity. If no major events are identified, a historical context-based relationship assessment is provided. The output is a single JSON object.

Relationship Assessment Framework. The bilateral relationship for {target_year} is classified into one of the following mutually exclusive categories, reflecting interaction intensity and nature:

- **State of War / Active Conflict**: Sustained, large-scale armed conflict.
- **Crisis / Intense Confrontation**: High tension with disputes or limited clashes, short of war.
- **Hostile / Antagonistic Relationship**: Animosity marked by sanctions or diplomatic friction.
- **Competitive / Rivalrous Relationship**: Strategic competition with limited cooperation.
- **Limited Contact / Cool Relationship**: Minimal, neutral interaction.
- **Selective Cooperation / Transactional Relationship**: Cooperation on specific interests amid competition.
- **Broad Cooperation / Partnership**: Extensive sectoral cooperation with regular dialogue.
- **Strategic Partnership**: Deep coordination on strategic issues with high trust.
- **Alliance**: Formal treaty-based mutual support, often military.

Analytical Steps. The prompt specifies five steps for event identification, classification, and assessment:

a. **Verify Political Entities**: Confirm {country_1} (code: {country_1_code}) and {country_2} (code: {country_2_code}) existed in {target_year}. If not, identify the primary political entity controlling the relevant territory (e.g., Soviet Union for Russia before December 26, 1991; Russian Federation thereafter). Ambiguities are noted in the evaluation summary, with analysis using the best-identified entities, reflected in JSON fields `country1` and `country2`.

b. **Data Collection**: Use search tools to compile interactions between verified entities during {target_year}.

c. **Identify Significant Political Events**: List all reasonably significant political events in {target_year} that involve direct interaction between the two entities and affect or indicate the state of the bilateral relationship, verified by reliable sources, searching across the six categories of Table A.1 (Appendix C.2).

d. **Event Analysis**: For each event:
   i. Assign `country1` and `country2` as initiator/target or participants.
   ii. Provide `event_name` and `event_description`.
   iii. Classify using CAMEO (see Appendix C.3): `CAMEO_quad_class` (Verbal/Material Cooperation/Conflict), `CAMEO_root_code` (e.g., 04), `CAMEO_event_code` (e.g., 043), emphasizing economic actions (e.g., 163 for sanctions) and mediation (e.g., 045).
   iv. Assign `Goldstein_Scale` (−10.0 to +10.0, see Appendix C.3), reflecting intensity, adjusted for bilateral context but consistent with CAMEO.
   v. Assign `event_category`, the primary subcategory listed in Table A.1. Record the event date at the finest verifiable resolution and any involvement of international organizations.
   vi. Provide `evaluation_summary` justifying classifications and scores.
e. **Overall Relationship Assessment**: Select one `relationship` category for {target_year}, integrating event patterns (via CAMEO/Goldstein) and historical context. If no events are found, assess based on interaction absence and historical trends.

JSON Output. The output is a JSON object with key `historical_political_events`:

- If events found: List of objects with fields:
  - `year`, `country1`, `country2`, `event_name`, `event_description`,
  - `event_category`, `event_exact_date`, `event_month`, `event_quarter`, `event_year`,
  - `CAMEO_quad_class`, `CAMEO_root_code`, `CAMEO_event_code`, `Goldstein_Scale`,
  - `international_org_involvement`, `relationship`, `evaluation_summary`.
- If no events found: Single object with `event_name` = "No Significant Bilateral Events Found," null category, CAMEO, and Goldstein fields, and context-based `relationship`.

The scores in this paper use only the event-level `Goldstein_Scale` values. The `relationship` field is not used.

C.2. Event Identification Details

This section lists the event categories and the inclusion threshold that the prompt fixes for Step 3 of Appendix C.1.

Event Verification. The prompt instructs the model to check potential events against reliable sources and to exclude unverified or fabricated events. When events span multiple categories, the primary classification reflects the dominant mechanism or domain of impact, with secondary dimensions noted in the event description.

Event Dimensions. Events are identified across six major dimensions of bilateral relations. Each event is assigned one primary subcategory (`event_category`). Table A.1 lists all 25 subcategories, and the descriptions below summarize the main ones.

Economic Relations. This dimension covers economically mediated bilateral interactions, both cooperative and coercive.

- *Trade Policy and Market Access*: Imposition, adjustment, or removal of tariffs; non-tariff barriers such as technical standards, sanitary measures, and import licensing; trade agreement negotiations, signings, ratifications, withdrawals, or dispute settlement proceedings.
- *Financial and Monetary Relations*: Financial sanctions including asset freezes, transaction bans, and SWIFT exclusions; currency swap agreements and bilateral payment arrangements; foreign investment restrictions or liberalizations; bilateral investment treaty developments.
- *Economic Coercion and Inducements*: Comprehensive trade embargoes and sectoral sanctions; export controls and entity list designations; technology transfer restrictions; foreign aid packages, development finance, and debt relief programs.
- *Strategic Economic Sectors*: Energy supply agreements and pipeline projects; telecommunications and digital economy restrictions (e.g., 5G bans); critical resource arrangements covering rare earth elements, strategic minerals, and food security.
- *Economic Integration and Infrastructure*: Bilateral infrastructure initiatives; supply chain arrangements including friend-shoring and critical supply agreements; regional economic arrangement participation.

Diplomatic and Political Relations. This dimension captures formal state-to-state interactions and official communications that shape the bilateral relationship.

- *Formal Diplomatic Engagement*: Embassy or consulate openings and closures; ambassador appointments and recalls; diplomatic staff expulsions; formal protests, démarches, and official condemnations or commendations.
- *High-Level Political Interactions*: Presidential or prime ministerial visits and bilateral summits; ministerial meetings across foreign affairs, defense, and trade portfolios; joint commission sessions and strategic dialogues.
- *Public Diplomacy and Rhetoric*: Major policy speeches affecting bilateral relations; parliamentary resolutions; government white papers on the bilateral relationship; significant public campaigns or propaganda efforts.

Security and Defense. This dimension encompasses military cooperation, competition, and security-related incidents between the two countries.

- *Military Cooperation and Competition*: Defense cooperation agreements and military alliances; status of forces agreements; major weapons sales or cancellations; arms embargoes; joint military exercises and military-to-military exchanges.
- *Security Incidents*: Border skirmishes and territorial claim assertions; airspace or maritime violations; naval encounters and air intercepts; demarcation agreements.
- *Intelligence and Cyber Operations*: Publicly revealed espionage scandals; intelligence officer expulsions; intelligence sharing agreements or suspensions; state-sponsored cyber attacks and cyber security cooperation.

Legal, Territorial, and Movement. This dimension covers international legal proceedings, sovereignty disputes, and policies governing the movement of people.

- *International Legal Actions*: Bilateral disputes before international courts such as the International Court of Justice or International Tribunal for the Law of the Sea; WTO disputes with significant political dimensions; contentious extradition cases.
- *Territorial and Maritime Issues*: Exclusive economic zone disputes; continental shelf claims; fisheries agreements or conflicts; freedom of navigation operations; border demarcation and sovereignty recognition.
- *Movement of People*: Visa regime changes and travel bans; visa-free agreements; guest worker programs; readmission agreements; refugee and asylum policy changes.

Multilateral and Global Governance. This dimension captures bilateral dynamics manifested through international organizations and global issue areas.

- *International Organizations*: United Nations Security Council confrontations; General Assembly coalition building; specialized agency disputes; regional organization membership changes.
- *Global Issues*: Climate agreement positions and environmental cooperation; pandemic response coordination and vaccine diplomacy; human rights criticism or defense in international fora.

Other Significant Events. This residual category encompasses additional politically consequential interactions.

- *Historical and Symbolic*: Apologies for historical wrongs; memorial visits; monument disputes; anniversary commemorations with bilateral significance.
- *Humanitarian and Disaster Response*: Aid offers or rejections following natural disasters; joint rescue operations; humanitarian access disputes.
- *Technology and Space*: Joint space missions; satellite cooperation; technology theft accusations; research collaboration terminations.
- *Communications and Media*: Journalist expulsions; broadcasting restrictions; undersea cable disputes.

Inclusion Threshold. The prompt defines the threshold for inclusion in advance. Significant events include regular high-level meetings, formal agreements, trade actions, diplomatic communications, and other events that characterize the bilateral relationship, with a focus on events that have a demonstrable effect on the relationship's trajectory. When economic tools are employed for political purposes, events are classified under economic categories.

C.3. Conflict and Mediation Event Observations and Goldstein Score

Our analysis employs the Conflict and Mediation Event Observations (CAMEO) framework (Schrodt and Yilmaz 2012) to classify bilateral political events. CAMEO categorizes international political

actions along two primary dimensions: the nature of the interaction (cooperation versus conflict) and the form of action (verbal versus material).

CAMEO Classification Structure. The two dimensions define four quadrant classes:

- **Verbal Cooperation**: Diplomatic statements, consultations, expressions of intent to cooperate, and formal diplomatic cooperation including treaty signing and public endorsements.
- **Material Cooperation**: Tangible cooperative actions such as economic aid provision, military cooperation, intelligence sharing, and policy concessions.
- **Verbal Conflict**: Critical statements, accusations, demands, rejections, threats, and public protests that express disagreement or hostility.
- **Material Conflict**: Concrete hostile actions including economic sanctions, military demonstrations, coercive measures, and various forms of violence.

Within each quadrant, CAMEO provides a hierarchical coding system with root codes (two-digit) representing general action categories and event codes (three-digit) specifying precise actions. For example, root code 16 (REDUCE RELATIONS) includes event codes 163 (impose economic sanctions) and 161 (reduce diplomatic relations).

Implementation in Our Analysis. The prompt walks the model through CAMEO classification. The process begins with identifying the core bilateral action in each event description, determining its cooperative or conflictual nature, and assessing whether the action is primarily verbal or material. The LLM then selects the most appropriate root code within the identified quadrant class and chooses the specific event code that best fits the action.

We pay particular attention to economic statecraft. For instance, we distinguish broad economic sanctions (code 163) from targeted administrative sanctions (code 172). We also distinguish routine consultations (code 040) from formal agreement signing (code 057).

Integration with Goldstein Scale Scoring. CAMEO classifications provide the first reference point for Goldstein scale scoring (Goldstein 1992), which assigns numerical values from −10.0 (maximum conflict) to +10.0 (maximum cooperation). After classifying an event, the LLM evaluates its conflict–cooperation intensity against the scale's coding conventions, intensity bands, and action-based anchor cases, then allows limited adjustments for the event's specific bilateral context and severity. The anchors are the same for all dyads. The taxonomy and threshold define which events are recorded, and the Goldstein score reflects each event's intensity of conflict or cooperation.